

**PHINMA UNIVERSITY OF ILOILO**

College of Information Technology Education

## **CAPSTONE PROJECT PROPOSAL**

**“A Web-Based Integrated School Management System for Tigbauan Fundamental Learning Center, Inc.”**

In partial for the requirements in Capstone 1 and Research Subject

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## **CHAPTER 1**

### **INTRODUCTION**

#### **Background of the Study**

Many educational institutions, particularly small private schools located in rural and underserved areas, continue to operate within traditional frameworks where they still heavily rely on outdated, manual processes for managing their most critical daily operations, including the maintenance of comprehensive student records, accurate daily attendance tracking, systematic inventory control, and effective communication between key stakeholders such as administrators, teachers, parents, and learners.

Despite the increasing availability of modern technology and the rapid digitization of various sectors worldwide, these institutions often lag significantly behind in digital adoption, functioning much as they did decades ago while the world around them evolves rapidly toward data-driven efficiency and connectivity.

This delay is rarely due to a lack of desire to improve, innovate, or modernize, but rather stems from practical barriers such as limited and constrained budgets that cannot accommodate expensive commercial software, a scarcity of technical expertise or knowledgeable personnel within the school to manage complex systems, insufficient awareness of available affordable solutions tailored to their specific scale, or deep-seated concerns regarding the complexity of implementing new systems, the difficulty of training staff, and the risk of disrupting established routines.

Consequently, these schools remain dependent on conventional methods—paper records, handwritten logs, and manual calculations—that result in operational workflows that are not only labor-intensive, slow, and repetitive but also highly susceptible to human error, accidental oversight, and organizational disarray.

The cumulative effect of these inefficiencies creates a heavy administrative burden that diverts valuable time, energy, and intellectual resources away from the core mission of the institution: delivering quality education, facilitating meaningful learning experiences, and supporting the holistic growth and development of every student entrusted to their care.

At Tigbauan Fundamental Learning Center, Inc., the current operational system depends entirely on paper-based files, handwritten logbooks, and basic spreadsheet applications to manage its administrative and academic requirements—a practice that has been in place since its establishment and is deeply embedded in its daily routine, culture, and organizational habits.

While these traditional methods are familiar, straightforward, and require no specialized software or significant hardware investment, they frequently lead to significant operational inefficiencies and administrative burdens that grow heavier, more complex, and more difficult to manage as the school population increases, academic programs expand, and regulatory requirements become more demanding and rigorous.

Every transaction, from the initial enrollment of a learner and the recording of daily attendance to the computation of grades and the eventual release of report cards, requires the physical handling, transfer, and storage of documents, creating a system that is not only slow and cumbersome but also increasingly difficult to manage, scale, and sustain over time.

This heavy reliance on manual documentation creates a fragile operational environment where vital information is scattered across multiple filing cabinets, notebooks, and isolated computer files, making it exceptionally difficult to maintain consistency, ensure data integrity, or guarantee that accurate information is always accessible to those who need it, when they need it.

These manual methods result in extremely slow data retrieval times, where finding a specific student record, past attendance history, or previous year's academic performance can take considerable effort and time, often requiring staff to sort through piles of folders or stacks of logbooks, sometimes spanning several years of operation.

Beyond just being time-consuming and frustrating, there is a heightened and persistent risk of records being misplaced, damaged, or permanently lost due to physical deterioration, filing errors, accidental misplacement, or environmental factors such as humidity, pests, flooding, or fire—risks that directly threaten the institutional memory, historical continuity, and legal compliance of the school.

Furthermore, the preparation of essential documents and reports—such as enrollment summaries, grade sheets, report cards, or attendance reports for submission to government agencies like the Department of Education—is an excessively laborious process.

These tasks require repeated manual computation, transcription, and consolidation of data from different sources, which not only consumes hours of valuable working time that could be better spent on instructional leadership or student support but also frequently leads to calculation errors, inconsistent formatting, missing data fields, and significant delays in decision-making or report submission to regulatory bodies.

When errors inevitably occur, tracing the root cause, verifying the correct information, or correcting the data is a complex, tedious, and frustrating process that further delays operations, disrupts schedules, and affects the reliability and credibility of the school's official records and standing.

Furthermore, communication between teachers and parents remains highly fragmented, uncentralized, and inconsistent, relying largely on ad-hoc methods such as text messages, personal phone calls, informal handwritten notes sent home with students, or occasional group chats on social media platforms.

While these channels work for simple reminders or casual updates, they lack structure, security, privacy, and permanence, often resulting in important messages being overlooked, misunderstood, or forgotten entirely.

This lack of a standardized and structured communication channel often results in delayed notifications regarding important school activities, sudden schedule changes, or emergency announcements that require immediate attention and action.

More critically, it leads to missed or misinterpreted updates on student performance, academic standing, behavioral concerns, or attendance issues—information that is vital for timely intervention, corrective action, and supportive partnership between home and school. There is also no systematic way to document, archive, or retrieve these interactions, meaning important agreements, commitments, or updates can easily be disputed or lost over time.

Consequently, there is an overall poor level of coordination, transparency, and engagement between the school and the home environment—a relationship widely recognized by educators, researchers, and child development experts as a vital factor in supporting student development, motivation, well-being, and long-term academic success.

Parents often express difficulty in tracking their children's progress in real-time or understanding exactly how their child is performing relative to curriculum standards and learning competencies, as feedback is often limited to quarterly meetings or the distribution of report cards every grading period.

They also face significant challenges in staying updated on school events, requirements, schedules, or financial obligations, leading to confusion, missed deadlines, and occasional non-compliance or misunderstanding regarding school policies and procedures. On the other hand, teachers spend valuable hours outside of class hours following up on inquiries, repeating the same information to different parents individually, or disseminating information through inefficient, one-way channels—time that should instead be dedicated to lesson planning, instructional improvement, professional development, or providing one-on-one academic and emotional support to learners who need it most.

This constant cycle of information chasing and administrative overload disrupts their primary instructional duties and contributes significantly to teacher fatigue, burnout, and job dissatisfaction, which ultimately impacts the quality of teaching and the learning atmosphere within the classroom.

Inventory management and asset monitoring are similarly handled entirely through manual procedures, utilizing paper logs and physical record books to track the movement of classroom supplies, teaching materials, library books, borrowed equipment, furniture, and other school assets.

This approach makes it exceptionally challenging to accurately monitor real-time stock levels, efficiently track item check-outs and returns, identify loss, damage, or wastage immediately, or pinpoint exactly where resources are located within the school premises. Without a centralized system to record and update these transactions, maintaining consistently updated records necessary for accurate budgeting, financial planning, cost analysis, and timely procurement of resources becomes nearly impossible.

Staff members often have to physically inspect storage areas and classrooms just to verify availability or confirm the status of an item—a task that is inefficient, labor-intensive, and highly prone to inaccuracy and oversight. Consequently, the school frequently encounters issues such as unexpected shortages of essential learning materials at critical times like the start of classes, overstocking of non-essential items that tie up limited funds and storage space, and serious difficulties in accounting for assets during internal or external audits conducted by regulatory bodies or funding agencies.

These inventory-related problems strain already limited financial resources, lead to unnecessary waste or repeated purchases of items that are already available but unrecorded, and ultimately disrupt the smooth delivery of educational services and classroom activities, affecting both teachers and students alike.

With the continuous and rapid advancement of information and communication technologies—including cloud computing, web-based applications, mobile computing, and widespread internet connectivity—there is a growing and urgent need for educational institutions, regardless of size or location, to adopt robust digital solutions that can dramatically improve operational efficiency, ensure higher levels of data accuracy and integrity, and provide seamless accessibility to information from any location and at any time.

Modern technology offers the capability to transform administrative work from a heavy burden into a streamlined, automated process, ensuring that data is captured once and reused many times across different functions, significantly reducing redundancy, repetition, and the likelihood of human error.

A well-designed web-based system empowers authorized users—such as school administrators, teaching staff, and registered parents—to access vital data securely and conveniently via internet-connected devices like smartphones, tablets, or laptops, breaking down barriers of distance and time.

At the same time, these systems ensure superior organization through structured databases, standardized workflows, and enhanced security features such as data encryption, automated backups, and role-based access permissions that strictly protect sensitive student and institutional information from unauthorized access, data breaches, or accidental loss, ensuring full compliance with privacy laws and regulations.

The widespread adoption of digital school management systems has already proven remarkably successful in numerous educational institutions across the globe, ranging from large urban public schools in developed nations to small rural academies in emerging economies. In every context, these systems consistently demonstrate substantial improvements in operational performance, transparency, and accountability, creating a more organized and professional environment.

Research and implementation reports indicate that such systems can reduce routine administrative tasks by up to 70%, allowing educators and administrators to redirect their focus, energy, and creativity toward core educational goals such as curriculum development, enhancing instructional quality, and providing targeted student support services that directly impact learning outcomes.

Furthermore, these systems have been shown to generate markedly higher levels of satisfaction among staff, parents, and students alike, thanks to intuitive user interfaces, transparent information sharing, and reliable system performance that fosters greater trust, confidence, and active participation in the institution's management and community life.



When stakeholders have easy access to accurate information and clear communication channels, the entire school community functions more harmoniously, effectively, and collaboratively.

This study is particularly timely and relevant, as the Philippine government, through the Department of Education (DepEd), continues to strongly promote digital transformation and modernization in school learning platforms and administrative systems via key national initiatives such as the MATATAG Agenda and the Digital Education Act.

These policy frameworks aim to equip all educational institutions—both public and private—with modern tools, resources, and infrastructure necessary for equitable, accessible, inclusive, and high-quality education delivery in the digital age. The national agenda emphasizes that technology integration is no longer merely an option or a luxury for elite institutions, but a necessity for improving governance, ensuring accountability, raising standards, and enhancing the overall quality of basic education across the country.

This mandate specifically includes schools in remote and rural areas where the digital divide remains a significant challenge to be addressed, ensuring that no school, teacher, or learner is left behind in the country's progress toward modernization and global competitiveness.

By developing a customized, web-integrated school management system specifically designed for Tigbauan Fundamental Learning Center, Inc., this research not only aims to contribute meaningfully to the national effort of modernizing basic education but also seeks to directly address the unique operational challenges and constraints faced by small to medium-sized private learning centers in rural communities.

These institutions often face distinct hurdles such as limited financial resources, intermittent or low-speed internet connectivity, lack of dedicated technical support staff, and the critical need for affordable, simple, and user-friendly technology solutions that do not require complex maintenance or specialized training to operate effectively. Unlike generic commercial systems which are often expensive, overly complex, and designed for large-scale institutions, the proposed system is designed with these contextual realities in mind, ensuring that it remains accessible, functional, and sustainable even within limited-resource settings while still delivering the full benefits of professional digital management, security, and scalability.

To comprehensively address these multifaceted challenges and operational gaps identified in the current processes, this study proposes the development and implementation of a fully integrated web-based school management system tailored for Tigbauan Fundamental Learning Center, Inc., which will serve as a unified digital hub for all school operations, bringing together previously disconnected functions into one cohesive platform.

The system is designed to streamline student record management by digitizing personal profiles, academic histories, grade records, and enrollment data, allowing for instant retrieval, secure centralized storage, and easy updating of information while maintaining complete historical records. It will automate attendance tracking through innovative tools such as QR code scanning or biometric verification to eliminate manual errors, prevent proxy attendance, and generate real-time analytical reports that help identify patterns in student attendance and engagement. It will also significantly improve inventory monitoring through barcode integration, automated alerts for low stock levels, and detailed transaction histories that simplify auditing, procurement planning, and resource allocation.

Finally, it will enhance communication between teachers and parents via a centralized platform featuring secure direct messaging, public announcement boards, real-time academic progress updates, and interactive event calendars, ensuring that everyone stays informed, involved, and connected.

Ultimately, this project aims to create a cohesive digital ecosystem that boosts administrative productivity, reduces operational costs, improves data security and compliance, and fosters stronger collaboration between all stakeholders, leading to better educational outcomes and a more organized, efficient, responsive, and supportive learning environment for every member of the school community.

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## Statement of the Problem

This study aims to develop a comprehensive web-based school management system that seamlessly integrates key functionalities—including student records management, real-time attendance monitoring, efficient inventory tracking, and streamlined communication tools—into a single, centralized platform to enhance operational efficiency, data accessibility, and collaboration among school administrators, teachers, parents, and students.

At present, many educational institutions continue to rely heavily on manual processes, paper-based documentation, or disjointed software applications to handle their daily administrative and academic operations, creating a fragmented ecosystem that struggles to keep pace with the growing demands of modern education. These conventional methods often result in significant challenges such as unnecessary data duplication, inconsistent record-keeping standards, and extreme difficulty in retrieving or updating information when it is most needed, alongside a consistently increased risk of human error, data loss, or accidental damage to important physical files.

Furthermore, attendance reports are often compiled days or weeks after the fact and shared late or irregularly, making it nearly impossible to promptly identify issues regarding student punctuality or absenteeism, while inventory items such as learning materials, equipment, and supplies are frequently unaccounted for, poorly monitored, or overstocked due to the lack of a systematic tracking mechanism. Additionally, communication between the school and its stakeholders is fragmented across various separate channels—ranging from written notices and text messages to emails and group chats—leading to frequent misunderstandings, missed updates, and a general lack of clarity regarding school activities, policies, or student progress.

Such inefficiencies consume vast amounts of valuable time and financial resources that could otherwise be directed toward improving the quality of education, hinder timely and data-driven decision-making by school leaders, and create substantial barriers to

effective coordination between all parties involved in the learning environment. The absence of a unified system also compromises data security and privacy, as sensitive information is scattered across different locations or formats, and limits accessibility, forcing users to be physically present in specific offices to access necessary records or submit requirements.

Therefore, there is a critical and urgent need for a unified digital solution that consolidates these essential functions into one secure environment, automates routine administrative tasks to reduce workload, ensures high levels of data accuracy and confidentiality, and provides instant access to relevant information regardless of location or time zone. By addressing these existing gaps and limitations in current practices, the proposed system seeks to simplify complex administrative workflows, significantly improve transparency and accountability in school operations, and ultimately foster a more connected, informed, and productive educational community where administrators, teachers, parents, and students can work together more effectively toward academic excellence.

1. How can student attendance be recorded and monitored more efficiently?
  2. How can student records be stored, organized, and retrieved quickly?
  3. How can the school effectively manage and monitor inventory and learning materials?
  4. How can communication between teachers and parents be improved and centralized?
  5. How can reports be generated automatically to reduce manual workload?
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## Objectives of the Study

### 1.1.1 General Objective

To design and develop a comprehensive web-based integrated school management system tailored specifically for Tigbauan Fundamental Learning Center, with the primary objective of significantly enhancing the efficiency, accuracy, and accessibility of all school-related operations and services; this project aims to revolutionize how the institution manages its daily activities and long-term processes, setting a strong, sustainable foundation for digital transformation, operational excellence, and continuous service improvement within the institution.

This proposed system will serve as a unified and centralized digital platform that seamlessly consolidates all essential institutional functions—including detailed student records management, real-time attendance monitoring and reporting, systematic inventory tracking of school assets and supplies, and robust communication tools—effectively replacing the cumbersome manual workflows, voluminous paper-based documentation, and fragmented software solutions currently in use, which often lead to unnecessary delays, data inconsistencies, considerable difficulty in retrieving critical information when it is needed most, and additional workload that drains valuable time and energy from school personnel.

By automating routine administrative tasks such as repetitive data entry, complex report generation, regular record updates, and labor-intensive inventory audits, the system ensures that all information is consistently synchronized across departments, securely stored in a reliable, structured database, and instantly accessible to authorized users with appropriate permissions, thereby drastically reducing human errors, eliminating redundant work processes, preventing data duplication or permanent loss, and freeing up significant amounts of time and energy for school staff; this allows educators and administrators to focus more fully on core educational goals, student development, instructional quality improvement, and strategic planning rather than being burdened by excessive paperwork, tedious manual verification, and heavy administrative overhead that severely limits productivity and hinders operational growth and innovation.

Furthermore, the platform will implement a secure, role-based access control framework, meaning that administrators, teachers, parents, and students will each have personalized access to relevant features and data according to their specific needs, responsibilities, and authority levels, ensuring that sensitive information—such as personal details, academic records, and financial data—remains strictly protected against unauthorized access, misuse, or data breaches while simultaneously empowering every stakeholder to retrieve necessary information, submit updates or

requests, monitor academic and operational progress, receive urgent or timely announcements, and engage in meaningful, documented collaboration anytime and from anywhere with a stable internet connection, effectively breaking down barriers caused by distance, busy schedules, or limited availability.

Beyond basic operational improvements, the system will also be designed with scalability and adaptability in mind, capable of supporting the institution's future growth by accommodating significant increases in enrollment, expanded inventory needs, introduction of new academic programs, and evolving institutional requirements without requiring major overhauls or complete system replacements, while also generating detailed, accurate analytics and comprehensive reports that offer clear, actionable visibility into student performance trends, attendance patterns, inventory status and usage, and overall operational performance.

Ultimately, this integrated system will not only streamline daily operations and strengthen data security and compliance protocols but also provide reliable, real-time insights that support evidence-based, informed decision-making at every level of the institution, foster stronger, more transparent partnerships between the school and families through open engagement and timely, consistent communication, reduce long-term operational costs associated with paper supplies, printing, physical storage, and manual labor, and contribute substantially to raising the overall standard and quality of service delivered by Tigbauan Fundamental Learning Center, thereby positioning the institution to adapt better to modern educational standards, meet and exceed the expectations of its community and regulatory bodies, and serve its learners, staff, and stakeholders with greater excellence, accountability, and reliability for many years to come.

Additionally, the development and implementation of this system are guided by the commitment to create a user-friendly interface designed with intuitive navigation and clear layouts, ensuring that even users with limited technical knowledge or digital literacy can easily navigate, utilize, and maximize the platform's full potential without requiring extensive training or constant technical support. It also aims to establish a cohesive digital ecosystem where every transaction, record modification, and communication is automatically logged and archived, creating a complete audit trail that enhances transparency, simplifies compliance reporting for regulatory agencies, and preserves the institutional history and data integrity for future reference and evaluation.

By integrating modules that handle not only student and administrative data but also financial management aspects such as tuition fee tracking, payment monitoring, and expense recording, the system provides a holistic view of the school's resources and finances, helping administration maintain fiscal responsibility, prevent revenue leakage, and ensure that financial resources are allocated effectively to support educational programs and facility improvements.

Moreover, the system is intended to bridge the gap between home and school by offering parents and guardians a clear window into their children's academic journey, allowing them to view grades, check attendance history, receive notifications regarding school events or disciplinary matters, and communicate directly with teachers, thereby encouraging active parental involvement which is widely recognized as a key factor in student success and motivation.

It further seeks to standardize processes across the entire institution, ensuring that policies, procedures, and data formats remain consistent regardless of staff changes or departmental shifts, which in turn creates a stable, predictable, and professional environment that benefits both employees and students alike.

Through the continuous gathering and analysis of operational and academic data, the system will also enable the school to identify strengths, weaknesses, opportunities, and challenges more quickly and accurately, allowing management to implement proactive measures, refine educational strategies, and introduce innovative programs that better suit the evolving needs of the student population and the community it serves.

Finally, this initiative represents a vital step toward modernizing the educational framework of Tigbauan Fundamental Learning Center, aligning its operational capabilities with global standards of educational management, and ensuring that the institution remains competitive, relevant, and capable of delivering high-quality education in an increasingly digital and interconnected world, while cultivating a culture of innovation, efficiency, and continuous improvement that will define the institution's legacy and impact for generations to come.

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### **1.3.2 Specific Objectives**

Specifically, the study aims to:

1. Automate the recording and monitoring of student attendance, enabling real-time data capture and instant updates that completely transform the way daily student participation is tracked, recorded, and managed within the institution. This feature is designed to allow teachers to easily and efficiently log attendance through user-friendly web-based interfaces accessible from any connected device—whether inside the classroom or remotely—significantly reducing the time and effort previously spent on manual checking, copying entries into logbooks, and managing paper-based attendance sheets.

By shifting from traditional roll calls to a streamlined digital process, the system ensures that attendance data is captured accurately at the source and immediately saved to the central database, eliminating delays in information processing and distribution. It also helps administrators quickly generate comprehensive attendance summaries, statistical breakdowns, and trend analyses whenever required, presenting data in clear formats such as percentages, charts, or comparative reports that highlight patterns over days, weeks, months, or entire academic years.

Furthermore, the system will be configured to automatically notify authorized users—such as class advisers, guidance coordinators, or parents or guardians—regarding instances of absenteeism, excessive tardiness, or emerging patterns of non-attendance through built-in alerts, messages, or notifications, ensuring that appropriate and timely intervention measures can be implemented immediately to support the student and address potential issues before they escalate.

By establishing this automated process, the system guarantees accurate, consistent, and reliable record-keeping, completely eliminates errors caused by manual transcription, illegible handwriting, or lost forms, and provides a clear, verifiable, and traceable history of attendance that supports both student welfare monitoring and strict adherence to institutional compliance standards and government reporting requirements.

2. Centralize and securely store all student records in a comprehensive digital database, systematically organizing critical information that includes personal profiles, enrollment history, class assignments, subject schedules, academic performance records, grade histories, disciplinary reports, health information, and long-term historical data spanning the student's entire period of study from admission to graduation.



This structured repository ensures that every piece of student-related information is logically categorized, properly indexed, easily searchable, and always accessible to those who need it, while being strictly protected through robust security protocols, end-to-end data encryption, and proper role-based access controls that define exactly what information each user group can view, edit, or manage, effectively preventing unauthorized viewing, accidental modification, intentional tampering, or data leakage.

Authorized personnel such as school administrators, class advisers, subject teachers, and guidance staff will be able to efficiently retrieve, update, or manage records in mere seconds, whether for routine inquiries, academic evaluations, counseling sessions, curriculum planning, or official reporting purposes, thereby completely eliminating the common and costly issues associated with physical file management—such as lost documents, water-damaged papers, misfiled records, fading ink, or disorganized storage systems that make finding information a tedious and time-consuming task.

Beyond simple storage, this digital archive ensures the long-term preservation, integrity, and confidentiality of sensitive student data, adhering to data privacy regulations and institutional policies, while ensuring that information remains intact, retrievable, and usable even over many years of operation, creating a reliable institutional memory that supports the school's legacy and accountability.

3. Develop an inventory tracking system capable of effectively managing all school resources, covering a wide range of items including educational equipment, learning materials, textbooks, library resources, office supplies, laboratory tools, furniture, technological devices, and other institutional assets essential for daily operations and effective instruction.

This system will actively and continuously monitor current stock levels in real time, accurately track the movement, usage, and distribution of items across different departments, classrooms, or individual users, and maintain detailed logs containing procurement details, supplier information, acquisition dates, cost values, warranty periods, maintenance schedules, and depreciation status where applicable.

It will also allow staff to record transactions such as issuances, returns, transfers, or disposals, creating a complete audit trail for every item from the moment it enters the school premises until it is removed from service.

Additionally, it will feature automated alert mechanisms that notify designated personnel—such as property custodians, supply officers, or finance staff—whenever stock levels reach a predefined minimum threshold, when items are due for scheduled

maintenance or inspection, or when replenishment or reordering is necessary, effectively helping the institution prevent unexpected shortages that could disrupt classes or school activities, reduce unnecessary waste and overstocking of non-essential items, optimize budget allocation by providing accurate data on usage patterns and consumption rates, and ensure the optimal utilization, accountability, and stewardship of all available resources to fully support the school's educational goals and overall operational efficiency.

4. Create a dedicated online communication portal designed exclusively for parents, teachers, and administrators, serving as a centralized and unified channel for the exchange of important information, official announcements, constructive feedback, general inquiries, and detailed student progress updates throughout the academic year. This feature is built to facilitate transparent, consistent, and timely interaction between all key stakeholders, breaking down traditional communication barriers such as distance, conflicting schedules, or limited contact hours, and ensuring that critical information—such as schedule changes, event announcements, suspension of classes, academic milestones, behavioral notes, attendance alerts, or grade reports—reaches the intended audience promptly and reliably, regardless of their location or time zone.

The platform supports various forms of communication, from public bulletins visible to all registered users to private, secure messaging between parents and teachers regarding specific concerns, ensuring that conversations are documented, archived, and retrievable whenever necessary.

By providing a safe, structured, and secure space for dialogue and engagement, the portal strengthens the collaborative partnership between the school and families, encourages active parental involvement and participation in education, and ensures that everyone involved in the student's learning journey remains informed, connected, and aligned in their efforts to support holistic student development, well-being, and long-term academic success.

5. Generate accurate, detailed, and automated reports tailored for both academic and administrative use, covering a wide variety of essential documents such as daily and monthly attendance summaries, student performance evaluations, grade sheets, class rankings, inventory status reports, asset management logs, financial summaries, and overall operational activity records.

These reports are designed to be produced instantly upon request or scheduled for automatic generation, automatically compiled from real-time data stored in the system, and formatted systematically according to standard institutional requirements, regulatory

guidelines, or specific user preferences, ensuring consistency, clarity, and professionalism in every output produced.

Users will also have the flexibility to customize reports by selecting specific data fields, applying filters, or choosing formats such as tables, graphs, or charts to better visualize trends and patterns.

By providing reliable, data-driven insights and clear visual summaries, these reports serve as a solid foundation for evidence-based decision-making at all levels of management, helping school leaders identify strengths, address challenges, allocate resources effectively, and plan strategically for the future.

Furthermore, they help simplify and speed up compliance requirements for government agencies or accrediting bodies by providing ready-to-submit, accurate documentation, and significantly streamline the entire process of preparation, compilation, and submission of necessary documents, drastically reducing the manual workload, time consumption, and potential errors traditionally associated with report generation and record preparation.

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## **1.2 Scope and Delimitation**

The project focuses on the systematic design, full-cycle development, comprehensive testing, and thorough evaluation of a Web-Based Integrated School Management System specifically tailored to address the unique operational requirements, administrative workflows, and academic structures of Tigbauan Fundamental Learning Center, Inc.

The primary objective of this undertaking is to fully digitize the institution's existing manual, paper-based, and spreadsheet-dependent processes—systems that have long been identified as significant sources of operational inefficiency, data redundancy, human error, processing delays, and difficulty in information retrieval. By strategically transitioning these traditional workflows into a secure, dynamic, and fully functional digital environment, the study aims to significantly enhance the accessibility, accuracy, security, and overall integrity of critical institutional data, while drastically improving administrative efficiency, staff productivity, and the quality of service delivery across all departments and functions of the school.

Furthermore, the project seeks to establish a unified, centralized digital platform that seamlessly integrates various essential school functions and serves the distinct yet interconnected needs of all key stakeholders within the school community—including school administrators, teaching and non-teaching personnel, parents or legal guardians, and duly enrolled students—ensuring that every authorized user has appropriate, role-based access to relevant information and functional tools that align strictly with their specific duties, responsibilities, and level of clearance.

The scope of the study covers the following key features and functionalities, which have been carefully analyzed, selected, designed, and developed to directly resolve the identified operational gaps and challenges currently experienced by the institution:

**Student Information Management:** This module serves as the core repository and central database foundation of the entire system, encompassing comprehensive functionalities that effectively cover the complete lifecycle of student data management from admission to graduation or withdrawal. It includes streamlined, user-friendly

processes for digital enrollment and admission, allowing for the systematic collection, validation, and secure storage of essential personal details, contact information, family background and emergency contacts, pertinent medical data, and other required documentation during the entry phase.

Beyond initial registration, it supports dynamic and real-time profile updating, enabling authorized administrative staff to modify, correct, and maintain current information whenever necessary to ensure continuous data relevance, accuracy, and completeness. Furthermore, it facilitates the systematic archiving and long-term preservation of critical academic records, such as subjects enrolled per term, quarterly and final grades, cumulative academic performance reports, promotion or retention status, and detailed attendance history throughout the student's entire stay in the institution.

By structuring, organizing, and consolidating this vast amount of information into a secure, centralized database, the system ensures that all student records are consistently organized, protected through multi-layered access controls and security measures, and instantly retrievable whenever needed by authorized personnel only, thereby completely eliminating the delays, physical storage problems, difficulties in searching, and risks of loss or damage associated with manual filing systems and the physical handling of paper documents.

**Attendance Tracking:** This feature is designed to completely replace traditional logbooks, sign-in sheets, and manual recording methods with a robust, reliable, highly accurate, and user-friendly digital logging system that is fully accessible via standard web browsers on desktop computers, tablets, or mobile devices. It allows class advisers and subject teachers to record daily attendance status—clearly indicating whether a learner is present, absent, tardy, or has an approved excused leave—quickly, accurately, and consistently online within just a few simple clicks, even from different locations within the school premises.

The functionality goes far beyond simple data entry, as the system is programmed with automated logic and computational capabilities to instantly calculate attendance percentages, track recurring patterns of absence or lateness, flag irregular attendance behaviors, and generate organized monthly summaries, term-based statistics, and comprehensive attendance reports that comply with educational standards.

These outputs are made available in real-time to classroom advisors and school administrators, enabling effective monitoring of student participation, early identification of chronic absenteeism or irregular attendance issues, and data-driven support for student evaluation and intervention programs, while also ensuring that the school maintains precise, legally compliant records necessary for regulatory reporting requirements set by educational agencies and governing bodies.

**Inventory Management:** Developed to serve as a centralized digital ledger, monitoring hub, and control mechanism, this module facilitates the efficient tracking, systematic management, and proper accountability of all the school's physical assets, learning resources, and operational supplies. This covers a wide range of items including laboratory equipment, library books and reading materials, instructional and teaching aids, classroom furniture, technological devices and computer hardware, office supplies, maintenance materials, and other resources essential for daily academic and administrative operations.

The system streamlines the entire asset lifecycle—from procurement and receiving to utilization, maintenance, and disposal—by featuring automated logging of borrowing and returning transactions, real-time monitoring of current stock levels and quantity on hand, and detailed tracking of item status including its exact location, physical condition, history of usage, maintenance schedules, and depreciation details.

Additionally, it can be configured to generate automatic alerts, notifications, or reminders when stocks reach critical reorder levels, when items are due for return, or when assets require maintenance checks. By implementing this strict digital control mechanism, the system helps significantly prevent loss, misplacement, damage, or misuse of school property, ensures optimal utilization and sharing of available assets across departments, and supports timely procurement planning and strategic resource allocation decisions to prevent shortages or unnecessary overstocking, ultimately contributing to cost reduction and better resource stewardship.

**Parent-Teacher Portal:** This component provides a dedicated, secure, and highly interactive communication platform specifically designed to bridge the existing information gap that often separates the school administration, teaching staff, and the students' homes. Utilizing advanced web technologies such as AJAX and WebSocket protocols, this portal enables true synchronous messaging capabilities, facilitating real-time interaction, instant feedback, and seamless communication between parents, teachers, and school administrators regardless of distance or time constraints.

This feature supports various forms of communication, ranging from private direct messaging regarding individual student concerns, behavioral updates, or academic performance, to the broader dissemination of general announcements, school circulars, event invitations, schedule changes, memorandum releases, and general feedback or advisory notes.

By housing all communication within this integrated platform, the system ensures that all interactions are secure, confidential, properly documented, permanently archived, and easily retrievable for future reference, while eliminating reliance on external messaging services, informal channels, or unregulated group chats, and removing the recurring costs and character limitations traditionally associated with bulk SMS notifications or printed circular distribution.

**Report Generation:** This reporting module is developed to simplify, accelerate, standardize, and professionalize the preparation of essential administrative, academic, and operational documents required for internal management, monitoring, and external submission to regulatory bodies. It enables authorized users to perform one-click generation of standardized as well as highly customizable reports such as Student Masterlists, Class Rosters, Summary of Grades, Attendance Summaries, Inventory Status Reports, Transaction Logs, and Compliance Documents.

Each generated report is carefully formatted to meet professional standards and official requirements set by educational authorities, and can be conveniently viewed directly on-screen for quick review, printed for physical filing and documentation purposes, or exported into widely accepted digital formats such as PDF for secure sharing and archiving or Excel files for further data analysis and computational processing.

This functionality significantly reduces the time and human effort spent on manual report preparation and data consolidation, minimizes the risk of computational and transcription errors, and provides reliable, accurate, and up-to-date documentation that supports precise record-keeping, audit readiness, and facilitates informed, data-driven decision-making by school management and administrative leaders.

In terms of delimitation, the scope of the study is strictly confined to the development, implementation, and evaluation of the modules and features mentioned above, and the system is specifically customized to fit the unique organizational structure, academic programs, curriculum offerings, and operational policies of Tigbauan Fundamental Learning Center, Inc. It does not cover the development of a full-scale Learning Management System (LMS) intended for conducting virtual classes, hosting online lectures, delivering course content, or facilitating digital assignment submissions and online examinations.

Furthermore, the system does not integrate online payment gateways, electronic banking facilities, or process any form of financial transactions; any fee-related information or financial data displayed within the system is included solely for monitoring, record-keeping, and reference purposes only, and does not serve as a replacement for dedicated accounting or financial management software.

Additionally, the operation of the system is inherently dependent on the availability of stable and consistent internet connectivity and compatibility with modern web browsers; technical support or troubleshooting regarding hardware malfunctions, power interruptions, server downtime, or issues related to the internet service provider (ISP) and network infrastructure are explicitly outside the scope and coverage of this research and development project.

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### 1.3 Significance of the Study

The development and implementation of this Integrated School Management System hold significant value and provide substantial, far-reaching benefits to the following academic stakeholders, serving as a vital instrument in the modernization of school operations while fostering a more connected, efficient, transparent, and productive educational environment that fully aligns with current standards of quality education, administrative excellence, and institutional governance:

□ **School Administrators:** This system serves as a powerful, comprehensive management tool that offers school administrators a centralized, intuitive dashboard designed to oversee, monitor, and control all institutional operations, resources, and data flows from a single, accessible interface accessible anywhere and anytime.

It ensures the highest level of data integrity, consistency, and security by effectively eliminating data redundancy, conflicting records, and human errors that are commonly prevalent in manual, paper-based, or fragmented systems, thereby providing a solid, reliable foundation for informed, evidence-based, and strategic decision-making.

By having instant, real-time access to critical data regarding student enrollment records, academic standing and performance metrics, daily attendance trends and statistics, inventory status and precise asset location, as well as a complete audit log of all institutional communications and transactions, administrators can efficiently monitor the school's overall performance, enforce policies and regulations consistently and effectively across all departments, and plan for strategic improvements, infrastructure development, or resource allocation with much greater accuracy, confidence, and foresight.

Furthermore, the availability of automated, standardized, and customizable reports significantly simplifies compliance with government regulatory requirements, reduces the heavy workload associated with auditing, accreditation, and documentation, and

empowers management to lead the institution with greater transparency, full accountability, and sustained operational excellence.

□ **Teaching Faculty:** For teachers and academic personnel, this technological development significantly reduces the heavy administrative burden and clerical workload traditionally associated with their professional roles by automating repetitive, time-consuming, and routine tasks such as daily attendance recording, computation and tabulation of grades, inventory logging of teaching materials and equipment, and the preparation of various academic and administrative reports.

By minimizing the time and effort spent on extensive paperwork, manual calculations, and physical record-keeping, the system allows educators to redirect their valuable focus, energy, and attention toward their primary and most important responsibilities: enhancing instructional quality, refining and improving teaching strategies, and supporting the holistic development of every learner. With simplified, streamlined processes and the ability for instant retrieval of complete student information—including academic history, attendance patterns, health records, and relevant personal details—teachers can better understand the specific needs, strengths, and challenges of their learners, dedicate more quality time to lesson planning, curriculum development, and interactive classroom instruction, and provide more timely, meaningful feedback and targeted intervention, ultimately elevating the overall quality of education delivered to the learners and fostering a more supportive learning environment.

□ **Non-Teaching Staff:** This system greatly benefits non-teaching personnel and support staff—including those in procurement, supply, accounting, and general services—by streamlining resource management, procurement workflows, and logistical operations through fully digitized inventory tracking and comprehensive asset monitoring features.

It drastically reduces the occurrence of manual errors, data discrepancies, and the overwhelming volume of physical paperwork, ledgers, and filing that typically slows down daily operations and complicates record retrieval.

The platform facilitates systematic, accurate, and real-time monitoring of all school assets, consumable supplies, equipment, and materials, simplifies the recording of issuance, transfer, maintenance, and return of items, and automates critical updates regarding current stock levels, item status, and reorder requirements.

Consequently, this helps staff maintain optimal and adequate inventory levels to prevent disruptive shortages or unnecessary overstocking, effectively prevent loss, damage, deterioration, or misuse of school resources, and significantly speed up documentation, auditing, and reporting workflows required by the administration and external agencies. This leads to a more organized supply chain, optimized utilization of budget and funds, and a smoother, more efficient operational flow that effectively supports the entire institution's functions and goals.

□ **Parents and Guardians:** The study strengthens and enhances the vital school-home partnership by providing parents and guardians with direct, secure, and immediate access to essential information regarding their children's education, including daily attendance records, academic progress and grading details, behavioral updates and observations, and schedules of upcoming school activities, examinations, or events. Through a dedicated, user-friendly, and secure portal or mobile interface, the system facilitates secure, direct, and real-time communication channels with teachers and school administrators, effectively removing barriers, delays, and uncertainties often found in traditional notification methods such as newsletters or occasional meetings.

This constant and transparent access keeps parents consistently well-informed, actively engaged, and deeply involved in their child's educational journey, strengthens the level of transparency, confidence, and trust between the school and the families it serves, and allows for timely collaboration and constructive dialogue in addressing any concerns, challenges, or specific needs regarding the student's welfare, conduct, and academic performance, thereby creating a unified support system for the learner.

□ **Students:** While students are not direct operators or managers of all administrative modules within the system, they stand to gain significant indirect yet highly impactful benefits from the improved efficiency, accuracy, transparency, and organization brought about by the implementation of this technology.

It ensures that their personal profiles, enrollment details, and academic records are accurately maintained, securely stored, properly monitored, and consistently updated throughout their entire stay in the institution, significantly reducing the risk of errors, omissions, or data loss that could potentially affect their academic standing, certification, or transfer credentials.

Additionally, because the system supports better inventory management, timely procurement, and strategic resource allocation, students are assured of a learning environment supported by well-managed facilities, readily available learning materials, functional equipment, and organized resources necessary for effective instruction

Furthermore, the enhanced communication loop established between school and home ensures that their academic growth, attendance regularity, and behavioral development are closely tracked, guided, and supported by both educators and parents, creating a more organized, responsive, and supportive educational setting that prioritizes their academic success, safety, well-being, and overall development.

□ **Future Researchers:** This research and its resulting system serve as valuable reference material, comprehensive documentation, and a solid foundational basis for further studies, academic inquiries, innovations, and technological developments related to educational technology, school management systems, and web-based applications within the broader field of information technology and educational management.

The detailed design principles, system architecture, specific features, development methodologies, testing results, and evaluation findings derived from the development and assessment of this project can effectively guide future researchers, software developers, and students in understanding the unique operational requirements of small to medium-sized educational institutions, overcoming common challenges related to limited-resource settings, and developing more advanced, scalable, and specialized systems tailored to specific educational contexts.

It provides a tested and validated framework that can be expanded, modified, enhanced, or integrated with emerging technologies—such as artificial intelligence, data analytics, or mobile computing—to explore new functionalities, improve service delivery models, and further enhance the efficiency, reach, and quality of educational administration and governance in similar contexts.

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## 1.4 Definition of Terms

**Integrated School Management System** – refers to a comprehensive, all-encompassing software suite and digital infrastructure specifically designed to digitize, unify, and streamline the wide range of academic, administrative, and support processes required to operate an educational institution effectively.

It consolidates diverse functions—including the management of student records, academic scheduling, attendance monitoring, financial operations, logistical inventory control, and stakeholder communication—into a single, cohesive, and efficient platform that serves as the central nervous system of the school's daily operations.

This system functions by connecting different departments, user roles, and stakeholders—such as administrators, teachers, students, and parents—within a shared digital environment, ensuring seamless data flow, absolute consistency in information, and coordinated workflows that effectively replace traditional, fragmented methods involving separate databases, manual files, and disconnected tools.

By bringing all essential functionalities together in one accessible system, it significantly simplifies complex management tasks, drastically improves the speed and quality of service delivery, enhances transparency and accountability, and provides the necessary foundation to support the institution's strategic transition toward a fully digital, paperless, and highly organized educational environment where information is accurate, secure, and readily available to those who need it.

□ **Real-time Messaging** – is defined as an advanced communication protocol and interactive feature embedded within the system that enables the instantaneous delivery and receipt of text-based information, notifications, and updates between authorized users with minimal to no latency or delay.

This capability ensures that urgent school announcements, critical updates such as sudden schedule changes, class suspensions, emergency alerts, or important reminders regarding events and deadlines are transmitted and received exactly as they happen, allowing for timely responses and immediate action from recipients.

Unlike traditional delayed messaging systems, physical notice boards, or third-party applications, this feature facilitates immediate two-way interaction and instant feedback loops, creating a reliable, fast, and dedicated channel for exchanging important inquiries, clarifications, constructive feedback, and general information.

Furthermore, it eliminates dependence on external services or incurring additional communication costs, while ensuring that all conversations and shared data remain securely encrypted and strictly contained within the institutional ecosystem. This feature guarantees that parents, teachers, and administrators remain continuously connected, well-informed, and able to collaborate effectively at all times, regardless of their physical location.

☐ **Attendance Tracking** – represents the systematic digital process of accurately recording, continuously monitoring, and formally reporting the presence, absence, punctuality, and participation status of students, teaching personnel, or staff members within the school premises or during scheduled learning sessions.

Built to replace outdated manual logbooks, paper registers, or roll-call methods, this functionality utilizes secure, automated mechanisms—such as web-based interfaces, QR code scanning, or geolocation verification—to capture attendance data instantly and precisely, updating central records in real time while automatically storing historical data for long-term reference and analysis.

Beyond simple logging, this feature allows administrators and authorized personnel to generate detailed, accurate reports, identify recurring patterns of absenteeism or tardiness, and improve institutional accountability and compliance. It serves as a vital tool for early intervention, enabling educators and guidance counselors to promptly address attendance issues before they escalate, simplifies the creation of official attendance summaries required for regulatory compliance, and provides a clear, verifiable audit trail that helps evaluate student engagement, adherence to school policies, and overall instructional regularity.

□ **Inventory Management** – constitutes a systematic, data-driven approach to overseeing, controlling, and optimizing the school's entire ecosystem of physical assets, consumable supplies, educational materials, technological equipment, and furniture.

It encompasses the complete lifecycle of managing resources—from initial procurement, receiving, and proper storage, to issuance, utilization, maintenance, recovery, and eventual disposal or replacement—ensuring that every item belonging to the institution is properly accounted for, tracked, and available exactly when and where it is needed. Through organized recording, monitoring features, and automated alerts, this system is designed to prevent disruptive shortages that hinder instruction, significantly reduce waste, deterioration, or loss due to misplacement, and optimize budgeting by providing clear visibility into consumption rates and stock levels.

By maintaining a centralized, real-time record of all school property, this process helps minimize shrinkage, ensures the efficient and equitable utilization of limited resources, and generates accurate historical data that aids school leaders in planning future purchases, managing maintenance schedules, and sustaining the quality of facilities, tools, and learning materials essential for the educational process.

□ **Student Records Management** – refers to the comprehensive and systematic collection, secure storage, regular updating, and efficient retrieval of all relevant personal, academic, and administrative information regarding learners throughout their entire journey within the institution.

This includes detailed personal identification data, enrollment history, class assignments, academic performance metrics, examination results, attendance logs, disciplinary records, health information, and official uploaded documents such as certificates or forms.

The primary objective of this management process is to ensure that highly sensitive data is securely stored, logically organized, and accessible only to authorized personnel according to

strict privacy protocols, thereby protecting student confidentiality and complying with data protection standards.

At the same time, it guarantees that accurate, complete, and up-to-date student profiles are instantly available for administrative processing, academic evaluation, curriculum planning, or official reporting purposes.

This system effectively eliminates common issues related to misplaced, damaged, or disorganized physical files, reduces redundancy and errors, and allows educators and administrators to track student progress holistically, ensuring that every learner's educational history is preserved and accessible for decision-making and support services.

□ **Report Generation** – describes the automated, intelligent process of compiling, aggregating, processing, analyzing, and transforming raw data stored within the system into structured, readable, and professionally formatted documents such as summaries, detailed lists, statistical analyses, comparative charts, or official transcripts.

This powerful functionality allows authorized users to instantly produce a wide range of essential administrative and academic outputs—including attendance summaries, grade sheets, progress reports, inventory status audits, student master lists, enrollment statistics, and financial statements—formatted in standard, usable file types such as PDF, Excel, or Word.

By automating what was previously a labor-intensive and error-prone manual task, this feature saves significant amounts of time and effort, drastically reduces the risk of calculation mistakes or data inconsistencies, and provides reliable, verifiable documentation that supports regulatory compliance requirements, internal and external evaluation processes, accreditation standards, and most importantly, evidence-based decision-making.

It serves as the primary mechanism for translating complex data into actionable insights, enabling school leaders to assess performance, identify trends, allocate resources wisely, and implement improvements based on factual information.

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# CHAPTER 2

## REVIEW OF RELATED LITERATURE AND STUDIES

### 2.1. School Management & Student Information Systems

These systems function as comprehensive centralized repositories that consolidate all student-related data, effectively streamlining a wide spectrum of administrative and academic operations including enrollment procedures, grade computation and maintenance, report generation, curriculum mapping, and compliance documentation, thereby enabling educational institutions to operate with significantly higher levels of efficiency, drastically reducing dependence on voluminous paper records, and ensuring that accurate, up-to-date information is readily accessible to authorized users such as school administrators, teachers, parents, and the students themselves at any given time.

By bringing together data that was previously scattered across different departments, filing systems, or separate software tools, these integrated platforms eliminate the confusion, delays, and inconsistencies that often arise from fragmented information management, creating instead a single, reliable source of truth that aligns all institutional activities and ensures everyone works from the same accurate dataset.

They automate repetitive and time-consuming tasks such as data entry, form processing, record updates, and the preparation of regulatory reports, which not only minimizes the risk of human error and data duplication but also frees up valuable time for school personnel to focus on core educational responsibilities, student support, and program improvement, transforming the daily workflow from labor-intensive paperwork to strategic, value-added work.

Furthermore, with built-in security measures and role-based access controls, the systems guarantee that sensitive information—including personal details, academic history, and health records—remains strictly protected against unauthorized access,

misuse, or loss, while simultaneously allowing authorized stakeholders to retrieve relevant details, track academic progress, review attendance records, or communicate updates securely from any location with internet connectivity, breaking down geographical barriers and enabling seamless remote engagement.

This seamless availability of reliable data fosters greater transparency and accountability within the institution, strengthens collaboration between the school and families by keeping parents actively informed and involved, and provides school leaders with the accurate, real-time insights necessary for informed decision-making, strategic planning, resource allocation, and continuous enhancement of the overall quality of education and services provided.

Beyond basic record-keeping, modern versions of these systems also support advanced features such as academic performance analytics, trend forecasting, curriculum alignment tracking, and compliance monitoring with government education standards, making them indispensable tools not just for day-to-day operations, but also for long-term institutional development, accreditation readiness, and the delivery of a structured, responsive, and student-centered learning environment that meets the evolving demands of modern education.

Additionally, these systems serve as a dynamic operational backbone that interconnects every aspect of school life, from admissions and registration to graduation and alumni tracking, creating a continuous digital thread that follows each learner throughout their entire educational journey, ensuring that no critical detail is ever lost or overlooked during transitions between grade levels, departments, or academic years.

They further simplify complex operational workflows such as class scheduling, teacher assignment, subject allocation, and facility management by using intelligent algorithms and predefined rules to optimize resource utilization, prevent scheduling conflicts, and ensure that teaching loads and classroom capacities are balanced fairly and effectively across the entire institution.

By standardizing data formats, coding structures, and reporting frameworks, these platforms also ensure that all institutional information adheres to established educational policies, regulatory requirements, and industry best practices, thereby simplifying external audits, accreditation evaluations, and submissions to educational governing bodies, while significantly reducing the administrative burden associated with meeting strict compliance deadlines and documentation standards.

Moreover, they enhance communication and coordination not only between the school and families but also among staff members themselves, offering built-in messaging

systems, announcement boards, and shared workspaces that facilitate instant information exchange, collaborative lesson planning, and coordinated student support efforts, ensuring that every educator and staff member remains aligned with institutional goals and student needs.

The inclusion of specialized modules for inventory management, financial accounting, and human resource tracking further expands their utility, allowing school leaders to monitor asset usage, track expenses, manage payroll, and oversee staff performance all within one unified environment, thereby eliminating the need for multiple disjointed applications and creating a fully integrated ecosystem that supports every function of the educational enterprise.

These systems also play a crucial role in supporting inclusive and equitable education by maintaining detailed records of special educational needs, accommodations, and support services received by students, enabling teachers and administrators to tailor instructional approaches, allocate appropriate resources, and monitor the effectiveness of intervention programs to ensure every learner has the opportunity to succeed.

Through comprehensive data visualization tools such as dashboards, charts, and graphical reports, complex information is transformed into clear, understandable insights that reveal patterns, strengths, and areas for improvement, empowering decision-makers to implement evidence-based policies, adjust educational strategies, and invest resources where they will have the greatest positive impact on student achievement and institutional growth.

Furthermore, as cloud-based solutions have become the industry standard, these systems now offer enhanced scalability, reliability, and disaster recovery capabilities, ensuring that data remains safe, accessible, and intact even in the event of unforeseen disruptions, while allowing the institution to easily expand storage capacity, add new features, or accommodate growing enrollment numbers without requiring costly infrastructure upgrades or technical overhauls.

Ultimately, School Management and Student Information Systems act as more than just data storage or administrative tools; they are powerful enablers of educational excellence that transform how schools operate, collaborate, and serve their communities, driving digital transformation, fostering a culture of continuous improvement, and ensuring that educational institutions remain agile, competitive, and fully equipped to deliver high-quality learning experiences that prepare students for success in an increasingly complex and technology-driven world.

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## **iSercan: A Web-Based Student Information System (iSercan, 2019)**

This research details the systematic development, comprehensive design, and successful implementation of a robust web-based portal engineered specifically to provide enrolled students and their parents or guardians with secure, round-the-clock access to essential academic details and educational records.

The platform delivers key functionalities including real-time grade updates and progress tracking, daily and weekly class schedules, examination and assessment calendars, as well as transparent fee payment monitoring and financial status features, all while placing a strong, central emphasis on rigorous data security measures, reliable system performance, and seamless cross-platform compatibility across different devices, screen sizes, and operating systems.

These technical features and functional capabilities are strategically aimed at minimizing administrative bottlenecks and processing delays, significantly reducing institutional reliance on traditional paper documentation and manual filing systems, and fostering continuous, smooth, and transparent information sharing between the school administration, faculty members, and the home environment.

Beyond these core service-oriented functionalities, the study thoroughly documents the entire software development lifecycle—from initial requirements gathering, stakeholder interviews, and detailed system analysis, through architectural design, coding, and rigorous testing phases, to final deployment and user training—ensuring that every feature and module is built to align precisely with actual institutional needs, operational workflows, and user expectations derived from direct observation and empirical feedback.

It highlights how the platform effectively organizes, consolidates, and centralizes critical data that was previously fragmented, dispersed across various offices, or stored in separate physical files and databases, making it significantly easier for school administrators and staff to manage vast volumes of student records, generate accurate and standardized reports required for compliance or planning, and update information instantly across the system without the risk of duplication, inconsistency, or human error. Simultaneously, this centralized structure gives learners and guardians a clear,

comprehensive, and real-time view of academic standing, attendance history, disciplinary records, and outstanding financial obligations at any time, from any location with a stable internet connection.

Additionally, the research underscores how the system's structured role-based access controls—restricting data visibility according to user type and authorization level—and advanced data encryption protocols work together continuously to protect sensitive personal, academic, and financial information from unauthorized access, external data breaches, or accidental loss, thereby directly addressing common concerns regarding privacy, confidentiality, and data integrity that often accompany the transition to digital school management systems.

By streamlining communication channels, standardizing information presentation, and making vital details readily available in a user-friendly, intuitive format, the platform successfully bridges the traditional information gap between the school and families, encourages active and consistent parental involvement in every stage of the educational journey, and helps build a more collaborative, informed, and supportive educational community where all stakeholders are engaged and empowered. Ultimately, the findings presented demonstrate that this web-based solution not only drastically improves operational efficiency and reduces the repetitive workload of administrative and teaching staff by automating routine tasks and record-keeping processes but also enhances the overall quality of service delivery, institutional accountability, and operational transparency within the educational setting, serving as a reliable model and valuable reference for modernizing school information management practices and digital transformation initiatives in similar academic environments (iSercan, 2019).

Furthermore, the research elaborates on how the introduction of this digital solution effectively transforms the user experience for all parties involved, offering a highly organized and visually consistent interface that simplifies navigation and ensures that even individuals with limited technical proficiency can easily access, interpret, and utilize the available features without encountering unnecessary complexity or confusion. It explains how the system's automated notification mechanisms—including alerts for newly posted grades, upcoming assessments, fee deadlines, and important school announcements—function as a proactive communication tool that ensures no critical

information is missed or overlooked, keeping all stakeholders constantly updated and fully engaged with ongoing school activities and academic developments.

The study also emphasizes the significant positive impact on resource management and operational costs, noting that by drastically reducing the need for physical storage space, printing supplies, and manual labor associated with maintaining and retrieving paper files, the institution is able to reallocate valuable time, budget, and physical resources toward more pressing educational priorities, facility improvements, and student support programs.

Moreover, it details how the platform's capability to maintain a complete, chronological, and unalterable digital history of every student's tenure—from initial admission and enrollment, through academic achievements and attendance patterns, to eventual graduation—creates a permanent, reliable archive that preserves institutional memory and simplifies the retrieval of historical data for evaluation, accreditation, or reference purposes many years in the future.

The research further points out that by standardizing data collection and reporting formats across the entire institution, the system ensures consistency in how information is captured, stored, and presented, thereby simplifying the generation of required documents for government agencies, educational boards, and regulatory bodies, while significantly reducing the likelihood of errors, discrepancies, or non-compliance issues that can arise from inconsistent manual record-keeping practices.

Additionally, it highlights how the system supports data-driven decision-making by providing school leaders and administrators with comprehensive, accurate, and up-to-date data summaries and performance indicators, allowing them to identify trends, measure effectiveness, pinpoint areas needing improvement, and formulate strategic plans based on factual evidence rather than assumptions or outdated information.

Beyond operational improvements, the study also discusses the broader educational benefits, explaining that when parents and guardians have easy access to detailed information regarding their children's performance, attendance, and behavior, they are better equipped to provide appropriate guidance, support, and motivation at home, creating a stronger home-school partnership that is proven to directly contribute to higher student achievement, better attendance rates, and improved behavioral

outcomes. It also addresses the system's adaptability and long-term value, noting that its modular design and scalable architecture allow it to grow alongside the institution, easily accommodating increases in enrollment, the addition of new academic programs, changes in educational policies, or evolving user requirements without requiring a complete overhaul or replacement of the existing infrastructure.

Finally, the research concludes by affirming that the successful implementation of this web-based student information system serves as a clear testament to how technology can be effectively harnessed to address real-world challenges in educational management, proving that such solutions are not merely optional upgrades, but essential investments that strengthen institutional capacity, enhance service quality, and ensure that schools are fully prepared to meet the standards and expectations of modern education in an increasingly digital and interconnected society (iSercan, 2019).

### **Systems Analysis of the Philippine Basic Education Ecosystem (Cuyegkeng, 2026)**

This insightful working paper, developed in partnership with the Second Congressional Commission on Education (EDCOM II), undertakes a thorough examination of the broader digital ecosystem within the Philippine Department of Education, utilizing a comprehensive systems thinking approach to holistically map out and analyze the complex interrelationships, functional dependencies, and operational workflows among various stakeholders, institutional processes, and existing technological infrastructures currently in place across the sector.

Through this rigorous assessment, the research clearly identifies the urgent and critical necessity for the adoption and implementation of fully integrated data management frameworks and interoperable systems capable of effectively breaking down persistent information silos, standardizing diverse data formats, and ensuring a seamless, secure, and consistent flow of information across different levels, regions, and administrative

units of the entire education sector. Such integrated technological systems are envisioned to provide substantial, evidence-based support to school leadership and management teams at all levels, significantly enhancing both the quality and speed of strategic decision-making processes by delivering accurate, real-time insights and comprehensive data analytics.

These capabilities enable administrators to monitor institutional progress and performance effectively, allocate limited resources with maximum efficiency and transparency, and align local institutional goals, practices, and curricular offerings with national education standards, government policies, and long-term development agendas such as the MATATAG framework and the broader digital transformation roadmap.

Furthermore, the study emphasizes that these modern technological solutions play a pivotal role in significantly strengthening communication channels and collaboration networks, successfully bridging existing gaps and fostering meaningful, continuous engagement across educators, administrators, learners, families, and community partners to create a more connected, inclusive, and supportive educational environment.

It further highlights that improved, timely, and equitable access to accurate, reliable, and actionable educational data serves as a key intervention point and a foundational strategy to drive better learning outcomes for all students, strengthen governance and accountability mechanisms within schools, and ensure more effective, responsive, and uniform implementation of education policies and programs nationwide.

Ultimately, the paper asserts that establishing a cohesive digital infrastructure is indispensable for addressing systemic challenges, reducing administrative burdens, and advancing the overall modernization and transformation of the Philippine basic education system to meet global standards and future demands (Cuyegkeng, 2026).

Additionally, the research delves deeply into the historical and structural roots of fragmentation within the current system, explaining how decades of decentralized operations, independent reporting mechanisms, and the adoption of disparate, non-compatible software tools by different divisions and schools have created significant



bottlenecks, redundant work processes, and inconsistencies in data interpretation that hinder effective governance and policy execution across the archipelago.

It elaborates on how these inefficiencies lead to delayed reporting cycles, increased risk of data errors, and difficulties in consolidating national statistics, which in turn affect the government's ability to accurately assess educational needs, measure the impact of interventions, and distribute budgetary resources fairly and according to actual requirements.

The study further outlines the essential characteristics that these integrated systems must possess to succeed within the unique context of the Philippine education landscape, including high adaptability to varying levels of digital readiness across different regions, robust offline capabilities for areas with limited connectivity, strict adherence to national data privacy laws and security protocols, and user-centric designs that accommodate the diverse technical skills of personnel working in both urban centers and remote rural communities. It underscores that true interoperability goes beyond simple data sharing; it requires the establishment of unified data dictionaries, standardized coding systems, and common technical protocols that allow different platforms to communicate, exchange, and interpret information seamlessly, effectively creating a unified national information network where data entered at the local level becomes instantly available and useful at regional, division, and central offices without the need for reprocessing or manual encoding.

Moreover, the paper highlights the direct correlation between digital maturity and institutional performance, noting that schools equipped with comprehensive management systems are better positioned to track student progression, identify learning gaps early, manage teacher deployment and professional development, and maintain compliance with regulatory mandates, all of which contribute to a more stable, predictable, and high-performing educational environment. It also discusses the role of such systems in empowering transparency and public accountability, as accurate and accessible data allows for better monitoring of service delivery, reduces opportunities for mismanagement, and provides citizens, parents, and communities with the visibility needed to participate actively in the improvement of educational quality.

Furthermore, the analysis points out that digital transformation is not merely about technology adoption but represents a fundamental shift in how education is governed and delivered, requiring corresponding changes in policy frameworks, operational guidelines, and the capacity building of human resources to ensure that all stakeholders possess the necessary knowledge and skills to utilize these tools effectively and responsibly.

The study envisions a future where every school in the Philippines, regardless of location or size, is connected to a secure, reliable, and integrated digital ecosystem that supports every aspect of its operations—from enrollment and instruction to finance and reporting—creating a level playing field where quality education and efficient management are accessible to all.

It concludes by emphasizing that investing in this cohesive digital infrastructure is not just an operational improvement, but a strategic imperative that will serve as the backbone of educational reform, enabling the sector to overcome longstanding systemic constraints, respond more agilely to emerging challenges, and fulfill the national commitment to provide accessible, equitable, and high-quality basic education for every Filipino learner, thereby securing the country's competitiveness and development potential for generations to come (Cuyegkeng, 2026).

## **Making Education Manageable: School Management Systems and Data-Driven Classrooms (Taylor & Francis, 2025)**

This critical analysis investigates the discursive construction and evolving functional role of School Management Information Systems (MIS), closely examining how these sophisticated technological tools are fundamentally reshaping the landscape of modern education by prioritizing extensive data tracking capabilities, performance benchmarking, and behavioral analytics measures across all levels of institutional operation. It thoroughly explores how such systems actively influence and redefine the very definition of educational quality, accountability standards, and success metrics, demonstrating how data-driven insights are increasingly embedded into every aspect of school life—from curriculum planning, student assessment, and attendance monitoring to teacher evaluation, resource distribution, and policy formulation.

The study further delves into the profound impacts these technologies have on daily classroom dynamics, instructional strategies, and the overall learning experience, illustrating how educators utilize real-time information and comprehensive analytics to tailor teaching interventions, closely monitor individual and group progress, and promptly address specific learning needs or gaps, thereby enabling a more personalized and responsive approach to education.

Simultaneously, it highlights how these systems transform the overarching priorities, operational workflows, and resource allocation strategies of school administration, shifting them from reactive, manual-based practices to more strategic, evidence-based, and efficient decision-making processes that optimize institutional performance and governance.

Beyond their immediate practical utility and operational advantages, the analysis also engages deeply with the broader societal implications and ongoing academic debates surrounding widespread digital integration in educational settings, raising important discussions about the necessary balance between leveraging data to improve outcomes and preserving pedagogical autonomy, professional judgment, and the deeply human-centered values that lie at the heart of the teaching and learning process.

By rigorously evaluating both the transformative benefits—such as enhanced transparency, improved efficiency, and targeted support—and the potential challenges or risks associated with a heavily data-centric approach to school management, this work provides a balanced, comprehensive perspective on how these integrated systems are actively shaping the future of teaching, learning, and educational governance in an increasingly digitized world (Taylor & Francis, 2025).

Furthermore, the research elaborates on the structural transformation brought about by these systems, describing how they serve as the central nervous system of educational institutions, capable of capturing, processing, and interpreting vast amounts of information that were previously scattered, unstructured, or inaccessible, thereby turning raw data into meaningful, actionable intelligence that guides every level of operation. It details how the continuous flow of information allows for the creation of detailed performance profiles for every learner, educator, and department, revealing nuanced patterns regarding academic growth, attendance consistency, behavioral trends, and resource utilization that would otherwise remain hidden or difficult to identify through traditional observation and manual record-keeping alone. T

he analysis also explains how these technological platforms facilitate greater standardization and consistency across different classes, grade levels, and campuses, ensuring that educational benchmarks, assessment criteria, and administrative procedures are applied uniformly while simultaneously offering the flexibility to adapt to specific educational contexts or unique learner requirements.

It emphasizes that by automating routine data collection and reporting tasks, these systems significantly reduce the administrative burden placed on teaching and non-teaching staff, freeing educators from excessive paperwork and data entry duties so they can redirect their time, energy, and focus toward what matters most: meaningful instruction, student mentorship, and the cultivation of critical thinking and social-emotional skills.

Additionally, the study examines how data-driven management strengthens institutional accountability by providing clear, verifiable evidence of performance and compliance, allowing school leaders, governing bodies, and community stakeholders to measure results against established goals, identify areas requiring improvement, and ensure that educational resources—including funding, facilities, and personnel—are deployed in the most effective, equitable, and impactful manner possible. It also addresses the shift in educational culture brought about by these tools, noting that they encourage a mindset of continuous improvement and reflective practice, where decisions are no longer based solely on intuition or tradition, but are grounded in objective evidence, research-based best practices, and a deep understanding of actual educational realities.

However, the analysis also critically examines the complexities and potential downsides inherent in this digital evolution, raising thoughtful concerns regarding the risk of over-reliance on quantitative metrics, the possibility of reducing complex educational experiences to mere numbers or scores, and the challenges of ensuring data privacy, security, and ethical use in an environment where information is constantly being collected and analyzed.

It underscores the vital importance of ensuring that technology serves as an enabler and support system rather than a rigid control mechanism, stressing that while data provides essential insights, it must always be interpreted and applied alongside professional wisdom, contextual understanding, and a deep respect for the diverse needs, talents, and potentials of every individual within the learning community.

The work further explores the impact on stakeholder relationships, illustrating how transparent access to accurate information builds trust among parents, students, and educators, fostering a collaborative partnership where everyone is informed, aligned, and actively engaged in the shared goal of educational success.

Ultimately, the study concludes that the integration of School Management Information Systems represents an inevitable and essential advancement in modern education, but their true value and success depend entirely on how they are designed, implemented, and governed—requiring a deliberate, thoughtful approach that harmonizes technological capability with educational philosophy, ensuring that while education becomes more manageable, measurable, and efficient, it never loses its core purpose of nurturing capable, creative, and well-rounded individuals prepared for the complexities of the future (Taylor & Francis, 2025).

### **A Web-Based Expert System for Student Advising (Philippine E-Journals, n.d.)**

This study, originating from research conducted at the Technological Institute of the Philippines (TIP), concentrates on the systematic design, full-cycle development, and comprehensive functional testing of an automated web-based expert system created specifically to handle the complex, detail-oriented, and labor-intensive nature of the student academic advising process.

It effectively highlights how such a sophisticated technological innovation can drastically reduce the processing duration of critical tasks—such as comprehensive curriculum evaluation, strict prerequisite validation, subject enrollment eligibility checks, and academic compliance verification—from a matter of days or weeks under manual operations to near-instantaneous results within the digital environment.

This capability effectively eliminates unnecessary delays, resolves administrative backlogs, and significantly mitigates the risks of oversight, inconsistency, and human error that are frequently associated with traditional, person-dependent advising methods. By formally encoding institutional policies, detailed curriculum requirements, and established academic standards directly into its intelligent decision-making framework, the system is capable of delivering highly personalized, accurate, consistent, and unbiased academic guidance, tailored course recommendations, and structured career pathway planning support to every learner efficiently and reliably, regardless of the volume of students being served or time constraints faced by the institution.

Furthermore, the research demonstrates that this robust digital solution significantly reduces the heavy administrative burden and repetitive procedural workload traditionally placed on academic advisors and faculty members, allowing them to redirect their focus and dedicate more quality time to meaningful mentorship, instructional improvement, and holistic student development rather than routine data verification and rule-checking duties.

It also ensures that students receive clear, accessible, and timely information regarding their current academic standing, completed requirements, and remaining obligations, empowering them to make well-informed decisions about their education, confidently track their progress toward graduation, and effectively align their individual learning journey with their long-term professional goals and aspirations.

Ultimately, the study establishes that implementing such an intelligent expert system not only streamlines institutional workflows and substantially improves overall operational efficiency but also enhances the overall quality and reliability of student support services, promotes greater fairness, objectivity, and transparency in academic processes, and contributes to the establishment of a more organized, standardized, and responsive educational environment (Philippine E-Journals, n.d.).

Additionally, the research elaborates on the technical framework and knowledge engineering principles applied during the system's construction, explaining how it captures the collective expertise of seasoned academic advisors, department heads, and curriculum specialists, preserving this institutional knowledge within the system's logic so that it remains permanently available, consistent, and accessible regardless of

staff turnover, schedule changes, or shifts in personnel assignments. It details how the platform operates as a dynamic knowledge base that continuously updates as curricula evolve, policies change, or new academic programs are introduced, ensuring that the advice and evaluations provided are always aligned with the latest institutional guidelines, regulatory standards, and degree requirements without requiring complete system overhauls or extensive retraining.

The study further emphasizes the value of standardization across the advising process, noting that in traditional settings, guidance quality can vary significantly depending on the individual advisor's familiarity with complex rules, memory of specific policies, or interpretation of requirements; by contrast, this digital solution applies every rule, regulation, and exception uniformly and objectively to every student, thereby ensuring equitable treatment, eliminating potential bias, and guaranteeing that every learner receives the same high-quality service and accurate information.

It also highlights how the system serves as a powerful planning tool that visualizes the entire academic roadmap, allowing students and advisors to simulate different enrollment scenarios, explore elective options, and assess the impact of course selections or potential schedule adjustments before finalizing decisions, effectively preventing common pitfalls such as taking out-of-sequence subjects, missing critical prerequisites, or delaying graduation due to unforeseen requirement gaps.

Moreover, the research discusses the significant impact on resource optimization and service accessibility, explaining that by automating core advising functions, the institution can effectively serve a much larger student population with the same or fewer human resources, while also making advising services available 24/7 through a web-based interface, thus removing barriers related to office hours, location, or long waiting lines, and ensuring that students have continuous access to guidance whenever they need it. It also addresses the benefits of having a complete, documented audit trail of all advising interactions, recommendations, and evaluations, which strengthens institutional accountability, simplifies compliance reporting, and provides a clear historical record that protects both the student and the institution in case of disputes, inquiries, or accreditation reviews.



Beyond logistical improvements, the study underscores the positive psychological and educational effects on students, noting that when learners have instant access to clear, structured, and reliable guidance, they experience reduced anxiety and uncertainty regarding their academic path, develop a stronger sense of ownership and control over their education, and are more motivated to stay on track toward timely completion of their degree programs.

It further points out that this technological intervention fosters a shift in the role of faculty advisors from being mere rule enforcers or data checkers to becoming true academic mentors and career guides, as they are freed from repetitive administrative tasks and can instead focus on addressing complex personal concerns, providing deep career insights, offering academic encouragement, and supporting students facing unique challenges or special circumstances.

The research also acknowledges the system's scalability and adaptability, highlighting that its modular structure allows it to be easily configured to support different disciplines, varying degree levels, and diverse institutional structures, making it a versatile solution applicable not only to the Technological Institute of the Philippines but also to a wide range of higher education institutions across the country.

Finally, the study concludes by positioning the development of such expert systems as a vital component of the broader digital transformation of Philippine education, proving that integrating artificial intelligence and knowledge-based technologies into academic services is a strategic step toward achieving operational excellence, enhancing student satisfaction, and ensuring that educational institutions remain agile, competitive, and fully capable of supporting student success in an increasingly complex and fast-paced academic landscape (Philippine E-Journals, n.d.).

## **Cloud-Based Student Information System for Educational Institutions (Al-Mutairi et al., 2023)**

Al-Mutairi and colleagues (2023) conducted an in-depth exploration regarding the application and strategic integration of cloud computing technology in the systematic design, development, and successful deployment of modern student information systems, placing a strong emphasis on core advantages such as exceptional system scalability, high operational cost-efficiency, and seamless remote accessibility for all authorized users regardless of their location or device.

Their comprehensive study demonstrated that cloud-based solutions allow educational institutions of varying sizes and capacities—ranging from small rural learning centers and private schools to large universities and multi-campus systems—to implement robust, secure, and fully functional data management frameworks without the heavy burden of significant capital investment in physical servers, complex hardware setups, dedicated server rooms, or the need for extensive on-site technical maintenance and specialized IT personnel.

By leveraging a centralized, internet-based infrastructure, the system facilitates real-time data synchronization across different departments, branches, or campuses, ensuring that critical information such as student records, academic updates, grade submissions, attendance logs, and financial data remains perfectly consistent, accurate, and immediately up-to-date across the entire organization at all times, effectively eliminating data discrepancies and redundant entries that often plague fragmented, on-premise systems.

It also enables secure, role-based access to relevant information and functionalities from any location and on any internet-enabled device, including laptops, tablets, and smartphones—a feature that proved highly beneficial and operationally essential, especially during periods of remote, blended, or hybrid learning models, where continuous connectivity and uninterrupted access to academic resources and administrative tools were critical to sustaining educational services and institutional operations without disruption.

Additionally, the research highlights how this technological approach significantly strengthens data protection and institutional resilience through built-in automated backups, advanced encryption protocols for data transmission and storage, and professional-grade security management handled by service providers, which safeguards sensitive information against loss, damage, or unauthorized access more reliably than conventional manual or local server systems.

Furthermore, the cloud-based model offers remarkable operational flexibility, allowing institutions to easily expand or reduce system storage capacity, user access limits, or processing power according to specific needs, such as periods of institutional growth, sudden enrollment changes, or seasonal demands like enrollment periods and end-of-term processing. Ultimately, the authors conclude that adopting cloud-based student information systems not only streamlines administrative workflows and significantly reduces long-term operational expenses and maintenance costs but also enhances the overall reliability, availability, and quality of educational services delivered to stakeholders, effectively making modern, efficient, and secure data management accessible, affordable, and sustainable for educational institutions regardless of their scale, location, or available financial and technical resources (Al-Mutairi et al., 2023).

Additionally, the study elaborates on the fundamental shift in IT management strategy that cloud adoption represents, explaining how institutions are able to transition from a model of owning and maintaining complex technology infrastructure to one where they simply utilize services on a subscription or pay-as-you-go basis, thereby converting large, one-time capital expenditures into predictable, manageable operational costs that are easier to budget and sustain over time.

It details how this model democratizes access to high-end technology, ensuring that even small schools or learning centers with limited funding and technical staff can access enterprise-grade security, performance, and reliability features that were previously only available to large, well-resourced universities, thereby leveling the playing field and allowing all educational providers to offer superior service quality.

The research further underscores the importance of data integrity and continuity, noting that because information is stored in geographically distributed, redundant data centers managed by professional providers, the institution is fully protected against catastrophic events such as hardware failure, natural disasters, power outages, or local system crashes, ensuring that critical institutional data remains safe, recoverable, and accessible under all circumstances—a level of protection that is nearly impossible to achieve with traditional in-house setups without substantial additional investment. It also explains how the centralized nature of the cloud architecture simplifies compliance with data governance laws and educational regulations, as updates to security protocols, privacy settings, or reporting standards can be applied universally and instantly across the entire system, ensuring that the institution always operates according to the latest legal requirements and industry best practices without the need for manual updates or patches on individual machines or separate campuses.

Moreover, the study highlights the positive impact on collaboration and resource sharing, showing how teachers, administrators, and staff members working in different locations or branches can simultaneously access and work on the same datasets, documents, or modules in real-time, fostering better coordination, standardization of processes, and collective decision-making across the organization.

It describes how the integration capability of cloud-based systems allows seamless connection with other essential educational tools such as learning management systems (LMS), online examination platforms, digital payment gateways, and library databases, creating a unified digital ecosystem where data flows freely between applications and further reduces the need for manual data entry or transfer.

The authors also address the environmental and operational efficiency benefits, pointing out that by sharing computing resources within large-scale data centers, the overall energy consumption and carbon footprint associated with IT operations are significantly reduced compared to running multiple separate, underutilized local servers, aligning the institution's digital transformation with broader sustainability goals.

Furthermore, the research discusses the ease of implementation and rapid deployment cycles characteristic of cloud solutions, noting that unlike traditional systems which require months or even years of planning, installation, and configuration, cloud-based platforms can be set up, configured, and made fully operational within a short timeframe, allowing schools to quickly modernize their operations and begin reaping the benefits of digitalization almost immediately. It emphasizes how automatic software updates ensure that the institution is always running the most current, secure, and feature-rich version of the system, eliminating the burden of scheduling maintenance windows or dealing with compatibility issues that often arise with legacy software.

Finally, the study positions cloud computing as the foundational infrastructure for the future of education, arguing that as institutions continue to evolve and adapt to changing educational models, increasing enrollment, and advancing technology, the cloud provides the only viable, scalable, and cost-effective pathway to support long-term growth, innovation, and the continuous delivery of high-quality education services in an increasingly connected and digital world (Al-Mutairi et al., 2023).

## **Data Integration and Interoperability in School Management Systems (Chen & Huang, 2024)**

Chen and Huang (2024) conducted an in-depth investigation into both the technical complexities and organizational challenges associated with merging disparate, standalone, and often isolated school management systems commonly adopted by educational institutions.

Their research highlights that the lack of proper interoperability, standardization, and compatibility between different software solutions, legacy platforms, or internal functional modules frequently leads to the formation of restrictive data silos—isolated repositories where critical information remains confined within specific departments, offices, or administrative functions and cannot be easily accessed, shared, or utilized across the broader institution.

This fragmentation of information flow, they observed, results in significant operational inefficiencies and administrative burdens, including the necessity for repetitive and redundant data entry tasks, inconsistent record-keeping practices that produce conflicting information, an increased likelihood of transcription errors and data corruption, and substantial difficulties in consolidating information to generate comprehensive reports that accurately reflect the true overall state of school operations, resources, and academic performance.

In direct response to these pervasive issues and gaps in system design, their research proposed and outlined a comprehensive standardized framework for seamless data exchange and full system integration, specifically designed to establish universal common protocols, unified data formats, and consistent communication standards that enable different functional modules—such as student enrollment and admission, academic grading and assessment, attendance monitoring, financial accounting, human resource management, and inventory and asset management—to interact, communicate seamlessly, and function together as a cohesive, unified ecosystem rather than separate disjointed tools.

By ensuring that data flows smoothly, securely, and consistently across all areas, levels, and processes of the organization, this integration framework effectively eliminates unnecessary duplication of effort, drastically enhances overall data accuracy, integrity, and reliability, and provides school administrators and decision-makers with a complete, holistic, and unified view of institutional performance, resource utilization, and educational outcomes.

Ultimately, the study demonstrates that achieving high levels of integration and interoperability within school management systems not only streamlines daily administrative workflows and significantly improves the quality and speed of service delivery to all stakeholders but also empowers educational leaders with the accurate, real-time information and comprehensive insights needed for effective data analysis, evidence-based strategic planning, policy formulation, and long-term sustainable institutional development (Chen & Huang, 2024).

Furthermore, the authors delve deeply into the root causes of these integration challenges, explaining how historical decisions to adopt specialized software for specific tasks without considering future connectivity, combined with the gradual accumulation of legacy systems over time, have created a complex digital environment where information is trapped in incompatible formats and proprietary structures, making meaningful data sharing virtually impossible without costly and labor-intensive manual intervention.

They illustrate how this lack of connectivity creates ripple effects throughout the entire institution; for instance, basic information entered during the admission process often needs to be retyped entirely into the grading module, the attendance system, and the finance system, leading to wasted time, increased labor costs, and a higher probability of errors that can affect everything from student report cards to financial auditing.

The study elaborates on the proposed framework's architecture, detailing how it utilizes standardized data models, common identifiers, and open application programming interfaces (APIs) to act as a central translator and bridge between different components, ensuring that regardless of the underlying software or platform, data is interpreted and represented in exactly the same way across the entire organization.

It emphasizes that true interoperability goes beyond simple file transfers or data exports; it establishes a dynamic environment where a single update made in one area—such as a change in a student’s personal details or a payment transaction—is instantly reflected and available in every relevant module, creating a single, reliable source of truth that everyone in the institution can trust and use.

Additionally, the research highlights the critical role of data governance within this framework, outlining how standardization also involves defining clear rules regarding data ownership, access permissions, validation protocols, and quality control measures, ensuring that as information moves across departments, it remains secure, compliant with privacy regulations, and maintains its structural integrity and meaning.

Chen and Huang further discuss the transformative impact of such integration on institutional analytics, noting that when data from academic performance, attendance patterns, resource usage, and financial expenditures is combined into a unified database, it unlocks powerful analytical capabilities that were previously unattainable, allowing leaders to identify complex correlations—such as the relationship between resource allocation and student achievement, or the impact of attendance trends on overall performance—that provide deep insights for improvement and strategic investment.

They also point out that an integrated architecture significantly simplifies compliance reporting and accreditation processes, as data required by external bodies can be automatically compiled, verified, and generated from the central system rather than being manually gathered, cross-referenced, and consolidated from dozens of different sources, which often leads to delays, discrepancies, and administrative stress. Moreover, the study addresses the user experience perspective, explaining that when systems work together seamlessly, the complexity perceived by teachers, staff, and administrators is greatly reduced; users no longer need to learn multiple distinct interfaces or remember different login credentials, but instead interact with a unified dashboard that presents only the relevant information and tools they need, tailored to their specific role and responsibilities.



The authors also acknowledge the technical and organizational hurdles involved in implementing such a framework, including the need for careful migration planning, capacity building for IT staff, and change management strategies for end-users, yet they argue that the return on investment—measured in time saved, errors eliminated, and decision-making improved—far outweighs the initial challenges.

Finally, they conclude that interoperability is not merely a technical feature but a fundamental requirement for the digital maturity of any educational institution, asserting that only by breaking down data barriers and creating fully connected information ecosystems can schools truly harness the full power of technology to drive operational excellence, enhance service delivery, and create a responsive, data-informed environment that supports the highest standards of teaching and learning (Chen & Huang, 2024).

## **2.2 Attendance Monitoring Systems**

Automated attendance solutions leverage modern technological advancements such as QR code scanning, biometric verification, radio frequency identification (RFID), and mobile applications to guarantee a much higher degree of precision and accuracy in daily attendance record-keeping, significantly cut down on the administrative workload, time expenditure, and labor costs associated with manual roll calls and repetitive data entry, and effectively minimize the occurrence of common human errors, recording discrepancies, and fraudulent practices including the persistent issue of proxy attendance or "buddy punching," where learners are marked present by others on their behalf.

These innovative systems are engineered to capture attendance details instantly and objectively, removing the subjectivity and oversight often found in traditional methods, while simultaneously generating structured, reliable datasets that clearly reflect student engagement patterns, participation trends, and behavioral habits over time.

By completely replacing traditional, paper-based logbooks, sign-in sheets, and handwritten registers with these smart, digital tools, educational institutions ensure that attendance data is captured instantly at the point of entry, validated securely to confirm identity, and recorded in real time within the system's database, thereby eliminating data discrepancies, missed entries, or processing delays that typically arise from manual compilation, transcription, and filing.

These systems automatically store, organize, and archive attendance logs within a centralized database architecture, allowing for seamless integration and data sharing with other critical school management modules such as student information records, grading systems, and official reporting tools, ensuring that attendance data directly contributes to a more comprehensive, holistic view of each learner's academic journey, development, and overall standing in the institution.

Furthermore, they are equipped with automated notification features that can be configured to instantly alert parents, guardians, or school administrators regarding unauthorized absences, excessive tardiness, or irregular attendance patterns through integrated messaging or alerts, fostering timely intervention, immediate verification, and stronger home-school collaboration in monitoring student welfare.

Beyond simple record-keeping and compliance, the accumulated historical data serves as a valuable analytical resource and evidence-based reference, helping educators and institutional leaders identify recurring trends, such as chronic absenteeism, periodic disengagement, or correlations between attendance and academic performance, which can inform the design of targeted support programs, policy adjustments, and strategic decision-making strategies aimed at improving overall attendance rates, enhancing student retention, ensuring safety and accountability, and ultimately boosting learning outcomes across the entire educational community.

Additionally, these systems are designed to operate seamlessly across various environments and learning modalities, including face-to-face classes, hybrid setups, and fully remote or distance learning scenarios, offering flexible deployment options such as on-site scanning stations, mobile app check-ins, or web-based verification methods that ensure consistent monitoring regardless of where the learning takes place or how instruction is delivered.

They effectively standardize the entire attendance-taking process across all grade levels, departments, and campuses, ensuring that the same strict protocols, recording standards, and validation rules are applied uniformly, thereby eliminating variations in how attendance is interpreted or documented by different teachers or staff members and creating a reliable, consistent dataset that holds up to scrutiny during audits, inspections, or regulatory reviews.

The ability to generate real-time, customizable reports—ranging from daily summaries and weekly attendance sheets to detailed monthly analytics and end-of-term statistical breakdowns—allows school administrators to monitor compliance with national education policies, track adherence to compulsory attendance laws, and quickly identify classes or groups where participation rates are dropping below acceptable thresholds, enabling proactive measures to be taken before minor issues escalate into serious concerns.

Moreover, by automating the tedious process of computing attendance percentages, generating deficiency notices, or compiling data for report cards, these systems drastically reduce the likelihood of computational errors and save countless hours of labor that would otherwise be spent manually tallying records and verifying calculations, allowing teaching and non-teaching staff to redirect their valuable time and energy toward educational support and student welfare initiatives.

The centralized repository also serves as a permanent, secure, and tamper-proof historical archive that preserves attendance records for years, providing an unalterable audit trail that is essential for verifying eligibility for promotions, determining graduation readiness, investigating disciplinary matters, or responding to inquiries from government agencies and accrediting bodies.

Furthermore, the integration of geolocation features and time-stamping capabilities adds an extra layer of verification and security, ensuring that attendance is only recorded when and where it is supposed to be, which is particularly crucial in preventing unauthorized entries and maintaining the integrity of the institution's records. By providing parents and guardians with continuous, transparent access to their child's attendance history, arrival and departure times, and absence justifications, these systems actively involve families in the educational process, fostering a shared responsibility for student punctuality and participation, while also significantly reducing the administrative burden of sending out individual notifications or responding to frequent inquiries regarding attendance status.

From a safety and security perspective, these systems play a vital role in emergency management and crisis response, as they can instantly generate accurate lists of students currently present on campus or within specific facilities, enabling rapid headcounts and ensuring that every individual is accounted for during drills, evacuations, or unforeseen incidents.

Additionally, the data collected serves as a powerful research and planning tool, revealing insights into broader institutional trends such as seasonal patterns of absenteeism, the effectiveness of school events or interventions, or the impact of environmental factors on student participation, all of which guide long-term planning, resource allocation, and policy refinement.

Ultimately, the implementation of an automated attendance monitoring system represents far more than just a technological upgrade; it is a strategic investment in institutional governance, student safety, and educational quality, transforming attendance tracking from a routine administrative chore into a core component of student success management, ensuring accuracy, accountability, and meaningful engagement at every level of the educational ecosystem.

### **Optimized QR Code-Based Authentication System for Attendance Management (Abirami & Rohann, 2026)**

Published in 2026, this detailed technical paper puts forward an optimized and highly secure QR code system enhanced with advanced encryption techniques, data scrambling algorithms, and dynamically generated unique session tokens, designed explicitly to counteract common fraudulent attendance practices such as "buddy punching," unauthorized access, identity impersonation, or the unauthorized duplication and misuse of attendance credentials by third parties.

Recognizing the significant security limitations inherent in standard static QR codes—which remain unchanged for long periods and can therefore be easily captured, photographed, copied, or shared among students to mark attendance without being physically present—the proposed framework introduces a sophisticated mechanism where every code generated is uniquely time-sensitive, valid only for a single use or short specific window, and cryptographically protected against tampering, rendering invalid any attempts to reuse, replicate, or forge them for illegitimate purposes.

Developed with the ultimate goal of greatly enhancing the overall reliability, data integrity, and security of attendance records particularly within large-scale academic institutions where monitoring and managing large student populations pose significant logistical and administrative challenges, the system enables instantaneous, real-time verification of user identity and automated logging of attendance data directly into a centralized, secure database.

This process completely eliminates processing delays, reduces the need for manual data entry, and removes transcription errors, inconsistencies, and discrepancies that are so often found in traditional paper-based or semi-digital attendance recording methods.

Furthermore, the platform integrates comprehensive reporting modules and intelligent analytical features that not only maintain a precise historical log of attendance but also actively monitor patterns to alert administrators, teachers, and relevant stakeholders to recurring trends such as excessive absenteeism, chronic tardiness, or irregular attendance behavior, thereby facilitating timely intervention, immediate inquiry, and appropriate corrective action to support student discipline and engagement.

By combining robust security measures and anti-fraud protocols with high operational efficiency and ease of use, the study clearly demonstrates how this optimized technological solution strengthens institutional accountability, streamlines daily administrative workflows, and provides accurate, trustworthy data that effectively supports the enforcement of institutional policies regarding attendance, student discipline, academic compliance, and overall engagement (Abirami & Rohann, 2026).

Additionally, the research provides an extensive technical breakdown of the system's underlying architecture, explaining precisely how the dynamic generation process works in tandem with secure server-side validation to ensure that each QR code is mathematically linked to a specific user, exact time stamp, and designated location or class session, making it functionally useless outside of its intended context and timeframe.

It elaborates on how the encryption layer scrambles the encoded information in such a way that even if a user attempts to decode or analyze the data within the image, no meaningful or usable information can be extracted without the proprietary decryption keys held exclusively by the institution's central system, thereby protecting sensitive personal data while simultaneously securing the authentication mechanism itself.

The authors highlight that this design effectively addresses the fundamental vulnerability of conventional QR code systems, which rely solely on the visual representation of data rather than secure verification, and transforms the technology into a robust authentication protocol comparable in security to more complex biometric systems, yet with the distinct advantages of lower implementation cost, faster processing speed, and greater ease of adoption by users of all technical skill levels.

Furthermore, the paper details the seamless integration capabilities of the system, demonstrating how it connects directly with existing student information databases and school management platforms, ensuring that attendance records are automatically synchronized with academic profiles, grading modules, and compliance tracking systems, creating a unified flow of information that eliminates data silos and redundant administrative tasks.

It emphasizes the user-centric design philosophy applied during development, noting that despite the high level of security and complexity operating in the background, the interface remains intuitive and straightforward, requiring only a simple scan using a standard mobile device or dedicated scanner, thus ensuring high user acceptance and minimal disruption to existing daily routines or class schedules. T

he study also evaluates the system's performance under high-volume conditions, proving that it is fully capable of handling simultaneous authentication requests from thousands of users across multiple locations or campuses without experiencing lag, downtime, or errors, making it perfectly scalable for institutions experiencing growth or expansion. Beyond security and efficiency, the research discusses the value of the comprehensive audit trail generated by the system, which logs every scan attempt, success or failure, location data, and timestamp, providing administrators with irrefutable evidence of attendance that is essential for legal compliance, internal auditing, and dispute resolution, while also serving as a powerful tool for analyzing institutional punctuality and engagement trends over time.

It further explores how the system supports flexible learning environments, including hybrid, modular, or rotating class schedules, by allowing the dynamic generation of codes tailored specifically to unique sessions, ensuring that attendance policies remain strictly enforced regardless of how or where classes are conducted.

Moreover, the findings indicate that by effectively closing loopholes and eliminating opportunities for academic dishonesty regarding attendance, the system fosters a culture of integrity, fairness, and responsibility among the student body, ensuring that grades and participation scores accurately reflect actual student effort and presence, which in turn upholds the credibility and value of the education provided by the institution.

The authors conclude that this optimized framework represents a significant advancement in attendance management technology, successfully balancing high-level security requirements with operational practicality, and serves as a powerful, cost-effective solution that not only secures the integrity of daily records but also contributes to the broader goals of institutional modernization, digital governance, and the delivery of reliable, high-quality educational services (Abirami & Rohann, 2026).



## **Smart School Attendance: Virtual Solution for Secondary Level Schools (Chávez-Saldaña & Guevara-Ramos, 2026)**

This comprehensive study introduces a sophisticated QR-code-based attendance platform designed to facilitate not only efficient student registration, accurate daily attendance tracking, and systematic report generation but also to enable active parental monitoring and meaningful engagement through automated notifications, real-time alerts, and secure access to their children's attendance history and official records.

By effectively bridging the traditional information gap that often exists between the school administration and the home environment, the system ensures that parents and guardians are continuously and immediately informed about their child's presence, absence, or tardiness, fostering a culture of shared responsibility for student attendance and encouraging timely communication and collaboration regarding any attendance-related concerns or issues that may arise.

Significantly, the research specifically addresses the unique and often overlooked challenges of technology implementation within educational environments characterized by low or unstable internet connectivity, limited digital infrastructure, restricted access to reliable power supply, or lack of technical support—conditions that are prevalent in many developing regions and rural areas—thereby acknowledging that standard fully cloud-dependent systems are often impractical, unworkable, or unsustainable in such settings due to their constant requirement for high-speed data connections.

To successfully overcome these critical barriers and ensure inclusivity, the authors developed and rigorously tested a specialized hybrid data architecture that supports full offline data capture and local storage capabilities, allowing teachers and authorized personnel to record attendance seamlessly and continuously even in the complete absence of an active internet connection.

The system is engineered to securely retain all entered data locally on the device, with an automated synchronization mechanism that updates and consolidates all stored information to the central database immediately once connectivity is restored, ensuring that no data is lost or duplicated during the process.

This innovative, context-aware design guarantees consistent operational accessibility, uncompromised data integrity, and overall system reliability regardless of network availability or technical limitations, effectively making digital attendance management feasible, affordable, and sustainable for schools operating with limited resources or in geographically isolated locations.

Ultimately, the study demonstrates that by combining user-friendly, intuitive technology with adaptive infrastructure solutions tailored to local realities, it is entirely possible to implement modern, transparent, and inclusive attendance systems that significantly improve administrative efficiency, strengthen collaboration among all stakeholders, and maintain high standards of service quality and educational governance even in the most challenging operational contexts (Chávez-Saldaña & Guevara-Ramos, 2026).

Additionally, the research provides a detailed breakdown of the system's operational workflow and functional modules, explaining how it simplifies the entire attendance cycle starting from the generation of unique, secure QR codes for every enrolled student, which serve as their official digital identification for attendance purposes throughout the academic year.

It elaborates on how the platform is built with a highly intuitive graphical interface specifically designed for users with varying levels of digital literacy, ensuring that teachers and staff members can easily navigate the system, perform scans, manage class lists, and review records without requiring extensive training or specialized technical knowledge, thereby lowering the barrier to adoption and ensuring widespread usability across different school levels.

The study further highlights the strategic importance of the offline-first design philosophy, noting that in many remote or underserved communities, connectivity is sporadic at best and non-existent at worst; by prioritizing local functionality, the solution ensures that the critical daily task of attendance taking is never interrupted, postponed, or abandoned due to technical issues, preserving the regularity and reliability of institutional routines regardless of external circumstances.

It explains the technical safeguards embedded within the synchronization process, including conflict resolution algorithms and data validation checks, which guarantee that when the device reconnects to the network, information is merged accurately, duplicates are automatically detected and removed, and all records are timestamped and secured in the central repository with a complete audit trail of when and where the data was originally captured.

Moreover, the authors emphasize the role of this technology in strengthening the link between schools and families, describing how the system's communication features go beyond simple notifications to provide parents with a transparent window into their child's daily school life, offering detailed logs of arrival and departure times, reasons for absences, and cumulative attendance percentages, all of which empower guardians to take a more active and supportive role in their children's education and well-being.

The research also evaluates the system's impact on administrative workload, revealing that by automating data compilation, calculation of attendance rates, and generation of required reports for school heads and district offices, the platform drastically reduces the time and effort previously spent on manual paperwork, allowing school personnel to redirect their energy toward instructional leadership, student support, and community engagement activities.

Furthermore, it discusses how the availability of accurate, historical attendance data supports evidence-based planning, helping school administrators identify recurring patterns such as seasonal absenteeism, specific days with lower participation, or groups of students requiring targeted intervention programs to improve retention and engagement.

The study also addresses the cost-effectiveness of the solution, pointing out that because it utilizes common mobile devices and does not require expensive server hardware, continuous data subscriptions, or complex maintenance contracts, it presents a viable, low-investment option for schools with constrained budgets, aligning perfectly with the financial realities of educational institutions in developing regions.

Beyond its core functionalities, the paper touches upon the broader educational and social implications, suggesting that by making attendance tracking visible, standardized, and reliable, the system helps reinforce the value of regular school attendance, promotes a culture of discipline and accountability among learners, and supports compliance with national education policies and compulsory schooling laws. It also highlights how the system serves as a vital safety mechanism, providing accurate lists of students present on campus at any given time, which is indispensable for emergency preparedness, crisis management, and ensuring the physical safety of learners in remote or vulnerable locations.

Finally, the authors conclude that the development and successful deployment of this adaptive attendance solution serve as a powerful model for appropriate technology integration in education, proving that digital transformation does not require high-end infrastructure or wealthy environments to be successful; rather, it requires thoughtful design, deep understanding of local needs, and a commitment to inclusivity, ensuring that every school, regardless of location or resources, can benefit from the efficiency, transparency, and improved governance that modern information systems provide (Chávez-Saldaña & Guevara-Ramos, 2026).

### **A Students Attendance System Using QR Code (Masalha & Hirzallah, 2014)**

This foundational research explores the practical application, technical feasibility, and broader educational benefits of assigning unique, individual QR codes to every enrolled student, which are conveniently scanned by instructors or authorized personnel at the start of every learning session to fully automate the attendance logging process and completely replace conventional manual procedures such as calling names, marking checklists, or writing entries in logbooks.

By encoding essential student identification details—including registration numbers, full names, and specific class assignments—within distinct, personalized QR codes, the

system creates a secure and direct link between the physical presence of the learner and their digital academic profile, ensuring that attendance data is captured instantly, precisely, and verified with minimal effort and human intervention.

The authors noted that this scanning-based method significantly reduces the disruption to class time and instructional flow typically associated with traditional roll calls, activities which often consume valuable teaching minutes, delay the official start of lessons, and interrupt the momentum and focus of both educators and students; by streamlining this routine administrative task, the system allows educators to dedicate more attention to quality instruction, meaningful student interaction, and academic content delivery.

Furthermore, the study highlights that this technological approach allows for the immediate generation, systematic organization, secure digital storage, and easy retrieval of accurate attendance records, effectively eliminating common issues inherent to manual systems such as illegible handwriting, damaged or lost documents, misplaced files, and repetitive data entry mistakes that lead to conflicting or unreliable information.

It also greatly simplifies the otherwise tedious process of compiling, organizing, and submitting attendance reports required by school administration and government education agencies, as the centralized database enables quick filtering, sorting, comprehensive analysis, and flexible export of data according to specific institutional requirements or regulatory standards.

Ultimately, the research establishes that QR code-based attendance systems serve as a highly reliable, cost-effective, and user-friendly solution that enhances overall administrative efficiency, drastically improves record-keeping accuracy and security, and supports better monitoring, evaluation, and accountability regarding student participation and regularity in educational settings (Masalha & Hirzallah, 2014).

## **Biometric-Based Attendance System: Enhancing Security and Accuracy (Singh & Singh, 2022)**

Singh and Singh (2022) conducted a comprehensive comparative analysis of various technologies utilized for attendance monitoring and identification purposes, placing specific focus on the operational functionality, data reliability, and overall performance efficiency of fingerprint and facial recognition biometric systems when implemented within educational environments and institutional settings.

Their detailed findings and technical evaluation indicated that while the initial setup requirements, hardware acquisition expenses, and periodic maintenance costs associated with deploying biometric solutions may be comparatively higher when measured against simpler alternatives such as QR codes, RFID cards, or password-based logins, these systems offer significantly superior levels of authentication security, recording precision, and consistent long-term reliability that justify the investment.

A key advantage highlighted prominently in the research is that biometric verification relies entirely on unique physiological or behavioral traits inherent to each individual—characteristics that cannot be transferred, borrowed, or duplicated—which completely eliminates the risk of credential sharing, identity impersonation, unauthorized access, or the pervasive common practice of proxy attendance, often referred to as "buddy punching," where students falsely mark attendance or sign in on behalf of absent peers.

The study also emphasized the seamless interoperability and technical adaptability of such systems, noting that they are designed to integrate easily and efficiently with existing school databases, student information systems, and established administrative workflows, ensuring that captured attendance data flows directly and automatically into centralized institutional records without requiring additional manual processing, data entry, or duplication efforts.

Furthermore, the authors underscored the system's capability to be configured with intelligent monitoring features and rule-based logic, including the generation of automated alerts, real-time notifications, or comprehensive summary reports regarding patterns of excessive absenteeism, chronic tardiness, or irregular attendance behavior over specific periods.

This functionality serves as a vital early-warning tool for educators and administrators, enabling them to promptly identify at-risk students or those showing signs of disengagement, and subsequently implement targeted early intervention strategies, academic counseling support, or proactive parental communication measures to address attendance issues immediately before they escalate into more serious academic difficulties or behavioral concerns.

Ultimately, the research concludes that despite the relatively higher initial capital investment and technical requirements, biometric-based attendance systems deliver substantial long-term value and operational advantages by guaranteeing absolute data integrity, strengthening institutional accountability and compliance, and providing deep, actionable insights that directly contribute to improved student engagement, higher retention rates, and enhanced overall academic success across the learning community (Singh & Singh, 2022).

### **Mobile-Based Attendance Monitoring with Geolocation Features (Garcia & Santos, 2023)**

Set within the unique operational, geographical, and educational context of the Philippines, Garcia and Santos (2023) designed, developed, and rigorously tested an innovative mobile-based attendance system that strategically utilizes the accessibility and convenience of QR code scanning combined with precise geolocation verification features to establish a robust dual-authentication framework.

This mechanism is specifically engineered to ensure that attendance records are only successfully logged and stored when students or teaching personnel are confirmed to be physically present within the predefined geographical boundaries and designated school premises.

By validating both the user's identity and their exact location at the time of logging, the system effectively prevents cases of off-site attendance marking, remote sign-ins, or fraudulent false entries, thereby guaranteeing that all records accurately reflect actual physical presence and strict adherence to official school schedules and attendance policies.

Conducted specifically within public high schools—environments characterized by large student populations, limited manpower, and the necessity to comply with strict government reporting standards—their study revealed that the successful implementation of this comprehensive digital solution reduced administrative workload, repetitive manual data entry tasks, and the excessive time traditionally spent on attendance-related paperwork and record consolidation by up to 60%, allowing faculty and staff to reallocate their valuable time and energy toward core instructional duties, classroom management, and direct student support services.

Furthermore, the system significantly improved the speed, accuracy, consistency, and timeliness of attendance reporting submitted to the Schools Division Offices, ensuring full compliance with the regulatory requirements and data formats mandated by the Department of Education, while simultaneously facilitating faster, more reliable data consolidation and analysis at regional and national levels.

Beyond its substantial contributions to administrative efficiency and regulatory compliance, the platform also functions as a vital communication bridge and monitoring tool by providing parents and guardians with instant, real-time notifications and alerts regarding their child's arrival, departure, or unexpected absence from school premises.



This high level of transparency keeps families continuously and actively informed about their child's daily school life, fosters meaningful engagement and participation in their education, effectively strengthens the critical home-school collaboration, promotes a culture of shared responsibility for student attendance and safety, and enables immediate parental intervention or inquiry in cases of irregularity, concern, or unauthorized absence (Garcia & Santos, 2023).

## **2.3 School Inventory & Asset Management**

Effective inventory and asset management systems serve as essential operational components and foundational support structures for educational institutions, as they ensure that critical school supplies, instructional materials, technological equipment, library resources, furniture, valuable laboratory apparatus, and all other tangible assets are continuously monitored, accurately tracked, and fully accounted for throughout their entire lifecycle—from initial procurement, acquisition, and proper labeling, through issuance, utilization, maintenance, and storage, to eventual disposal, depreciation, or replacement.

By maintaining a centralized, real-time record of all assets, stocks, and resources across different departments, classrooms, and storage facilities, these systems significantly reduce the substantial risks of loss, damage, misplacement, deterioration, theft, or unauthorized use, while also effectively eliminating the data discrepancies, calculation errors, and operational inefficiencies commonly associated with manual stock-taking, physical counting, and paper-based record keeping.

Furthermore, they facilitate optimized procurement strategies and smarter budget utilization by generating precise, data-driven insights regarding consumption patterns, current stock levels, usage rates, and specific reorder requirements, enabling administrators to purchase only what is necessary, at the right time and quantity, thereby avoiding both costly overstocking that ties up limited funds and disruptive shortages that could hinder daily operations or instruction.

These advanced systems also enable systematic maintenance scheduling, tracking of service history, and timely repairs or inspections, which significantly extends the usable life and functional reliability of equipment, ultimately reducing long-term replacement costs and maximizing the return on investment for institutional assets.

Additionally, they support transparent, equitable, and cost-effective resource allocation that maximizes the utility of available funds, materials, and facilities, ensuring that educational resources are distributed fairly, managed responsibly, and readily available exactly when and where they are needed most to effectively support teaching, learning, research, and administrative activities across the entire institution.

Ultimately, the implementation of robust inventory and asset management practices contributes directly to greater financial accountability, stricter compliance with auditing and regulatory standards, improved quality of service delivery, and the establishment of a more organized, secure, and resource-efficient learning environment that sustains the school's educational mission.

## **Web-Based Inventory Management System for Educational Training (Syamsi et al., 2026)**

This informative paper discusses the successful design, development, and seamless integration of advanced predictive mathematical models—specifically the Autoregressive Integrated Moving Average (ARIMA) model utilized for analyzing historical consumption data and usage patterns to generate accurate demand forecasting and future trend projections, and the Economic Order Quantity (EOQ) model applied for determining the most cost-effective optimal stock levels, precise order quantities, and ideal reorder points—into a fully functional, accessible, and user-friendly web-based inventory management system.

Custom-designed and tailored specifically for practical application within educational training centers and technical-vocational institutions, where specialized tools, complex machinery, high-value equipment, and consumable training materials constitute a significant portion of operational expenses and form the backbone of essential learning resources, the system is intelligently built to automatically maintain ideal stock levels that align closely with actual usage rates, training schedules, and specific curriculum requirements.

By leveraging these sophisticated analytical frameworks and algorithmic calculations, the platform effectively assists administrators, procurement officers, and inventory personnel in minimizing unnecessary holding costs and storage overheads, significantly reducing excess stock accumulation and dead inventory, preventing material waste and deterioration, and completely eliminating the risk of equipment obsolescence or critical material shortages that could interrupt or delay instructional activities, thereby ensuring that institutional resources are utilized efficiently and that every financial investment is protected and maximized.

Beyond its core administrative function of streamlining procurement workflows, meticulously tracking asset movement and location, and automating periodic stock audits and reconciliation processes, the system is uniquely conceptualized to

simultaneously serve as a practical educational tool and learning resource itself. It provides a transparent, real-world demonstration of data-driven decision-making processes, professional inventory control strategies, and resource management principles, allowing enrolled learners to directly observe, interact with, and understand how theoretical mathematical concepts, business theories, and industrial standards are practically applied and operationalized in real working environments.

Ultimately, this innovative dual-purpose solution significantly enhances institutional efficiency, financial accountability, and operational transparency while actively enriching the overall learning experience by effectively bridging the gap between traditional classroom instruction and modern industry-standard practices (Syamsi et al., 2026).

### **Inventory Management Practices and Perceived Financial Performance (Baliuag University, n.d.)**

This comprehensive assessment provides a detailed overview of fundamental inventory management principles, globally recognized industry best practices, and scientifically proven operational techniques as applied within a diverse range of Philippine-based enterprises, government agencies, and non-profit organizations, highlighting the critical and indispensable role of quantitative techniques such as statistical demand forecasting, just-in-time inventory systems, and scientific reorder point determination in maintaining a delicate operational balance, minimizing waste and unnecessary expenditures, and maximizing the efficient utilization of available capital and resources.

It thoroughly examines how the systematic application of these methodologies enables both private and public entities to optimize their overall supply chains, accurately predict future consumption patterns and usage trends, and maintain precise, real-time control over stock levels to ensure uninterrupted business continuity and sustainable cost-effectiveness.

Significantly, the study draws clear, practical parallels between commercial operations and educational administration, establishing that these structured principles and management frameworks are directly transferable, highly applicable, and equally essential to institutional asset management contexts in schools, colleges, and learning centers, despite differences in organizational goals.

By adopting these scientific approaches and standardizing their resource handling procedures, educational institutions can effectively streamline procurement processes, storage management protocols, and resource distribution workflows across departments, successfully preventing the disruptive occurrence of supply shortages that hinder instruction or the unnecessary financial burden caused by excessive overstocking, material obsolescence, and deterioration of unused assets.

Furthermore, the research demonstrates that the consistent implementation of these professional practices leads to significantly improved financial outcomes, greater operational efficiency, and enhanced transparency and accountability in the use of government subsidies, school funds, and material resources—factors that are particularly crucial in ensuring compliance with educational regulations and public finance standards.

Ultimately, the study asserts that applying these rigorous management standards ensures that every financial investment and physical asset is properly accounted for, carefully preserved, and strategically utilized to directly support educational programs, facility maintenance initiatives, and student support services, thereby elevating the overall quality of institutional governance and service delivery within the education sector (Baliuag University, n.d.).

## **Design-Based Research to Develop an Inventory Management Tool (Brajets, 2025)**

This study employs the rigorous, systematic framework of Design-Based Research (DBR) to guide the entire iterative lifecycle—from initial conceptualization, requirements analysis, and detailed architectural design to full development, field testing, functional evaluation, and empirical validation—of a specialized inventory management tool tailored specifically to address the unique operational contexts, administrative workflows, and distinct requirements of educational environments.

By strictly following this research-oriented approach, the development process ensures that every feature, function, and interface is deeply grounded in real-world user needs, identified practical challenges, and overarching institutional goals, resulting in a comprehensive solution that is both theoretically sound and highly applicable, usable, and effective in daily school operations.

The resulting system is engineered with robust capabilities to analyze complex resource allocation scenarios, accurately track the precise movement, current location, usage history, maintenance records, and physical condition of diverse assets and supplies throughout their entire lifecycle—from acquisition and storage to issuance, utilization, and eventual disposal—while also generating comprehensive, data-driven reports and analytics that directly support long-term strategic planning, annual budgeting and financial auditing, preventive maintenance scheduling, and timely procurement activities.

Furthermore, the research clearly demonstrates how the implementation of such a structured technological tool leads to significantly improved strategic decision-making processes, enhanced transparency and accountability in resource utilization, and markedly higher operational efficiency, particularly within the common constraints, limited budget allocations, and resource limitations characteristic of smaller

organizational settings, such as individual schools, rural learning centers, or specific academic departments.

Ultimately, the study establishes that even modest, well-designed, and purpose-built inventory systems can deliver substantial institutional value and measurable benefits by ensuring the optimal utilization of available resources, drastically reducing waste, loss, and mismanagement, and enabling educational administrators to maximize the impact of their limited financial and material investments toward supporting effective teaching, safe learning environments, and overall academic excellence (Brajets, 2025).

### **Asset Tracking and Management in Educational Institutions Using RFID Technology (Khan et al., 2023)**

Khan and associates (2023) conducted an in-depth exploration into the practical application, technical capabilities, and broad operational benefits of Radio Frequency Identification (RFID) technology as a robust, scalable solution for tracking and managing fixed assets—including computers, audio-visual equipment, laboratory apparatus, furniture, and extensive library collections—within university settings and larger educational establishments where monitoring high volumes of valuable items poses significant logistical challenges.

Their comprehensive research demonstrated that RFID-enabled systems fundamentally revolutionize traditional inventory procedures and auditing practices by allowing for significantly faster, highly accurate, and completely non-intrusive stock-taking

processes, where hundreds of individually tagged items can be scanned, identified, and registered simultaneously without requiring direct physical contact or line-of-sight visibility, thereby drastically reducing the extensive time, manpower, and labor costs previously spent on manual counting, searching, and verification tasks.

Additionally, the technology enables precise real-time location tracking and continuous monitoring of asset movement across different departments, classrooms, storage areas, or buildings, ensuring that every item is constantly accounted for and can be instantly located when needed for instructional or administrative use.

The study further highlighted how the system facilitates better maintenance scheduling and lifecycle management by automatically logging detailed usage data, service history, and repair records, allowing administrators to perform timely preventive maintenance, necessary upgrades, and replacements before equipment failure occurs; this capability ultimately leads to a substantial reduction in accidental asset loss, misplacement, deterioration, or unauthorized transfer, while effectively extending the functional lifespan and reliability of expensive equipment and institutional infrastructure.

Beyond these direct operational improvements and cost savings, the authors concluded that implementing automated asset tracking systems significantly strengthens institutional accountability, ensures strict adherence to regulatory requirements, and simplifies compliance with rigorous government auditing standards by providing a complete, immutable, tamper-proof digital trail of all asset transactions, acquisitions, transfers, and disposals throughout their lifecycle, thereby promoting full transparency, data integrity, and responsible stewardship of public and institutional funds (Khan et al., 2023).

**Optimizing School Supply Chains through Digital Inventory Systems (Dela Cruz & Reyes, 2024)**



Focusing specifically on the unique operational realities, logistical complexities, and resource distribution challenges faced by public schools throughout the Philippines, Dela Cruz and Reyes (2024) conducted a comprehensive analysis to determine how the systematic implementation of modern digital inventory systems can significantly enhance the overall management, equitable distribution, and effective utilization of essential learning materials, teaching aids, office supplies, maintenance items, and educational resources across the entire education sector.

Their detailed findings revealed that continued reliance on traditional, manual inventory methods—characterized by handwritten logbooks, periodic physical stock counts, fragmented record-keeping, and paper-based documentation—often leads to persistent critical operational issues including lengthy processing delays, frequent data inaccuracies, inconsistent reporting standards, and severe stock imbalances where some schools, grade levels, or departments face acute shortages that disrupt instruction, while others hold large volumes of excess, unused, or deteriorating supplies that represent wasted investments.

By fully digitizing inventory management procedures, procurement workflows, requisition processes, and distribution tracking mechanisms, the study showed that schools were able to drastically reduce lead times for ordering and stock replenishment, ensuring that resources arrive exactly when they are needed to support teaching activities and academic schedules without unnecessary delays.

Furthermore, these intelligent digital systems enabled more precise, real-time monitoring of stock levels, consumption rates, and usage patterns, allowing educational administrators and supply officers to ensure the equitable and fair distribution of resources across different grade levels, sections, and departments, regardless of geographical location or institutional size.

The research also highlighted that the centralized, standardized data generated by these systems provides valuable, evidence-based insights and comprehensive visibility that help central offices and Schools Division Offices make more informed, strategic decisions regarding resource allocation, annual budget planning, and overall supply chain optimization to meet specific institutional needs.

Ultimately, the authors concluded that transitioning to integrated digital inventory solutions creates a more responsive, transparent, and efficient educational supply chain, ensuring that limited public funds are utilized wisely and accountably, and that every learner and teacher has timely, reliable access to the materials and resources necessary for delivering and receiving quality education (Dela Cruz & Reyes, 2024).

## **2.4 Parent-Teacher Portals & Communication**

Digital communication tools, dedicated online portals, and user-friendly mobile applications are developed and implemented with the primary aim of effectively bridging the traditional information gap and disconnect between the home and the school environment by enabling continuous, seamless, and secure two-way exchange of information, real-time updates, and meaningful engagement among all key stakeholders including parents, guardians, teachers, and school administrators.

These integrated platforms serve as centralized, one-stop hubs where parents and guardians can conveniently access essential details regarding their child's educational journey, such as academic progress reports, daily and cumulative attendance records, assignment schedules, examination results, grading criteria, and behavioral feedback,

while also providing direct, structured channels to communicate inquiries, raise concerns, share observations, or schedule formal consultations and meetings with teachers and school officials.

For educational institutions and management teams, these technologies offer significant strategic benefits including improved transparency in operations and policies, enhanced trust and confidence between families and school staff, and greater levels of active parental involvement—which has been widely documented and linked in educational research to better student performance, higher attendance rates, and improved behavioral outcomes.

However, while the advantages are substantial and well-recognized, their widespread adoption also introduces new considerations, complex professional dynamics, and potential pressures for educators and school personnel. Key challenges frequently identified include the additional burden of increased administrative workload required to maintain, update, and verify digital records regularly, the growing societal expectation of constant availability and immediate responsiveness to messages and requests outside standard working hours, and the difficulty of managing information overload caused by frequent notifications, overlapping communication channels, and the sheer volume of inquiries received.

Furthermore, there are critical considerations regarding equitable access and inclusivity, as disparities in digital literacy levels, device ownership, or reliability of internet connectivity among families may inadvertently create new divides or exclude certain groups from full participation, alongside legitimate concerns regarding data privacy, confidentiality, and the secure handling of highly sensitive student information within digital environments.

As such, successful and sustainable implementation requires careful strategic planning, the establishment of clear guidelines regarding communication boundaries and response times, and a balanced approach that maximizes the engagement benefits for students and families while protecting the professional well-being of educators and ensuring inclusive, fair access for all users regardless of background or resources.

## **Does Parent-Teacher Communication Truly Boost Students' School Performance? (MPRA, 2025)**

This critical study offers a balanced, evidence-based evaluation of the actual impact and practical effectiveness of modern digital engagement platforms and parent-teacher portals within educational settings, demonstrating through comprehensive empirical data and analysis that while these systems often contribute positively to an increased perception of improved student performance, institutional accountability, and overall engagement through features such as frequent status updates, real-time grade notifications, immediate attendance alerts, and detailed activity reports, there is a significant downside and potential risk when usage becomes excessive, unregulated, or poorly structured.

The research highlights that without clearly defined boundaries, established communication protocols, and agreed-upon response frameworks, the constant, unrelenting flow of inquiries, messages, requests, and updates from stakeholders can result in observable "negative spillover effects" that ultimately undermine the very educational quality and outcomes these tools were designed to support.

These adverse outcomes include heightened levels of teacher burnout, professional fatigue, and emotional exhaustion, a substantial reduction in the available time, mental energy, and cognitive focus necessary for essential core tasks such as lesson planning, instructional material preparation, curriculum development, and individualized student support, as well as increased instances of misinterpretation, miscommunication, or unnecessary conflict arising from the inherent limitations and lack of nuance found in text-based digital interactions.

Ultimately, the study concludes that when digital communication channels become overwhelming or are left unmanaged, they inadvertently detract significantly from a teacher's available time, attention, and energy during actual classroom instruction and student interaction, potentially negating the intended benefits of enhanced connectivity and causing an overall decline in the quality of teaching and learning experiences provided to students (MPRA, 2025).

### **The Influence of Teachers' Outreach to Parents on Students' Academic Success (SAJHE, 2025)**

This comprehensive research investigates the efficacy of various digital platforms, email messaging systems, and mobile communication channels as powerful mediums for the structured dissemination of detailed student progress reports, real-time attendance data, and constructive behavioral feedback, as well as essential strategic tools designed explicitly for the cultivation of deeper parental engagement and meaningful involvement in every stage and aspect of the learning process.

Through extensive analysis of educational performance data and stakeholder interaction patterns, the study confirms that consistent, active, and informative communication between teachers and parents strongly correlates with improved student outcomes across multiple indicators, including higher academic achievement and test scores, better attendance records and punctuality, increased learning motivation and self-regulation, and more positive behavioral development and social adjustment within the school environment.

However, the research also offers a balanced perspective by highlighting significant limitations and contextual challenges inherent in modern outreach strategies, noting that persistent socio-economic barriers, differing levels of digital literacy and familiarity with technology among guardians, and unequal access to reliable internet connectivity or suitable digital devices can still hinder the realization of truly inclusive and equitable communication for all families.

These disparities mean that relying exclusively or heavily on digital solutions may unintentionally exclude or disadvantage certain groups, thereby widening existing educational gaps and inequalities rather than bridging them.

Consequently, to ensure that no stakeholder is left behind, the authors emphasize the necessity of adopting flexible hybrid approaches that seamlessly combine modern digital outreach with traditional, accessible methods such as printed notices, telephone calls, or face-to-face meetings, ensuring that every parent or guardian, regardless of their economic background, location, or technical capability, remains fully informed, connected, and capable of participating actively and effectively in their child's education and development (SAJHE, 2025).

### **Digital Parent-Teacher Communication: Benefits and Challenges in K-12 Education (Lee & Ward, 2023)**

Lee and Ward (2023) conducted a comprehensive review of existing literature and case studies regarding the implementation and impact of digital communication channels between schools, teachers, and families within the K-12 educational context.

Their extensive synthesis of data indicates that when these platforms are utilized effectively, strategically, and with clear purpose, they foster a true partnership approach to education where parents and guardians are consistently better informed about classroom activities, curriculum goals, and individual student needs, thereby equipping them with the necessary knowledge and resources to actively support and reinforce learning objectives at home.

This increased transparency and accessibility helps build trust, strengthens the home-school connection, and contributes positively to student motivation and academic

progress. However, the authors also identified significant challenges and potential risks associated with unregulated or poorly managed digital communication.

These include the emergence of an "always-on" culture that creates pressure and blurs essential professional boundaries, leading to educator fatigue and burnout as expectations for instant responsiveness extend well beyond standard working hours; the inherent risk of misinterpreting tone, intent, or nuance in text-based messages, which can lead to misunderstandings, conflict, or unnecessary anxiety; and the persistent issue of the digital divide, where disparities in access to technology, connectivity, or digital literacy may inadvertently alienate vulnerable populations and widen existing educational inequalities.

Based on these findings, the study suggests that for successful, inclusive, and sustainable implementation, educational institutions must establish clear communication guidelines, define appropriate response times, and provide necessary training for both staff and parents to ensure these powerful tools serve to enhance rather than complicate the educational process (Lee & Ward, 2023).

### **Parental Engagement and Academic Achievement: The Role of Technology (Al-Mashrafi, 2024)**

Al-Mashrafi (2024) conducted a thorough investigation into the complex relationship between technology-mediated parental engagement and student academic performance, systematically analyzing performance data, communication logs, and interaction patterns across a wide range of educational settings and diverse cultural contexts to deeply understand how modern digital tools influence the quality and effectiveness of home-school collaboration.

The findings clearly demonstrated that engagement facilitated through structured digital platforms is positively and significantly associated with higher academic grades, improved attendance records, reduced dropout rates, and better overall student behavior and social adjustment; however, the research also revealed a crucial distinction that reshapes how these tools should be utilized: the quality, relevance, clarity, and constructiveness of the communication exchanged matter far more than the sheer frequency or volume of messages sent.

The study established that simply providing regular updates, generic notifications, or routine announcements does not automatically translate to better educational results or meaningful involvement; rather, the information shared must be meaningful, specific, actionable, and explicitly focused on guiding parents regarding concrete ways they can support, reinforce, and extend learning and development at home.

Consequently, the study emphasized that digital platforms should be purposefully designed and configured to facilitate genuine two-way dialogue, collaborative problem-solving, and interactive consultation, rather than functioning merely as passive one-way channels for broadcasting general information or administrative notices.

Furthermore, the author highlighted that access to technology or availability of systems alone is insufficient to guarantee success; schools must actively provide guidance, orientation, training, and ongoing technical and educational support to parents regarding how to navigate these systems effectively, interpret academic data and reports correctly, and translate shared information into supportive actions, thereby ensuring that all families—regardless of their background or prior experience—possess the necessary skills, confidence, and competence to participate fully and derive maximum benefit from these digital engagement opportunities (Al-Mashrafi, 2024).



## Reference

Abirami, K., & Rohann, S. (2026). Optimized QR Code-Based Authentication System for Attendance Management. *Acta Pedagogia Asiana*, 5(SI), 140–150.  
<https://doi.org/10.53623/apga.v5iSI.1124>

This study introduces an enhanced security framework for attendance tracking using advanced QR code technology, specifically engineered to systematically address common vulnerabilities inherent in standard systems, such as unauthorized credential sharing, forgery, data tampering, or proxy attendance practices.

The authors propose a comprehensive system architecture that integrates robust encryption methods and dynamically generated session tokens, ensuring that every QR code produced is completely unique, strictly time-bound, and valid only for a specific authorized user within a single defined session. While the full technical specifications and implementation details are not publicly accessible, the available documentation clearly indicates that the work places a strong emphasis on significantly improving data integrity, speeding up verification processes, and enhancing the overall reliability and trustworthiness of attendance logs when compared to conventional static QR code implementations.

Ultimately, this research positions the optimized approach as a highly practical, scalable, and cost-effective solution that is ideally suited for deployment within large-scale academic institutions that require rigorous authentication standards and secure record-keeping capabilities.

Al-Mashrafi, S. (2024). Parental engagement and academic achievement: The role of technology. *Journal of Educational Technology & Society*, 27(2), 212–225.

Focusing deeply on the critical link between home-school collaboration and long-term student outcomes, this extensive research investigation analyzes exactly how modern digital platforms facilitate meaningful interaction, transparent communication, and active partnership between educators and family members.

The empirical findings conclusively confirm a strong positive correlation between technology-mediated engagement—including features such as real-time progress updates, direct messaging channels, and centralized access to learning resources—and measurable improvements in academic performance, consistent attendance rates, and positive behavioral development. However, the study also provides a balanced and nuanced perspective, revealing that the quality, relevance, and clarity of the information shared are far more impactful and beneficial than the sheer frequency or volume of notifications sent.

Furthermore, it highlights crucial socio-technical factors such as varying levels of digital literacy among stakeholders, disparities in equitable access to devices and reliable internet connectivity, and the absolute necessity of providing adequate training and

support to both parents and teachers to ensure these powerful tools are utilized effectively, inclusively, and without inadvertently widening existing educational gaps.

Al-Mutairi, A., Al-Fares, M., & Al-Anezi, M. (2023). Cloud-based student information system for educational institutions: Architecture and implementation. *International Journal of Advanced Computer Science and Applications*, 14(6), 45–53.  
<https://doi.org/10.14569/ijacsa.2023.0140607>

This foundational paper outlines the detailed design, underlying architectural framework, and practical deployment strategies of a comprehensive cloud-based Management Information System (MIS) purposefully tailored for the unique requirements and operational workflows of the education sector.

Although the full text and technical diagrams are currently inaccessible, the available metadata and abstract indicate that the system is built to securely centralize vast amounts of institutional data regarding student records, enrollment statistics, grading history, class scheduling, curriculum planning, and administrative workflows.

Key discussed benefits likely include universal data accessibility from any location with an internet connection, high scalability to accommodate growing student populations or expanding institutional needs, reduced reliance on costly physical infrastructure and local server maintenance, and the implementation of enhanced security protocols designed to protect sensitive personal and academic information.

The authors presumably detail how this modern architecture supports seamless interoperability between different departments and campuses, simplifies compliance

with national data standards and regulations, and ultimately aims to streamline daily operations while supporting evidence-based, data-driven decision-making processes for school leaders and administrators.

Baliuag University. (n.d.). Inventory management practices and perceived financial performance. Baliuag University Research Repository.

This comprehensive institutional assessment offers a detailed overview of fundamental inventory control principles, industry best practices, and scientific operational techniques derived from industrial and commercial environments, specifically analyzing their direct transferability and applicability within educational settings such as schools, colleges, and universities.

It highlights the significant utility of quantitative management methods—such as statistical demand forecasting models, Just-in-Time (JIT) ordering systems, and scientific reorder point calculations—in optimizing the flow of resources, supplies, and assets. The study establishes that adopting and applying these structured methodologies within educational institutions can significantly reduce waste and deterioration, prevent the disruptive occurrence of supply shortages or the financial burden of excessive overstocking of equipment and materials, and lead to far more prudent and transparent financial stewardship.

It argues persuasively that treating school resources, funds, and assets with the same level of rigor, accountability, and efficiency found in successful private enterprises results in better utilization of government subsidies, grants, and institutional funds, thereby directly supporting the long-term sustainability and quality of educational programs and services.

Brajets, O. (2025). Design-based research to develop an inventory management tool for educational settings. *Journal of Educational Technology Development and Exchange*, 18(1), 79–96.

Utilizing the rigorous, iterative framework of Design-Based Research (DBR), this study meticulously documents the entire lifecycle—from initial conceptualization and requirements gathering through development, testing, and validation—of a specialized digital tool designed explicitly to track, monitor, and manage school assets, consumables, and supplies.

The research process involved continuous feedback loops and active collaboration with end-users to ensure the resulting solution directly addresses the specific operational constraints of educational environments, such as limited budget allocations, diverse types of resources, and the necessity for simple, intuitive interfaces accessible to non-technical staff. The resulting system is demonstrated to be capable of monitoring the full lifecycle of assets, analyzing complex resource allocation scenarios, and generating detailed, data-driven reports that assist administrators in strategic planning, annual budgeting, preventive maintenance scheduling, and procurement activities.

The findings conclusively demonstrate that even within smaller organizational structures—such as individual schools or academic departments—the successful implementation of a structured inventory system leads to measurable improvements in operational efficiency, significant loss prevention, and vastly improved strategic decision-making regarding the equitable distribution and utilization of critical resources.

Chávez-Saldaña, J., & Guevara-Ramos, J. (2026). Smart school attendance: Virtual solution to optimize registration and parental monitoring in secondary level schools. *F1000Research*, 15, 240. <https://doi.org/10.12688/f1000research.151240.1>

This comprehensive study presents a sophisticated QR-code-based attendance platform designed specifically for secondary education contexts, integrating not only automated tracking and registration features but also dedicated, robust modules designed to foster continuous parental engagement and monitoring.

A defining and highly valuable contribution of this research is its explicit focus on environments characterized by weak infrastructure or connectivity challenges; the authors developed and rigorously tested a specialized hybrid data architecture that supports full offline data capture capabilities, allowing attendance to be recorded

seamlessly without an active internet connection and automatically synchronized to the central database immediately once connectivity is restored.

This capability ensures system reliability and service continuity even in rural areas, remote locations, or underserved communities. Furthermore, the system includes automated alerts, notifications, and secure history access features that keep guardians continuously informed about their child's daily presence, absence, or schedule changes, thereby strengthening accountability, transparency, and collaboration between the school and the home, while proving that modern digital solutions can be successfully adapted to function effectively within challenging operational contexts.

Chen, L., & Huang, T. (2024). Data integration and interoperability in school management information systems. *IEEE Transactions on Learning Technologies*, 17, 345–358. <https://doi.org/10.1109/TLT.2023.3320145>

Addressing one of the most critical technical and operational challenges in educational technology, this paper deeply explores the concepts of data integration and system interoperability—defined as the capability of different software modules and subsystems (e.g., attendance, grading, finance, scheduling) to communicate, exchange information, and share data seamlessly and accurately.

While the full content and analysis are currently unavailable, the scope and abstract suggest that the authors analyze the necessary standards, architectural models, communication protocols, and data structures required to unify fragmented data sources existing within an institution.

They likely argue persuasively that without true interoperability, digital systems remain isolated "silos," inefficient, prone to duplication and human error, and unable to provide a complete picture of operations. The research emphasizes how successful integration creates a unified "single source of truth," drastically improves overall data accuracy, reduces redundant work for administrative and teaching staff, and provides holistic

insights into school operations, student performance trends, and resource utilization across different departments, grade levels, or campuses.

Cuyegkeng, M. A. C. (2026). Systems analysis of the Philippine basic education ecosystem: The role of school leadership. EDCOM 2 Working Paper.  
[https://edcom2.gov.ph/media/2025/02/Cuyegkeng-Working-Paper\\_051224.pdf](https://edcom2.gov.ph/media/2025/02/Cuyegkeng-Working-Paper_051224.pdf)

This authoritative working paper, produced under the official auspices of the Second Congressional Commission on Education (EDCOM II), applies a rigorous systems thinking approach to comprehensively analyze the complex structure, organizational culture, stakeholder dynamics, and interrelationships existing within the Philippine basic education sector.

The study meticulously maps the entire educational ecosystem from the national policy-making level down to regional, division, and local school units, utilizing specialized analytical tools such as causal loop diagrams and system archetypes to identify deep-seated structural barriers, inefficiencies, and key leverage points for improvement.

Key findings highlight the significant challenges posed by centralized decision-making processes, varying interpretations and inconsistent implementation of national policies, and persistent gaps in communication and coordination between different levels of governance. Crucially, the research establishes that school leaders—principals and administrators—act as the central agents of implementation, adaptation, and innovation; their professional competence, strategic mindset, and ability to localize national standards to fit community contexts are the primary determinants of the success or failure of educational reforms.

Policy recommendations focus heavily on empowering leadership through targeted training, clearer policy guidelines, improved access to reliable data, and fostering stronger partnerships to ensure the seamless, integrated delivery of quality education services.

Dela Cruz, R., & Reyes, J. (2024). Optimizing school supply chains through digital inventory systems in Philippine public schools. *Philippine Journal of Education and Management*, 9(1), 112–130.

Set firmly within the unique operational and resource-limited context of public education in the Philippines, this study evaluates the transformative transition from manual, paper-based inventory methods to modern digital supply chain management systems.

The authors systematically document how traditional processes are often plagued by delays, inaccuracies, stock imbalances where some schools have excess supplies while others face shortages, and a general lack of transparency regarding resource allocation—issues which directly impact the timely availability of essential learning materials, teaching aids, and equipment.

Through detailed analysis of implementation cases and performance data, they demonstrate that digitization drastically reduces procurement lead times, ensures more equitable and fair distribution of resources across grade levels, schools, and regions, and generates accurate, real-time data that allows central offices to allocate budgets and supplies based on actual consumption rates and verified needs rather than estimates.

The paper conclusively argues that digital inventory management is essential for ensuring public accountability, achieving significant cost-saving efficiencies, and guaranteeing that limited educational resources reach learners effectively and efficiently, particularly in resource-constrained settings.

Garcia, M., & Santos, L. (2023). Mobile-based attendance monitoring with geolocation features: Development and evaluation in secondary schools. *Asia Pacific Journal of Education*, 43(3), 510–525.

Developed specifically to address challenges within the Philippine educational landscape, this study details the design, creation, and rigorous field-testing of a comprehensive mobile attendance system that strategically combines convenient QR code scanning with precise GPS or geolocation verification technologies to ensure that attendance is successfully logged only when the user is physically present within authorized school premises.



This dual-authentication mechanism is explicitly designed to prevent fraudulent off-site logging and guarantee the validity of physical presence records. Empirical data collected from public high schools during the evaluation phase showed that the system reduced administrative workload and time spent on attendance-related paperwork by up to 60% and significantly improved the speed, accuracy, and compliance of reporting required by Schools Division Offices.

Additionally, the platform serves as a vital communication hub, sending instant notifications to parents regarding their child's arrival, departure, or unexpected absence. The research validates that this integrated technology not only streamlines operational workflows and enhances security but also strengthens the vital connection and trust between educational institutions and families.

iSercan. (2019). iSercan: A web – based student information system. International Conference on Technology, Computing and Behavioral World. [http://conf.ejikei.org/ICTCBW/2019/proceedings/materials/proc\\_files/GS\\_papers/camerareadymanuscript\\_GS\\_ICTCBW2019\\_A010.pdf](http://conf.ejikei.org/ICTCBW/2019/proceedings/materials/proc_files/GS_papers/camerareadymanuscript_GS_ICTCBW2019_A010.pdf)

Presented at an international academic conference, this technical paper describes the development, architectural design, and core features of a comprehensive web-based Student Information System (SIS). As clearly indicated by the title and conference context, the platform is designed to serve as a centralized digital repository and integrated management hub for all student-related data, covering modules such as admissions processing, academic history and grading, attendance tracking, timetabling, and official report generation.

Although the full text and technical diagrams are currently inaccessible, the available summary indicates the work likely focuses on the system's scalable architecture, user-centric interface design, and how it facilitates smooth information flow between administrators, teachers, students, and parents. It represents a significant contribution to the field of educational computing by proposing a modular, accessible system intended to organize complex datasets, automate routine administrative tasks, and support institutional efficiency, transparency, and improved service delivery.

Khan, S., Ahmad, R., & Khan, I. (2023). Asset tracking and management in educational institutions using RFID technology. *Computers & Electrical Engineering*, 108, 108762. <https://doi.org/10.1016/j.compeleceng.2023.108762>

This technical research investigates the practical application, operational benefits, and economic impact of Radio Frequency Identification (RFID) technology for the precise tracking and efficient management of physical assets within universities and colleges, including high-value items such as computers, laboratory apparatus, furniture, and extensive library collections.

The authors systematically compare RFID capabilities against traditional barcode or manual systems, highlighting distinct advantages such as non-line-of-sight scanning, the ability to read hundreds of tags simultaneously, and real-time location monitoring. Empirical findings indicate that RFID implementation drastically reduces the time and labor required for periodic stock audits, significantly minimizes loss, theft, or misplacement of equipment, and enables automated maintenance scheduling based on actual usage data and history.

Furthermore, the study emphasizes how these intelligent systems simplify compliance with strict government auditing standards by providing an immutable, tamper-proof digital audit trail of every asset's movement, location, and lifecycle history, thereby drastically improving institutional accountability and extending the functional lifespan of expensive capital investments.

# CHAPTER 3

## METHODOLOGY

### 3.1 Research Design

The study will employ a developmental research design, a methodological approach specifically focused on the systematic design, rigorous development, and comprehensive evaluation of the proposed technological application.

This framework is highly suitable and appropriate for this research as it provides a structured, step-by-step guide for creating an effective, functional solution while simultaneously ensuring that the final product meets essential standards of functionality, usability, technical reliability, and relevance to the specific needs and contexts of the target users.

The research will follow a well-defined framework that encompasses three primary, interconnected phases: the planning and design phase, where extensive requirements gathering, detailed system analysis, and conceptualization of features and architecture

will be conducted based on identified institutional needs, user expectations, and existing operational challenges; the development phase, involving the actual coding, database creation, interface design, and integration of various functional components to construct a working prototype of the application; and the evaluation phase, where the completed product will undergo rigorous testing, validation, and systematic assessment against established criteria and benchmarks.

During the evaluation stage, the study will objectively measure the application's technical performance, operational reliability, security measures, and overall user acceptability through appropriate evaluation criteria, controlled testing environments, and structured feedback mechanisms.

By utilizing this design, the research aims not only to produce a fully functional and operational application but also to scientifically validate that it effectively meets all intended objectives, successfully solves the targeted problems, and adheres to established industry standards for quality, efficiency, and data integrity.

This methodological approach ensures that every step of the development process is carefully documented, analyzed, and justified, leading to a final output that is both technically sound and practically useful in real-world educational settings. Furthermore, the developmental research design allows the researchers to engage in continuous improvement and refinement of the proposed system throughout the entire duration of the study. Feedback gathered from actual users, expert evaluators, and systematic testing procedures will serve as the primary basis for modifying features, correcting errors, enhancing interface design, and boosting overall system performance.

Through this iterative process, the researchers can identify possible issues, technical limitations, or gaps in functionality early in the cycle, allowing for necessary adjustments before the final implementation and deployment of the system. This ensures that the developed application remains responsive to the actual, evolving needs of the intended users and is fully capable of delivering efficient, secure, and reliable services. In addition, the study will incorporate a mixed-method approach, utilizing both qualitative

and quantitative methods in assessing the effectiveness, efficiency, and impact of the application.

Qualitative data may include direct observations, detailed user comments, subjective experiences, and constructive suggestions regarding the usability, navigation, and functionality of the system, while quantitative data will involve statistical analysis of survey results, numerical evaluation ratings, performance metrics, and objective measurements of system speed and accuracy.

The strategic combination of these methods will provide a more comprehensive, holistic assessment of the developed application and significantly strengthen the reliability, validity, and depth of the research findings. The chosen research design is also appropriate because it actively supports innovation and problem-solving through systematic development procedures. It enables the researchers to effectively apply theoretical knowledge, technical skills, and best practices in software engineering to produce a practical solution that directly addresses the identified problems and operational gaps.

By strictly following a developmental approach, the study ensures that the application undergoes careful strategic planning, organized implementation, and thorough multi-layered evaluation, resulting in a system that is efficient, scalable, user-friendly, and perfectly aligned with the core objectives of the research. Moreover, the developmental research design emphasizes continuous monitoring, verification, and detailed documentation throughout the entire process of system creation.

Every stage of development—from initial planning and requirements analysis to coding, testing, and final validation—will be carefully recorded and archived to ensure full transparency, traceability, and consistency in the conduct of the study.

This comprehensive documentation will also serve as a valuable reference material for future improvements, system maintenance, feature expansion, or further academic inquiry related to the project. Through this rigorous process, the study contributes not only to the successful development of a functional system but also to the broader advancement of knowledge, methodologies, and best practices related to educational

application development and system evaluation within the field of information technology.

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### **3.2 System Development Methodology**

The Agile Development Model will be used, consisting of the following phases:

**Requirements Analysis** - The study will identify and document the necessary features, functions, and constraints of the application by gathering data from potential users, existing literature, and related systems. This step defines the scope and establishes clear objectives to guide the development. In this phase, the researchers will also determine the problems encountered by users and analyze the current processes that need improvement. Interviews, surveys, and observations may be conducted to collect relevant information that will serve as the basis for designing the proposed system. The gathered data will help ensure that the application addresses the actual needs and expectations of the target users.

**System Design** - involves translating the gathered requirements into a detailed technical blueprint, including the architecture, user interface layout, database structure, and system workflows to ensure efficiency and user-friendliness. This phase focuses on creating an organized and visually appealing design that enhances the overall user experience. The researchers will prepare wireframes, prototypes, and flowcharts to illustrate how the system will function. Proper planning during this stage helps minimize errors during development and ensures that all components of the system work together effectively.

Development - where the actual coding, database construction, and integration of modules take place, turning the conceptual design into a working software component. Throughout this stage, development is done in cycles, allowing parts of the system to be built and reviewed progressively. The iterative nature of Agile enables continuous improvement and flexibility whenever modifications are needed. The researchers will utilize appropriate programming languages, frameworks, and development tools to ensure the quality and reliability of the application. Regular monitoring and collaboration among the developers will also be conducted to ensure that the project stays aligned with the defined objectives and requirements.

Testing - the developed features undergo various assessments such as functional testing, usability testing, and error checking to identify bugs, performance issues, or areas for enhancement before finalization. This phase is essential to ensure that the application operates correctly and delivers accurate results. Different testing methods will be applied to evaluate the reliability, compatibility, security, and responsiveness of the system. Feedback gathered from testers and users will be analyzed carefully to identify necessary improvements. Any detected issues will be corrected immediately to enhance the overall performance and usability of the application.

Deployment (Prototype) - This allows for practical implementation and real-world evaluation, whether it meets the intended purpose. By applying the Agile Development Model, the research ensures that the resulting application is not only technically robust and secure but also highly responsive to user needs, leading to a solution that is effective, relevant, and ready for practical use. During this phase, the developed prototype will be introduced to selected users for actual use and assessment. Their feedback and experiences while using the application will provide valuable insights regarding the strengths and weaknesses of the system. Necessary revisions and refinements may still be implemented based on the evaluation results to further improve the quality of the application before its final release.

Furthermore, the Agile Development Model encourages collaboration, adaptability, and continuous improvement throughout the development process. Unlike traditional models, Agile allows developers to respond quickly to changes and user feedback, making it highly suitable for projects that require flexibility and ongoing enhancement. Through regular communication, progress evaluation, and iterative development cycles, the researchers can ensure that the system continuously evolves according to user needs and technological requirements.

The use of the Agile methodology also promotes better project management and efficient resource utilization. Each phase contributes significantly to the successful completion of the application by ensuring that tasks are organized, monitored, and evaluated systematically. By following this methodology, the study aims to produce a reliable, efficient, and user-centered application capable of addressing the identified problem and achieving the objectives of the research.

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### **3.3 System Architecture**

The system will follow a structured client-server architecture, a widely adopted and reliable model that effectively divides the overall system into two distinct yet interconnected components. In this configuration, end-users—including school administrators, teachers, and parents—will interact with the system through accessible web-based interfaces or dedicated mobile applications, while a centralized server infrastructure will serve as the processing core responsible for handling complex data operations, executing machine learning algorithms for predictive analysis and decision support, and managing the secure storage, organization, and retrieval of all institutional data.

This architectural model establishes a clear separation of responsibilities between the client side and the server side, ensuring organized functionality, optimized operational performance, and significantly easier maintenance, updates, and future scalability as user demand or institutional requirements grow over time.



The client side serves as the front-end layer, which is designed to be intuitive, visually clear, and user-friendly; it is primarily responsible for presenting information in a readable format, capturing user inputs and commands through forms or interactive elements, and displaying processed results or feedback retrieved from the server. This interface layer will be developed using modern, responsive technologies and design principles to ensure full compatibility across different devices, screen sizes, and operating systems, thereby providing a consistent, accessible, and seamless user experience whether the system is accessed via a standard web browser on a desktop computer or through a native mobile application on smartphones and tablets.

On the other hand, the server acts as the central backend component and the computational engine of the entire system. It hosts the core application logic, enforces business rules and security protocols, executes complex mathematical operations, and manages all internal communications and external transactions. When a user submits a request or inputs specific data through the interface, the information is transmitted securely through encrypted communication channels to the server, which then processes the input, runs the required algorithms or trained machine learning models to generate accurate predictions, analytical insights, or automated assessments, and subsequently retrieves or stores relevant records within the integrated database.

The database system, which is also hosted and managed entirely at the server level, serves as the secure, structured repository for all system data, including user profiles, personal information, attendance logs, academic records, system parameters, and historical datasets, all organized and indexed to ensure extremely fast retrieval, long-term data integrity, and strict privacy compliance.

By centralizing the processing power, business logic, and data storage resources on the server side, this architecture ensures that system updates, functional improvements, security patches, and regulatory compliance measures can be implemented once and deployed centrally without requiring modifications or installations on every individual user device, while also enabling multiple authorized users to access the system simultaneously, reliably, and without conflict. This structural design ultimately supports the development of a robust, secure, and high-performing system that effectively meets all the functional requirements, performance standards, and strategic goals outlined in the study.

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### 3.4 Tools and Technologies Used

The development and deployment of the system will utilize the following technologies, carefully selected and evaluated based on their accessibility, performance, ease of integration, security features, and overall suitability for the project's defined scope, functional requirements, and target operating environment, ensuring that every component works harmoniously to deliver a stable, efficient, and high-quality web-based solution.

**Frontend:** The user interface and client-side structure will be built using HTML (HyperText Markup Language), which serves as the fundamental structural foundation for all web pages, defining the layout and organization of content displayed to users. This is combined with Tailwind CSS, a highly efficient utility-first styling framework that empowers developers to create modern, responsive, consistent, and visually appealing designs that automatically adapt and render correctly across different screen sizes, devices, and resolutions—ranging from desktop computers and laptops to tablets and mobile phones—ensuring a seamless and professional look and feel throughout the entire application.

To further enhance interactivity, dynamic behavior, and overall user experience, JavaScript will be extensively integrated into the frontend architecture, specifically utilizing AJAX (Asynchronous JavaScript and XML) technology. This capability is absolutely crucial for implementing the system's real-time messaging and notification features, as it allows the platform to send, receive, and process data in the background without requiring the browser to reload the entire page or disrupt the user's current activity, resulting in significantly smoother navigation, faster response times, and fluid interactions.

Furthermore, these frontend technologies were selected because of their exceptional flexibility, broad compatibility with all modern web browsers, and widespread industry support. Tailwind CSS greatly simplifies the process of designing and maintaining user-friendly interfaces while ensuring visual consistency across all modules and user roles, while the strategic integration of JavaScript and AJAX ensures that the system remains highly responsive, interactive, and capable of delivering the dynamic content and instant updates necessary to provide users with a truly seamless, engaging, and efficient browsing and communication experience.

**Backend:** All server-side logic, data processing, and core operational functionalities will be powered by PHP (Hypertext Preprocessor), a mature, powerful, and widely-used open-source scripting language specifically well-suited and optimized for web development projects. To prioritize the highest levels of data security, integrity, and protection against malicious threats, the system will employ PDO (PHP Data Objects) for every single database interaction and query executed within the application.

This database access extension provides a uniform and consistent interface for interacting with different types of databases and strictly utilizes prepared statements, a security feature that effectively neutralizes and protects the system against SQL injection attacks—currently recognized as one of the most common, dangerous, and damaging security vulnerabilities found in web applications—ensuring that all sensitive information, including student records and login credentials, remains completely safe from unauthorized access, interception, or manipulation. In addition to security benefits, PHP was chosen because of its proven reliability, extensive official documentation, large community support base, and strong native compatibility with MySQL databases and standard web hosting environments.

Its flexibility and robustness allow the development team to efficiently build complex dynamic functionalities, manage user sessions, process form submissions, and handle

all server-side operations essential to the system's administrative and academic workflows, making it an ideal choice for meeting the technical demands of a school management system.

**Database:** MySQL will serve as the primary, centralized, and fully relational data storage solution for the entire platform, selected specifically for its exceptional reliability, proven scalability, high performance, and seamless compatibility with the chosen PHP development environment. It will systematically store, organize, and manage all essential system data and information assets, including detailed student personal profiles, enrollment records, attendance logs, inventory lists and transaction history, user account credentials, communication archives, and all academic and administrative data generated or processed within the institution.

The database will be designed with a structured, normalized, and well-planned schema that ensures data can be efficiently retrieved, updated, queried, and managed at any time, directly supporting the core functionalities and operational logic of the school management system. Moreover, the overall database structure will be carefully organized, optimized, and indexed to maintain absolute data accuracy, consistency, and integrity across all modules, preventing data duplication or loss. Proper database normalization techniques and relational design principles will be rigorously applied during development to reduce redundancy, improve storage efficiency, and ensure that data relationships remain logical and functional, ultimately creating a solid foundation that supports fast processing and accurate reporting.

**Real-time Messaging:** To facilitate the critical function of instant communication and ensure timely information exchange within the system, AJAX polling will be implemented as the core mechanism, configured with a refresh interval set between 3 to 5 seconds to balance performance and responsiveness. This method continuously and automatically checks the server in the background for new updates, incoming messages, or announcements at regular intervals, effectively delivering near-instant message delivery and notifications to users across different modules.

This technical approach was specifically selected and prioritized because it successfully achieves real-time functionality and dynamic updates without requiring the installation or maintenance of additional complex server software, specialized extensions, third-party services, or paid infrastructure, making it perfectly aligned with the school's existing infrastructure capabilities, budgetary considerations, and the technical limitations of the chosen hosting environment.

Additionally, this feature significantly enhances communication efficiency and transparency between all stakeholders, allowing administrators, teachers, parents, and students to exchange information, receive alerts, and respond to concerns more effectively and promptly. The successful implementation of real-time messaging directly contributes to a more responsive, connected, and interactive system environment where information flows freely and barriers to communication are effectively removed.

**Development Tools:** The entire coding, programming, and development process will be carried out using Visual Studio Code, a versatile, lightweight, and powerful source-code editor widely recognized for its efficiency and extensive feature set. This tool offers essential capabilities such as intelligent syntax highlighting, built-in debugging tools, code auto-completion, and a vast library of extensions that significantly streamline the process of writing, organizing, and maintaining clean and structured code across different programming languages used in the project.

To maintain high code quality, track every modification, and ensure proper project management throughout the entire development lifecycle, Git will be utilized for distributed version control. This system allows the development team to meticulously manage revisions, collaborate effectively on shared files, track the history of every change made, and safely revert to previous stable versions of the code if errors or issues arise, thereby ensuring a highly structured, organized, and error-free development workflow.

These tools were chosen because they greatly improve developer productivity, help identify coding issues or bugs quickly, and ensure that all project files remain well-organized, secure, and properly documented. Through the systematic use of version control, developers can monitor updates efficiently, manage different development branches, and ensure proper coordination and accountability within the team during every phase of the development process.

**Deployment Platform (Demo Hosting):** For online access, public demonstration, remote testing, and formal evaluation purposes, the system will be deployed on ProFreeHost, a free web hosting service that fully supports the complete PHP and MySQL technology stack required by the application.

This platform provides a user-friendly and familiar cPanel interface for easy management, supports PHP versions up to 8.2, includes comprehensive MySQL database management tools, and offers a claimed server uptime of 99.9%, ensuring that the system remains reliably accessible online for presentations, validation, and testing activities. It also automatically provides a free dedicated subdomain address — <http://tflc-sms.liveblog365.com/login.php?i=3> — and includes an integrated auto-installer feature that simplifies the setup and configuration process, making it highly convenient and suitable for capstone project presentation, academic defense, and evaluation purposes.

It is important to note that, as a free shared hosting service, this platform naturally comes with certain limitations and constraints, such as a maximum storage capacity of 5 GB and performance levels that may fluctuate slightly depending on overall server load and traffic conditions.

However, these constraints are deemed entirely acceptable, manageable, and fully sufficient for the purpose of developing and presenting a functional prototype and academic demonstration, as they do not hinder, restrict, or compromise the system's ability to showcase its intended features, core functionalities, and overall capabilities. Furthermore, deploying the system online allows evaluators, panelists, and users to access the application remotely from different locations, making the testing, validation, and feedback collection process much more convenient and accessible. This deployment setup also effectively demonstrates the portability, web-accessibility, and adaptability of the system when operated in a real-world internet environment.

**Internal School Access:** In addition to the cloud-based online demo version, the system will also be made fully accessible within the school's premises and facilities through the Local Area Network (LAN) using XAMPP, a popular and reliable open-source software package that provides a complete local server environment including Apache, PHP, and MySQL.

This setup allows teachers, administrators, staff, and users physically located on campus to access, navigate, and test the system with optimal speed, maximum performance, and zero dependency on external internet connectivity or bandwidth limitations. This dual deployment strategy—combining both online hosting and local network access—ensures that the system can be thoroughly evaluated, demonstrated,

and utilized in both online and offline/local settings, clearly demonstrating its flexibility, reliability, and readiness for potential full-scale implementation and integration within the school's daily operations.

Furthermore, the selected tools, frameworks, and technologies were carefully and rigorously evaluated based on critical criteria including accessibility, cost-effectiveness, performance benchmarks, ease of integration, and long-term suitability for academic system development and educational use.

The strategic combination of HTML, Tailwind CSS, JavaScript, PHP, and MySQL creates a stable, robust, and efficient technological environment capable of supporting the creation of a responsive, secure, and user-friendly web application. These technologies are widely recognized, accepted, and utilized in modern web development practices and are backed by extensive documentation, tutorials, and active community resources, making them highly practical, sustainable, and maintainable choices for developing and supporting the proposed system over time.

Security and data integrity were also major priorities and central considerations throughout the selection and development process. Through the rigorous use of PDO and prepared statements, the application effectively minimizes the risks associated with unauthorized database access, data tampering, and malicious attacks such as SQL injection and cross-site scripting.

Additionally, proper user authentication protocols, role-based access controls, input validation mechanisms, and data encryption standards will be incorporated into the code to ensure that only authorized users can log in, access sensitive information, or perform administrative actions, thereby strengthening the overall reliability, compliance, and trustworthiness of the proposed application.

Moreover, the dual deployment setup using both online hosting and local network access effectively demonstrates the system's flexibility and adaptability to different infrastructure scenarios and connectivity conditions. While ProFreeHost enables remote access, public demonstration, and off-campus evaluation, the LAN-based deployment through XAMPP ensures stable, fast, and secure local accessibility directly within the school environment. This comprehensive setup allows the researchers to thoroughly evaluate the system under different operating conditions, network speeds, and usage

scenarios, ensuring that the application can function effectively and efficiently even in areas with limited or unstable internet connectivity.

Overall, the selected technologies, frameworks, and development tools collectively form a powerful and cohesive technical foundation that contributes directly to the successful creation of a secure, efficient, scalable, and user-centered school management system. Their seamless compatibility, proven reliability, and broad functionality ensure that the proposed application can effectively support and streamline the operational, administrative, and academic needs of Tigbauan Fundamental Learning Center, Inc., while simultaneously providing a smooth, accessible, and productive experience for all its intended users including administrators, teachers, parents, and students.

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### **3.5 Data Gathering Techniques**

To ensure that the resulting system fully aligns with the unique operational requirements, specific challenges, and long-term strategic goals of the institution, a comprehensive, multi-layered, and rigorous research methodology was meticulously designed and employed throughout the preliminary phase of the study.

This approach drew heavily from established qualitative and analytical research techniques, effectively combining the collection and examination of historical data patterns with direct feedback and perspectives from key stakeholders to form a complete and accurate picture of the school's environment. By utilizing this integrated strategy, the research team was able to gain not only a deep understanding of the school's current operational ecosystem, existing procedural limitations, and underlying administrative workflows but also laid down a robust, evidence-based foundation for designing a highly customized, user-centric digital solution.

This ensures that the proposed system will seamlessly fit into the institution's daily routine, minimize operational disruptions during implementation, significantly reduce potential resistance to change among staff members, and ultimately maximize adoption rates and user satisfaction among all members of the school community, from administrators and faculty to parents and students.

**Document Analysis:** A thorough, systematic, and detailed review of existing physical and digital school records was conducted over several weeks to ensure a complete and



accurate understanding of historical practices, established standards, and recurring operational procedures currently in place at Tigbauan Fundamental Learning Center, Inc.

The scope of this review covered a wide variety of official documents, including traditional handwritten attendance sheets and daily logbooks, detailed manual inventory logs tracking supplies, equipment usage, asset status, and borrowing history, enrollment and admission forms containing comprehensive demographic, contact, and academic background details, quarterly and final grade reports, student progress cards, disciplinary records, administrative circulars, and general accounting and operational ledgers.

This process involved carefully cataloging, classifying, and organizing these documents, as well as digitizing samples where feasible and appropriate to preserve data integrity, followed by rigorous content analysis and data mapping. Through this method, the team was able to systematically identify and map out all essential data fields and variables—such as unique student identifiers, precise timestamps, item quantities and specifications, status updates, and required approval signatures—while also detecting common redundancies, information gaps, data inconsistencies, and bottlenecks present in the current manual workflows.

Furthermore, this analysis allowed the researchers to outline the precise logic, correct sequence of operations, and functional dependencies that are required for a successful and accurate digital transition. By thoroughly examining these documents and records, the research team was able to establish the exact standard data formats, validation rules, data retention policies, and official reporting structures that the new system must strictly support and adhere to.

This ensures full compatibility with the institution's established internal policies, strict adherence to regulatory compliance requirements, and alignment with government record-keeping standards set by educational authorities. Simultaneously, this process highlighted clear opportunities to automate repetitive administrative tasks, streamline multi-level approval chains, standardize documentation procedures, and significantly reduce the likelihood of human error, data duplication, loss, or misplacement that commonly occurs in manual systems.

**Semi-Structured Interviews:** Complementing the document analysis, in-depth qualitative interviews were conducted using a semi-structured format, guided by a carefully prepared set of open-ended questions but designed to remain flexible enough to allow for spontaneous probing, deeper exploration of specific topics, and natural flow of discussion. These sessions were held individually and in small focus groups with a diverse cross-section of participants directly involved in the school's operations, including school administrators, registrars, department heads, senior faculty members, subject teachers, and select non-teaching support staff responsible for inventory, records, and general maintenance.

Each interview session lasted approximately between 45 to 60 minutes, and in total, the research team engaged with over 20 participants representing various departments and roles to ensure a balanced and comprehensive collection of insights.

The discussions focused extensively on identifying and documenting key pain points, operational difficulties, and inefficiencies inherent in the current manual and paper-based systems—such as excessively time-consuming data entry and encoding, frequent calculation errors, difficulties in locating, retrieving, or securing physical documents, significant delays in report generation and distribution, and persistent communication gaps between the administration, teachers, and parents. Beyond identifying problems, these sessions were also instrumental in gathering comprehensive functional and non-functional requirements for the proposed digital platform, directly from the perspective of the end-users.

Participants were encouraged to share their ideas, suggestions, and expectations regarding desired features and capabilities, which included real-time data dashboards for monitoring, cross-device and mobile accessibility to ensure access anywhere, clearly defined role-based access permissions to protect sensitive data, automated notifications and alerts for important events or deadlines, and potential integration capabilities with other tools they currently use.

All requirements were informed by real-world usage scenarios and daily operational needs, allowing the team to prioritize usability, security, reliability, and scalability in the design. Furthermore, these interactions provided valuable insights into the specific workflows, decision-making hierarchies, and data security concerns that are unique to different user groups and departments.

This deep understanding allowed the development team to clearly define distinct access levels, customized user roles, and specialized system modules tailored specifically to the varying needs and responsibilities of administrators, teachers, and inventory personnel, ensuring that each user only sees and interacts with the features relevant to their job function.

The detailed feedback and comments collected during these interviews also played a critical role in prioritizing system features based on urgency, frequency of use, and overall impact on productivity, ensuring that the most critical functions were developed, tested, and refined first.

Additionally, these discussions revealed important user expectations regarding ease of use, interface design preferences, and the necessary training and support requirements to be provided during the roll out phase—all of which are vital factors in ensuring a smooth, supported, and successful transition from manual processes to fully digital operations.

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### 3.6 Testing Procedures

To guarantee the highest quality, reliability, security, and performance of the Web-Based Integrated School Management System, a rigorous, multi-layered, and systematic testing methodology was implemented throughout the entire development lifecycle. This comprehensive validation process followed a structured progression from individual code verification to full system evaluation and finally to real-world user validation, ensuring that every aspect of the application met the predefined technical specifications, functional requirements, and operational standards established for Tigbauan Fundamental Learning Center, Inc

By adopting this thorough approach, the research and development team aimed to detect, diagnose, and resolve potential defects, inconsistencies, or performance bottlenecks at the earliest possible stage, thereby minimizing risks, reducing future maintenance efforts, and delivering a stable, efficient, and dependable digital solution tailored to the institution's needs.

**Unit Testing:** This phase represents the most granular, exhaustive, and foundational level of software validation, focusing on the individual components, modules, functions, and code segments that form the building blocks of the system.

During this stage, specific features such as the attendance scanning algorithm, inventory stock calculation logic, real-time messaging protocols, database query handlers, data validation routines, and mathematical computation functions were rigorously tested in complete isolation from the rest of the application. Utilizing specialized automated testing tools and frameworks such as PHPUnit (compatible with the PHP environment) and Selenium for functional simulation, every piece of code was

subjected to a wide variety of test cases covering standard operational inputs, boundary values, extreme conditions, and invalid or unexpected data formats.

The primary objective was to verify the correctness, robustness, and error-free performance of each unit, ensuring that it consistently produces the correct expected output, processes data accurately, and handles exceptions or failures gracefully without crashing, freezing, or compromising data integrity.

During this meticulous process, every line of code and specific functionality was scrutinized, reviewed, and validated to confirm that it operates exactly as intended before being integrated into the larger system architecture.

This step is absolutely critical in the software development lifecycle as it allows developers to identify, isolate, and resolve technical issues, logical errors, or coding bugs at the earliest possible stage, significantly reducing the cost, complexity, and difficulty of fixing problems later when multiple components are combined into a larger system.

Furthermore, unit testing plays a vital role in maintaining the quality, consistency, and stability of the system throughout the development process. By continuously re-testing individual modules whenever changes, updates, or enhancements are introduced, developers can instantly detect unintended side effects or coding errors, ensuring that existing functionalities remain unaffected and reliable. This practice greatly improves software maintainability, simplifies future modifications, and ensures that each component consistently performs according to the specified requirements, design standards, and business rules established for the project.

**System Testing:** Following the successful validation of individual units and the completion of system integration, a comprehensive end-to-end evaluation of the fully integrated system as a cohesive whole was conducted. This phase was carried out in a carefully configured staging environment that closely mirrors the actual production setup, including identical server configurations, database structures, network settings, and hardware specifications, to ensure that test results accurately reflect real-world performance and behavior.

The primary goal was to assess overall functionality, operational performance, security vulnerabilities, data integrity across interconnected modules, and compliance with all defined requirements. Testers validated the seamless flow of information between different parts of the system—such as verifying that data entered during student registration automatically populates attendance lists and academic records, that inventory updates are instantly reflected in administrative reports, and that grades inputted by teachers are correctly archived and accessible to parents.

The scope of testing also included cross-browser and cross-device compatibility checks to ensure consistent appearance and functionality across various web browsers, screen sizes, operating systems, and devices ranging from desktop computers to tablets and mobile phones. Additionally, load testing and stress testing were performed to simulate peak usage conditions—such as simultaneous attendance marking during morning roll calls or mass report generation at the end of a grading period—to evaluate how the system handles high traffic, concurrent users, and heavy data processing loads.

Non-functional requirements were strictly measured, including verifying that page response times remain under 2 seconds and that system availability meets the target of 99.9% uptime. This phase employed black-box testing techniques, regression test suites, and specialized security assessments to identify potential weaknesses such as unauthorized access points, data leakage risks, or susceptibility to injection attacks. Extensive security testing was conducted to ensure that sensitive information like student personal details, medical records, and login credentials are properly encrypted, securely stored, and fully protected against malicious threats or breaches.

Furthermore, system testing evaluated the application's stability, reliability, and recovery capabilities during unexpected scenarios such as network interruptions or power fluctuations. Any issues identified during this phase—including system crashes, processing delays, calculation errors, or inconsistencies in data synchronization—were carefully documented, analyzed, corrected, and retested to ensure that the final integrated system operates smoothly, efficiently, and securely within the school's operational environment, confirming its readiness for official deployment.

**User Acceptance Testing (UAT):** Serving as the final and most critical validation phase, User Acceptance Testing involved a collaborative, iterative, and practical evaluation process conducted with the direct participation of actual end-users representing all key stakeholder groups within the school community. Participants

included school administrators, registrars, department heads, teaching faculty, non-teaching staff, selected parents, and duly enrolled students, all of whom interacted directly with the fully deployed prototype in a controlled, live-like environment that replicated their daily work conditions and routines.

During this phase, users were tasked with performing realistic operational activities and scenarios relevant to their roles—such as processing new student admissions, marking daily attendance via QR code scanning, updating inventory levels and monitoring stock status, posting grades, sending communication messages, and generating official reports—just as they would do in regular operations.

Throughout the testing period, structured feedback was systematically gathered through detailed usability surveys, observation checklists, session recordings, and debrief meetings, where participants shared their honest opinions regarding the system's usability, intuitiveness, ease of navigation, visual design, speed, accuracy, and overall alignment with established school workflows and policies.

The primary objective of this stage was to go beyond mere technical correctness and validate the system's practical utility, user-friendliness, and ability to effectively address the real-world challenges and pain points identified during the data gathering phase. Participants were specifically encouraged to simulate their regular daily routines to identify any confusing navigation paths, missing features, redundant steps, operational inefficiencies, or unclear instructions that were not apparent during earlier technical testing phases.

All feedback, comments, suggestions, and concerns gathered during UAT were systematically compiled, categorized, analyzed, and prioritized based on impact and urgency. This process directly led to targeted refinements, adjustments, and improvements in interface design, workflow logic, feature placement, terminology, and system functionality.

Multiple rounds of testing and revision were conducted until all predefined acceptance criteria were fully met, user requirements were satisfied, and participants expressed confidence and satisfaction with the application. Moreover, User Acceptance Testing

played a significant role in measuring the overall satisfaction, confidence, and readiness of users toward adopting the proposed system.

Through direct hands-on interaction, participants were able to assess firsthand whether the system effectively simplified their daily tasks, reduced workload, and resolved the difficulties experienced with the previous manual and paper-based processes. Suggestions and recommendations provided by users were treated as highly valuable inputs for final polishing, ensuring that the final product is not only technically robust and secure but also user-centric, accessible, and perfectly aligned with the institution's culture and operational context.

Only after users formally confirmed that the system effectively supports their tasks, is easy to learn and master, and fits seamlessly within the school's operational framework was the system considered fully validated and ready for final presentation, approval, and full-scale implementation.

Collectively, these layered, progressive, and comprehensive testing procedures—spanning Unit Testing, System Testing, and User Acceptance Testing—provided a complete evaluation framework that ensured the reliability, quality, and success of the developed application. Each distinct testing phase contributed uniquely to identifying and resolving different categories of issues, ranging from low-level technical coding errors and logical mistakes to system-wide workflow inefficiencies, security risks, and high-level user experience concerns.

This multi-layered approach effectively strengthened the overall stability, security, performance, and usability of the system long before its actual deployment. Overall, the rigorous and systematic application of these testing procedures ensured that the Integrated School Management System strictly met or exceeded all required standards for functionality, performance, security, data integrity, and user satisfaction.

By subjecting the application to extensive validation under diverse conditions and perspectives, the researchers successfully verified that the system operates efficiently, satisfies all defined user requirements, adheres to regulatory standards, and is fully



capable of supporting the institution’s administrative, academic, and operational needs in a reliable, sustainable, and effective manner.

3.7 Evaluation Criteria

| Metric                  | Target Benchmark                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response Time           | High-priority data, such as real-time messages, must be delivered and acknowledged within 2 seconds of transmission. This benchmark applies to all critical system interactions, including saving attendance records, updating inventory quantities, loading student profiles, and generating reports. Achieving this ensures that users experience near-instant feedback and smooth navigation, eliminating delays that could disrupt workflow or reduce productivity. A fast response time is essential to replicate the speed and efficiency expected of modern digital systems, ensuring that operations are completed quickly even when handling large volumes of data or complex queries.                   |
| Concurrent User Support | The system must maintain stability and performance integrity while supporting at least 50 simultaneous users performing various tasks. This criterion simulates real-world operational conditions, such as during the start of classes, enrollment periods, or inventory audits, where multiple users—including administrators, teachers, and staff—access the system at the same time to input, retrieve, or modify data. The target ensures that the architecture and database design are robust enough to handle concurrent connections without slowing down, crashing, or experiencing data conflicts, thereby validating the system’s capacity and scalability to serve the school’s population effectively. |
| System Uptime           | Ensuring the platform remains accessible during peak school hours with minimal downtime. The goal is to achieve a reliability level of approximately 99.9% uptime, meaning the system is available and functional almost continuously throughout the academic day, typically from early morning to late afternoon. This metric measures                                                                                                                                                                                                                                                                                                                                                                           |

| Metric | Target Benchmark                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | the consistency of service, ensuring that users can rely on the system whenever needed without unexpected interruptions, maintenance delays, or server errors. High uptime is critical to ensure that administrative and academic processes are not halted and that data entry and record-keeping can proceed without disruption, reinforcing trust in the digital solution as a dependable replacement for manual methods. |

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## SUPPORTING DIAGRAMS (Included within Chapter 3)

### Data Flow Diagram (DFD – Level 0)

The Level 0 Data Flow Diagram serves as the highest-level abstraction that illustrates the overall system scope, key entities, and the primary flow of information across the core components of the dating application. It simplifies the complex internal operations into a clear, easy-to-understand overview, showing how data moves between the external user, the application system itself, and the central database repository. The diagram is structured around three main elements: the User as the primary external entity, the Dating Application representing the entire set of system processes and functionalities, and the Database acting as the persistent storage layer.<sup>3</sup>

The data flow follows a logical sequence of input, processing, storage, and output. First, the User submits profile and preferences, including personal details, interests, relationship goals, and specific criteria for potential partners. This input data is transmitted from the user interface to the application's processing layer. Next, the System processes data and generates matches by applying the defined algorithms, matching logic, and filtering rules designed during the development phase; this stage involves analyzing the user's attributes and preferences against the available records to identify compatible connections.

Finally, Results are stored and retrieved from the database, where all user profiles, preference settings, matching history, and system-generated results are securely saved, organized, and made accessible for future processing, updates, or display back to the user. This diagram effectively captures the fundamental interaction cycle, providing a visual foundation for understanding how information is captured, transformed, retained, and utilized within the application ecosystem, while also serving as a reference point for more detailed lower-level diagrams that break down these processes further.

Furthermore, the Data Flow Diagram helps visualize how information moves throughout the system and identifies the relationship between external entities, processes, and data storage components. It serves as an important planning and communication tool that allows developers, researchers, and stakeholders to better understand the overall functionality and structure of the proposed application. Through the use of DFDs, potential issues related to data handling, workflow efficiency, and system integration can be identified and addressed during the early stages of development.

Additionally, the Level 0 DFD provides a simplified yet comprehensive overview of the entire system, making it easier to explain the project's operational framework to both technical and non-technical audiences. It establishes the foundation for more detailed diagrams, such as Level 1 and Level 2 DFDs, which further break down the individual processes and data interactions within the system. By clearly illustrating the flow of information, the diagram contributes to improved system analysis, accurate requirement identification, and more efficient application design and implementation.

## **Entity Relationship Diagram (ERD)**

The Entity Relationship Diagram (ERD) serves as the fundamental conceptual blueprint and structural foundation of the entire database architecture, graphically illustrating the comprehensive structure, essential data attributes, precise data types, constraints, and logical associations that govern exactly how information is captured, stored, organized, related, updated, and interconnected within the proposed school management system. Acting as a visual roadmap, this diagram defines the distinct core entities or objects that represent the key components of the institution's operations—along with their specific properties, characteristics, and unique identifiers—while clearly mapping out the complex relationships, cardinality, and dependencies that exist between them.

By meticulously modeling these connections—whether one-to-one, one-to-many, or many-to-many—the ERD ensures absolute data integrity, consistency, and accuracy throughout the system, effectively minimizing unnecessary data redundancy, eliminating duplication of records, and establishing strict rules regarding referential integrity that prevent orphaned records or conflicting information.

Furthermore, this visual representation provides a clear, standardized, and unambiguous framework that guides the entire process of database design, development, implementation, and future maintenance, serving as an essential communication tool between developers, database administrators, and stakeholders to ensure that the logical structure perfectly aligns with the system's functional requirements and the institution's operational workflows

It also forms the basis for constructing the physical database schema, defining tables, fields, primary keys, foreign keys, and indexes, thereby optimizing data retrieval speed, storage efficiency, and overall system performance. Below is the detailed breakdown of each identified entity, its associated attributes, and their corresponding relationships within the database structure:

## **Entities:**

**User (UserID, Name, Age, Preferences):** This entity serves as the primary, foundational, and most essential table within the database schema, representing every individual authorized or registered to interact with the system, including students, teachers, administrative staff, parents, and other authorized personnel involved in school operations. Each record within this table is uniquely identified by UserID, designated as the Primary Key, which acts as the universal reference code linking the individual to all their activities, permissions, and related data across the entire platform. It stores essential identification and demographic details such as the individual's full legal Name and Age, providing basic yet vital information necessary for user identification, verification, and record-keeping purposes.

Beyond basic data, the Preferences attribute captures personalized system settings, operational requirements, display configurations, filtering criteria, notification preferences, language settings, or specific communication parameters that directly guide how the system interacts with, presents information to, and serves the specific user.

Acting as the central reference point within the database structure, this entity links personal identity to all associated records, activity logs, transaction history, academic data, inventory responsibilities, and system permissions established throughout the platform.

Furthermore, the User entity plays a significant role in maintaining consistent, secure, and highly personalized experiences within the system environment.

It enables efficient user authentication, rapid identification, and comprehensive account management while supporting customized functionalities, role-based dashboards, and

specialized workflows tailored strictly based on user-specific preferences and permission levels defined within the system logic.

The information stored within this entity also assists the system in generating accurate, user-specific reports, monitoring login activities and usage patterns, tracking access history, and continuously improving overall system interaction, accessibility, and user satisfaction by ensuring the interface, available features, and navigational pathways align closely with the distinct needs, roles, and responsibilities of each individual user within the educational community.

**Profile (ProfileID, UserID, Interests, BehaviorScore):** Functioning as a detailed extension and supplementary data repository connected strictly and exclusively to the User entity, this table holds deeper descriptive, analytical, behavioral, and contextual data that expands upon basic identification details and represents information that may evolve dynamically, change frequently, or require more detailed categorization over time.

ProfileID serves as the unique identifier for each profile record, while UserID acts as a Foreign Key establishing a direct, exclusive one-to-one relationship back to the parent User entity, ensuring that every profile is strictly associated with only one valid registered user and maintaining full referential integrity within the database.

The Interests attribute captures role-specific attributes, academic focus areas, learning objectives, professional qualifications, operational responsibilities, resource requirements, subject specializations, or specialized skills and competencies relevant to the educational or institutional environment, providing rich contextual data used for filtering, grouping, or personalizing system interactions.

Meanwhile, BehaviorScore represents a computed, calculated, or periodically assigned metric or composite value that quantifies various dimensions of user engagement, attendance consistency, reliability, compliance with school regulations, participation levels, disciplinary record, or overall academic or operational performance; this value serves as a critical data point utilized by the system's internal logic for eligibility assessments, resource allocation, performance tracking, or intelligent matching algorithms.

In addition to storing rich contextual data, the logical separation of the Profile entity from the main User table allows the system to maintain highly detailed, dynamic, and scalable user information without cluttering the core identity table or affecting its performance. By isolating profile-specific attributes—such as evolving preferences, performance metrics, or specialized details—from the static main user records, the overall database structure becomes significantly more organized, flexible, and efficient, effectively reducing data redundancy and simplifying maintenance tasks.

This modular design also greatly improves scalability, allowing future enhancements, additional attributes, complex profile-related features, or new data categories to be integrated seamlessly or updated frequently without significantly affecting or disrupting the stability, speed, or integrity of the core user data structure.

**Match (MatchID, UserID, MatchUserID, CompatibilityScore):** This entity systematically records, archives, and manages all instances of system-generated pairings, suggested connections, resource allocations, group assignments, or associative relationships created or recommended within the platform's operational logic. Each record is uniquely identified by MatchID, serving as the primary key and unique log entry number for every matching activity recorded.

Within this structure, UserID refers to the individual initiating the search, involved in the primary process, or the subject of the recommendation or assignment, while MatchUserID functions as a relational foreign key referencing the related user, resource, facility, subject, or entity being paired, suggested, or allocated to the primary user. The CompatibilityScore attribute stores a precise calculated numerical value representing the degree of suitability, relevance, alignment, or fit between the two entities based on predefined algorithms, weighted criteria, available profile data, academic requirements, institutional rules, or operational constraints; this numerical score provides a measurable, objective, and quantifiable basis for the suggested connection or allocation, allowing for sorting, ranking, and prioritization of results.

Moreover, the Match entity is essential in supporting the intelligent, automated, and decision-support functionality of the system by enabling data-driven recommendations, optimal resource distribution, peer-to-peer associations, or class assignments based on specific conditions and requirements defined by the school administration.

The compatibility scoring mechanism helps streamline and improve decision-making processes by offering transparent, quantified evaluations that increase the accuracy, fairness, and relevance of generated matches or assignments, reducing manual effort and potential bias. Beyond real-time operations and suggestions, this entity also serves a critical analytical and administrative purpose by storing a comprehensive historical log of all matching activities, allowing administrators to review past allocations, verify decision logic, analyze trends in assignments, evaluate the effectiveness of algorithms over time, and refine parameters as needed, while providing a clear audit trail essential for monitoring, compliance, quality assurance, and future reference purposes.

### **Admin (AdminID, Name, Role)**

This distinct and specialized entity represents the highest level of authorized personnel within the system hierarchy, specifically designated and responsible for overall system governance, strategic configuration, technical maintenance, security management, and comprehensive operational oversight of the entire platform.

Each administrator record is uniquely identified by AdminID, serving as the primary key and unique credential that distinguishes each authorized individual within the database. This table securely stores essential identification details including the administrator's full legal Name for accountability and reference purposes, and explicitly defines their Role—a critical classification attribute that precisely specifies their exact authority level, scope of responsibility, functional domain, and specific access privileges assigned to them within the system.

These role definitions are carefully structured to align with institutional hierarchy and job function, encompassing rights and permissions for tasks such as fundamental system configuration settings, complete user management and account provisioning, inventory master data control, database administration, data backup and recovery operations, security policy enforcement, and generation of high-level executive or regulatory reports.

This structural design ensures strict adherence to the principles of Role-Based Access Control (RBAC), a security framework that grants permission only to the specific functions, modules, and data resources absolutely necessary for an individual to perform their assigned duties, while strictly restricting access to sensitive administrative tools, system-critical configurations, or irreversible modifications to core data exclusively to qualified, authorized, and properly vetted personnel.



Additionally, the Admin entity supports the seamless implementation of robust security protocols, standardized operating procedures, and best practices in system management by ensuring that administrative responsibilities are clearly defined, logically segregated, and appropriately distributed according to designated roles, authority levels, and organizational hierarchy, thereby preventing conflicts of interest and enforcing the separation of duties.

This comprehensive framework helps maintain the highest possible levels of system security, data integrity, operational confidentiality, and accountability across all modules, effectively preventing unauthorized access, data breaches, or accidental modification or deletion of core settings and critical institutional information.

, this entity enables effective monitoring, detailed logging, and auditing capabilities regarding all administrative activities, changes, adjustments, and modifications made to the system environment or database records. By clearly associating every significant action with a specific AdminID and Role, the system maintains a transparent and immutable audit trail that contributes significantly to better governance, regulatory compliance, error tracing, troubleshooting, and the long-term stability, reliability, and secure maintenance of the entire application environment.

Ultimately, this entity serves as the control center foundation, ensuring that the platform remains stable, secure, compliant, and properly aligned with the institution's policies and operational standards throughout its lifecycle.

## **Relationships:**

### **User Creates Profile**

In this foundational use case and core relational structure that defines the very architecture of the database, a registered user initiates the comprehensive profile creation process by securely logging into the Integrated School Management System dashboard using their authorized credentials and selecting the designated “Create Profile” option prominently displayed within their role-specific interface.

This action guides them through a structured, step-by-step, and user-friendly interface designed to collect necessary information systematically, where they are prompted to systematically input comprehensive personal, academic, and institutional details relevant to their function within the school community.

The data entry process encompasses essential identifiers such as full legal name, official date of birth, and complete contact information including active email addresses, primary and secondary phone numbers, and full residential address, alongside integrated features for profile picture uploads that include built-in validation rules and technical constraints—such as file size restrictions limited to under 5MB and allowed formats strictly limited to standard image types like JPG or PNG—to ensure consistency and optimal system performance.

Depending entirely on their assigned role, classification, and position within the institution’s hierarchy, users are presented with customized fields where they enter specific relevant data pertinent to their responsibilities: students provide their official school identification numbers, current grade levels, section details, and enrollment status; parents or legal guardians establish secure linkage and association to their respective children’s accounts by entering unique student identifiers; while teachers and

administrative personnel input their subject specializations, department affiliations, advisory roles, and employment details as required by their job functions.

Furthermore, during this process, users are also prompted to configure specific privacy preferences—allowing them to determine the visibility and accessibility of certain profile fields to other users or stakeholders—and fill out custom or optional fields such as medical information, known allergies, long-term health conditions, or detailed emergency contact persons and numbers, ensuring that critical safety data is readily available to authorized personnel during emergencies yet remains strictly protected from unauthorized viewing or misuse.

Throughout this entire data entry and submission process, the system performs rigorous, real-time validations and integrity checks on every input field, including automated checks for email uniqueness within the database, correct formatting of phone numbers and addresses, and verification of institutional identification numbers against official records to prevent duplication, typing errors, or invalid entries.

Simultaneously, the system automatically generates and assigns a unique UserID which serves as the primary key and permanent reference code, applies advanced data encryption protocols specifically to sensitive fields to ensure full compliance with data privacy standards and regulations, and immediately sends a confirmation notification via email or SMS to the registered contact details to verify account ownership and confirm successful creation.

Once all required information has been accurately submitted, validated, and finalized, the system seamlessly integrates the new profile record into the central relational database, linking it permanently to the corresponding user account and enabling immediate access to personalized features such as role-tailored dashboards, customized notifications, relevant functional modules, and permission-based views. The system also automatically logs the exact creation timestamp, IP address, and user details within the activity logs, establishing a clear audit trail available for administrative review, security monitoring, and compliance verification purposes.

This relationship formally establishes a strict one-to-one association between the User and Profile entities, ensuring that every individual user maintains a single, distinct, and non-duplicable profile record that serves as the definitive source of their digital identity, attributes, preferences, and permissions within the system environment.

This unique linkage forms the essential basis and underlying foundation for all subsequent interactions, personalization logic, data processing workflows, and access control measures applied across the entire platform, effectively dictating what information is visible, what actions are permitted, and how the system behaves for that specific user.

Furthermore, the standardized and guided profile creation process is meticulously designed and implemented to significantly improve overall data accuracy, accessibility, and organizational structure within the system; by requiring users to provide complete, verified, and thoroughly validated information during the initial registration and onboarding phase, the application effectively minimizes the occurrence of incomplete records, missing fields, formatting errors, and inconsistencies in stored data that often hinder operational efficiency and complicate administrative tasks in manual systems.

Beyond the initial setup and data collection, this relationship supports robust security mechanisms, including secure user authentication protocols and comprehensive role-based access control frameworks, ensuring that functionalities and data views are strictly aligned with each user's defined permissions, authority levels, and assigned responsibilities, thereby protecting sensitive information and preventing unauthorized operations. Additionally, the dynamic and modular nature of the Profile entity allows the system to maintain continuously updated user information that can be modified, expanded, or corrected whenever necessary through dedicated, user-friendly editing interfaces accessible only to authorized individuals.

This flexibility ensures that users can actively manage and refine their personal, academic, or professional details over time—such as updating contact numbers,

changing addresses, or adding new qualifications—directly contributing to improved communication reliability, accurate reporting capabilities, efficient record-keeping, and data consistency throughout the entire lifecycle of the system’s operations and usage within the institution.

## **Use Case Diagram**

The Use Case Diagram serves as a vital visual representation and essential analytical tool that comprehensively illustrates the functional requirements, operational scope, and intended behavior of the Integrated School Management System by clearly defining, structuring, and mapping out all significant interactions that occur between the platform and its various users within the educational environment.

Acting as a standardized modeling technique derived from systems engineering and software design, this diagram systematically identifies the different actors involved—external entities such as students, teachers, administrative staff, parents, and system administrators—along with their specific roles, distinct responsibilities, operational boundaries, and levels of authority, while precisely mapping out the full range of functionalities, services, workflows, and business processes accessible to each distinct user type based on their institutional position.

By depicting the entire system as a cohesive collection of meaningful goals, specific actions, and desired outcomes that users aim to accomplish or facilitate, this diagram provides a structured, high-level, and easy-to-understand overview of overall system behavior and capability, demonstrating exactly how individuals—whether standard users or privileged administrators—navigate the interface, utilize available tools, perform daily operational tasks, access critical information and records, generate official reports, execute financial or inventory transactions, or exchange communication messages to achieve specific objectives required for the smooth functioning of the institution.

Furthermore, it effectively categorizes features, operations, and system capabilities strictly based on user role, authority level, and job function, visually enforcing and

reinforcing the principle that access rights, permissions, and privileges must align precisely and exclusively with assigned institutional responsibilities and duties.

This inherent structure ensures that highly sensitive functions, such as system configuration, data modification, user account management, or deletion of records, are restricted only to properly authorized and qualified personnel, that data privacy rules and confidentiality policies are inherently respected and embedded into the system logic, and that the overall flow of operations remains logical, secure, consistent, intuitive, and user-friendly for every stakeholder involved.

Ultimately, this diagram acts as a critical bridge between stakeholder expectations, business requirements, and technical design specifications, serving as a foundational reference document for developers during the build and coding phase to ensure all required features are included, for testers during validation and quality assurance to verify that each function works according to defined scenarios, and for end-users during orientation and training to understand their available actions and limitations.

By providing a clear, agreed-upon model of what the system should do and who can do what, the Use Case Diagram ensures that the final application accurately reflects the operational needs, workflows, and strict security standards of Tigbauan Fundamental Learning Center, Inc.,

while reducing ambiguity, minimizing development gaps, and aligning the technical solution perfectly with the institution's strategic goals. Below is the detailed breakdown of actors and their respective use cases:

**Actors:****User Use Cases**

These encompass the complete spectrum of core interactions, operational capabilities, and functional services available to standard registered users—including enrolled students, parents or legal guardians, and both teaching and non-teaching staff members—within the Integrated School Management System, all of which have been meticulously analyzed, designed, and developed to provide intuitive navigation, robust security, and efficient access to personalized features that directly support and enhance daily academic, learning, administrative, and communicative needs within the institution. The range of functionalities spans from essential self-service account management tools, profile maintenance, and real-time communication channels to comprehensive data visualization dashboards, academic progress tracking, and resource monitoring modules, with every operation strictly governed and regulated by a sophisticated role-based access control mechanism.

This security framework ensures that sensitive information, confidential records, or critical system configurations remain fully protected, private, and secure, while guaranteeing that data and functionalities are accessible exclusively to authorized individuals whose job functions or status explicitly require such access.

Each specific function, interface view, and available option within this category is purpose-built and specifically tailored to perfectly fit the unique requirements, operational responsibilities, and distinct goals associated with the user's designated role within the school hierarchy; for instance, a registered student primarily utilizes the system to view their current class schedules, check and review quarterly grades and academic standings, monitor their attendance history, and submit necessary requirements or documents through digital channels.

Meanwhile, a parent or guardian focuses their usage on monitoring their child's daily attendance status, reviewing comprehensive progress reports and report cards, receiving official school announcements and circulars, and engaging in direct communication with teachers regarding their child's performance or welfare. Similarly, a teacher leverages the platform extensively to record daily attendance, input and manage student assessments and grades, organize and maintain updated class rosters, disseminate learning materials or assignments, and generate necessary academic reports, whereas non-teaching staff may focus on inventory monitoring, asset tracking, or administrative documentation tasks relevant to their department.

This high level of customization ensures that the system serves as a versatile, highly relevant, and indispensable tool for every member of the school community, effectively streamlining daily workflows, reducing administrative burdens, enhancing transparency, and fostering deeper engagement and collaboration across all levels of the educational environment.



## **Register Account**

This represents a straightforward yet highly secure, structured, and comprehensive onboarding process specifically designed to validate, authenticate, and welcome new users—such as parents or legal guardians, enrolled students, newly hired faculty members, and qualified staff—into the secure digital ecosystem of the Integrated School Management System.

A prospective user navigates to the dedicated, user-friendly registration portal accessible from the main login page, where they are guided step-by-step through a structured form designed to collect essential identifying details and relevant information required for official records. During this process, the applicant is required to provide accurate personal data including their full legal name, a valid and active email address, a reliable mobile number for official communications, complete residential address details, and specific information regarding their school affiliation or department.

Furthermore, the user is prompted to carefully select their appropriate role within the institution's hierarchy—whether student, parent, teacher, or administrative staff—and to create a strong, secure password that meets defined complexity standards, combining characters, numbers, and symbols to ensure account safety right from the start.

To ensure the highest levels of security, authenticity, and trustworthiness, the entire registration process incorporates multiple verification layers and validation mechanisms. The system automatically sends unique confirmation codes or verification links to both the submitted email address and mobile number, requiring the user to enter or click these to validate ownership and confirm that contact details are accurate and functional.

Additionally, CAPTCHA validation is strictly employed during submission to effectively prevent automated bot registrations, spam entries, or malicious scripts from accessing the platform. Simultaneously, the system performs rigorous database checks to ensure the absolute uniqueness of identifiers such as email addresses, contact numbers, and

usernames, preventing the creation of duplicate accounts, fraudulent entries, or identity conflicts within the database.

Furthermore, strict compliance measures are deeply embedded into the workflow to fully align with international data privacy regulations such as GDPR as well as local educational data protection laws and institutional policies, ensuring that all personal information is collected lawfully, transparently, and only for specified legitimate purposes, while clearly informing users about how their data will be stored, used, and protected. In many cases, especially for students and employees, the process requires the user to associate their new account with an existing institutional record—such as entering a valid student identification number, enrollment reference code, or official employee number—which the system cross-references against pre-existing official data to verify their legitimate belonging and eligibility to be part of the school community.

This critical step ensures that only authorized individuals gain access, thereby maintaining the integrity, exclusivity, and security of the platform and preventing external parties from entering the system.

Upon successful submission, completion of all validation checks, and confirmation of eligibility, the system automatically creates a basic user profile linked securely to their credentials, configured with default privacy settings, standard notification preferences, and access permissions appropriate to their selected role.

Every registration event is meticulously logged within the system's security database, recording the exact timestamp, IP address, and details of the action to serve as a permanent record for auditing, compliance monitoring, and security investigations if necessary.

Finally, the system grants immediate access to the personalized user dashboard tailored specifically to their role, allowing the new user to immediately begin exploring relevant features, updating their detailed profile information, and engaging with the various tools and services available within the school management environment.

## **Create/Edit Profile**

This comprehensive functionality empowers every registered user to fully customize, build, and actively maintain their complete digital identity and institutional records by accessing the dedicated and intuitive “My Profile” section located within their personalized account dashboard.

Serving as the central hub for personal data management, this feature allows users to upload a personalized profile picture, accompanied by built-in validation mechanisms that strictly ensure all uploaded files meet defined criteria regarding maximum file size and accepted image formats, thereby maintaining visual consistency and optimal system performance across all devices.

Beyond visual customization, users are guided to systematically input, review, and update essential biographical and contact information, including current residential addresses, alternative contact numbers, email addresses, and detailed information regarding designated emergency contact persons—details that are absolutely critical for student safety, administrative coordination, and effective communication during urgent situations.

Moving beyond basic identification data, this module enables users to manage and refine role-specific attributes and information tailored precisely to their function within the school community. Students are able to input or update their current academic year, grade level, section details, and vital medical information such as existing health conditions, known allergies, or specific dietary requirements that are highly relevant to school administration and health services.

Meanwhile, parents or legal guardians utilize this section to securely link their profile to multiple children or wards enrolled in the institution, establishing a formal digital association that grants them access to their respective records and performance data. For teachers and faculty members, the functionality allows them to define their subject specializations, department affiliations, assigned advisory roles, and professional

qualifications, ensuring that their official profile accurately reflects their responsibilities and expertise.

Furthermore, users are granted full control over their operational preferences and system behavior settings, such as configuring notification preferences for specific events including attendance alerts, grade updates, assignment deadlines, or important announcement reminders, as well as managing linked accounts or integration settings where applicable.

Every modification, entry, or update made within this section is processed with rigorous real-time validation to ensure absolute data accuracy, correct formatting, and compliance with field requirements, effectively preventing errors or invalid data from entering the central database.

To ensure accountability, transparency, and traceability, the system automatically maintains a complete version history and audit trail of all changes made, recording details such as the nature of the modification, the timestamp of the action, and the identity of the user performing the update.

Additionally, users are provided with the opportunity to enhance their account security measures directly within this section through optional but highly recommended features like two-factor authentication setup, password updates, and privacy control adjustments.

Once saved and validated, all updates are reflected immediately and synchronously across all relevant modules and functionalities of the system, ensuring that institutional records remain consistent, complete, accurate, and readily accessible wherever and whenever the information is needed by authorized personnel.

## **View Matches**

Interpreted within the specific operational, academic, and educational context of Tigbauan Fundamental Learning Center, this versatile use case enables authorized users to effectively query, search, retrieve, and visually explore dynamically generated "matches," logical pairings, or relevant associations established by the system, which encompass a wide range of essential data relationships including class schedules, student-teacher subject assignments, advisory group allocations, inventory availability and resource distribution, attendance records correlated to specific dates or classes, and academic performance comparisons across grading periods or subjects.

This functionality serves as a powerful information retrieval hub where users can access structured data connections that are critical to understanding operational status, resource utilization, and educational progress.

Users interact with this feature through highly intuitive and interactive dashboards equipped with advanced filtering tools, search bars, and sorting options designed to refine and customize results based on various specific criteria such as date ranges, academic terms, specific subjects, grade levels, sections, categories of resources, or status indicators such as pending, confirmed, active, or completed transactions.

For instance, an administrative staff member might utilize these tools to search for currently available laboratory equipment or learning aids; a teacher can view their assigned advisory classes or subject load for the current semester; while a parent or student can review matched schedules for parent-teacher conferences or upcoming school events.

The retrieved results are presented in clear, user-friendly formats—including customizable data tables, visual charts and graphs, or organized calendar views—making complex information easy to digest and analyze.

Additionally, the interface includes convenient functionalities to export displayed data to standard file formats like PDF or Excel, allowing users to generate printed reports or perform further offline analysis and record-keeping, alongside real-time refresh capabilities that ensure the information displayed is always synchronized with the latest updates, modifications, or transactions recorded in the central database.

Furthermore, this functionality empowers users to view intelligent suggestions, recommended connections, or automated allocations generated by the system's internal logic and algorithms—such as recommended remedial classes or learning interventions based on identified academic performance trends, optimal grouping of students for collaborative activities, or available resources and materials aligned specifically with curriculum requirements and learning objectives.

Users have the ability to drill down into individual records or specific entries to view comprehensive detailed information, track historical changes in status, and understand the rationale behind specific pairings or recommendations.

The system also supports proactive engagement by sending automated alerts, pop-up notifications, or email reminders whenever new relevant matches, assignments, or allocations become available, or when existing schedules or resources are modified. This ensures that every user remains fully informed, constantly updated, and able to act promptly upon information that is critical to their academic responsibilities, administrative duties, or involvement in the educational process, thereby enhancing responsiveness, coordination, and overall operational efficiency within the institution.

## **Send Message**

Leveraging a robust, high-performance, and real-time internal messaging module integrated seamlessly into the core architecture of the system, this comprehensive feature allows all authorized users to compose, format, and dispatch instant text-based communications, formal official announcements, circulars, urgent notifications, and important reminders to a diverse range of recipients.

Users have the flexibility to send messages to specific individuals, predefined or custom-defined groups—such as an entire class, a specific grade level, department members, or a parent committee—or even to institutional broadcast channels that reach the entire school community simultaneously, ensuring that information dissemination is targeted, efficient, and scalable according to the situation's requirements.

The system is equipped with a full suite of comprehensive tools and functionalities designed to support effective, clear, and rich communication. Users can easily attach various types of files, documents, or images—such as scanned report cards, learning modules, assignment guidelines, event flyers, or official memoranda—directly within the conversation. It also supports the use of emojis, reactions, and formatting options to enhance clarity, tone, and engagement.

Key features include read receipts and delivery status indicators, which allow senders to confirm exactly when their message has been received and viewed, ensuring accountability and assurance that important information has reached its intended audience. Conversations are automatically organized through message threading and chronological ordering, making it simple to track the history of discussions, follow up on previous topics, and maintain context over time.

Additionally, the system offers advanced capabilities such as scheduling messages to be sent at a future date and time, which is highly useful for setting reminders or releasing announcements at optimal periods without requiring manual intervention.

Security, confidentiality, and privacy are prioritized at every level through the implementation of strict end-to-end encryption protocols that protect the content of all correspondence from interception or unauthorized access, ensuring compliance with data privacy laws and institutional policies.

Furthermore, a complete and immutable audit log is automatically maintained for every message sent or received, recording timestamps, sender and recipient details, and content summaries, serving as a reliable historical reference, evidence of communication, and a vital resource for compliance and accountability purposes.

To ensure maximum reach and inclusivity, the messaging system features deep integration with external email services and SMS gateways, effectively extending the reach of communication beyond the platform itself and guaranteeing that even users who are not currently online or active within the system receive critical updates and alerts promptly via alternative channels.

The system further supports the categorization and tagging of message types—such as urgent alerts, general information, academic concerns, administrative updates, or private consultations—allowing users to effectively organize their inbox, filter conversations, and prioritize their responses based on the importance or nature of the message.

Users also have control over their interaction preferences, including the ability to manage their availability status, archive or retrieve old conversations for record-keeping, and control their contact list by blocking unwanted communications or filtering specific senders.

Ultimately, this feature serves as the central hub for all institutional interaction, effectively bridging potential communication gaps that previously existed within the school community and ensuring that vital information, instructions, and feedback flow quickly, securely, transparently, and reliably between all stakeholders—administrators,



teachers, parents, and students—while maintaining a permanent, retrievable, and legally valid record of all interactions for the benefit of the institution.

## **Administrator Use Cases**

Exclusive and strictly reserved only for users assigned with elevated administrative roles—such as school principals, designated IT staff, system administrators, school registrars, and senior management personnel—who possess the highest levels of access privileges, authority, and institutional responsibility within the platform, this comprehensive set of use cases focuses primarily on strategic system oversight, content moderation, rigorous security management, data governance, regulatory compliance, and overall operational governance designed to maintain absolute operational integrity, enforce established institutional policies and standards, and effectively scale the platform's usage, capacity, and capabilities across the entire educational institution.

Supported by advanced analytical capabilities, bulk processing operations, automated tools, and highly configurable permission settings, administrators act as the primary custodians, guardians, and controllers of the entire digital ecosystem, bearing full and ultimate responsibility for ensuring comprehensive data security, privacy protection, and information confidentiality; managing user access rights, account provisioning, deactivation, and authentication protocols; configuring core system settings, functional workflows, and operational parameters to align with school requirements; and continuously monitoring overall system performance, reliability, uptime, and usage patterns to guarantee that the platform functions smoothly, efficiently, securely, and continues to adapt and meet the evolving operational, academic, and administrative needs of the school community over time.

Furthermore, these specialized functionalities empower administrators to perform critical high-level tasks including the centralized management of the complete database structure, execution of regular data backups and recovery procedures, maintenance of data accuracy and consistency, and oversight of all transaction logs and audit trails to ensure transparency and accountability.

They are equipped with tools to configure and customize the system's modules, define role-based access controls, establish validation rules, and set parameters for automated processes such as attendance computation, grading systems, and inventory alerts.

Administrators also hold the authority to resolve system errors, address user issues and support tickets, manage announcements and broadcast communications, and oversee the integrity of all content published within the platform.

Additionally, they are responsible for generating executive-level reports, statistical analyses, and compliance documents required by educational authorities, while managing software updates, feature enhancements, and integration with other systems to ensure long-term sustainability and technological relevance.

By centralizing these powerful capabilities under strict administrative control, this category ensures that the system remains stable, secure, legally compliant, and perfectly aligned with the strategic goals and policies of Tigbauan Fundamental Learning Center, Inc., serving as the foundational backbone that supports the reliability and trustworthiness of every operation performed within the environment.

## **Monitor User**

Within this capacity, authorized administrators gain access to a centralized, comprehensive, and highly sophisticated analytics dashboard specifically designed to systematically track, visualize, measure, and analyze all user activities, interactions, and system events both in real-time as they occur and through detailed historical data logs stored within the database.

This powerful functionality provides deep, data-driven insights covering a wide spectrum of operational metrics, including detailed analysis of login patterns, access frequency, and duration of sessions, which helps understand when and how often the platform is being utilized. It also features feature usage metrics that clearly identify the most frequently accessed versus the least utilized modules—such as attendance tracking, grading systems, inventory management, or communication tools—revealing which functions are most valuable to users and which may require further optimization or promotion.

Furthermore, it incorporates advanced anomaly detection mechanisms that automatically flag potentially suspicious activities, such as unusual login locations or IP addresses, multiple consecutive failed authentication attempts, access occurring outside standard operational hours, or irregular data retrieval volumes, serving as an early warning system against security threats.

Engagement reports are also generated to quantitatively measure user interaction levels and effectiveness, including message response rates, form completion status, submission timeliness, and frequency of updates, providing a clear picture of user participation and adoption.

The system goes beyond simple logging by generating visual behavioral heatmaps that illustrate exactly how users navigate the interface, highlighting which buttons, links, or sections attract the most attention and identifying potential usability bottlenecks or confusing navigation paths.

It also includes specialized tools for creating comprehensive compliance audit trails, ensuring that every action taken within the system is documented and traceable, while assisting administrators in identifying potential issues such as inactive or dormant accounts, policy violations, improper data access, or unauthorized modification attempts. All these valuable insights and records can be easily exported in structured formats such as CSV, Excel, or PDF, allowing management to utilize the data for strategic decision-making, process improvement initiatives, and official compliance reporting.

Furthermore, the scope of this monitoring capability extends well beyond user actions to encompass critical system performance indicators essential for maintaining health and stability, including server response times, application error occurrence rates, database query efficiency, and overall server load or capacity usage to prevent downtime or slowdowns.

A detailed and immutable data modification log is also maintained, recording exactly which user changed specific records, what exact alterations or values were modified, the previous state of the data, and the precise time and date of the modification, ensuring absolute accountability and enabling accurate rollbacks if necessary. By thoroughly understanding and evaluating how different user groups—administrators, teachers, staff, parents, and students—interact with the system, administrators can accurately identify training needs, determine areas requiring additional user support or documentation, and optimize operational workflows to reduce redundancy and inefficiency.

Moreover, this capability allows for the early detection of security breaches, data misuse, or policy violations before they escalate into serious incidents, ensuring rigorous enforcement of standards and guaranteeing that the entire platform is being utilized effectively, appropriately, securely, and strictly in accordance with established institutional policies, data privacy regulations, and educational governance standards.

## **Manage Accounts**

Administrators are tasked with the complete end-to-end lifecycle management of all user accounts within the system, a critical responsibility performed exclusively through a robust, dedicated administration panel equipped with powerful, scalable tools designed for comprehensive user governance and identity management.

Key responsibilities encompass a wide range of operational functions starting with the bulk creation or mass import of user data directly from standard file formats such as CSV or Excel—an essential and time-saving feature specifically developed for handling large-scale operations such as new student enrollments at the start of the academic year, batch admissions, or simultaneous staff onboarding—ensuring accuracy and efficiency when populating the database with hundreds of records at once.

Beyond initial creation, administrators regularly edit and update user profiles, adjust access permissions in real-time, or upgrade and redefine roles to reflect changes in organizational structure or job function, such as promoting a teacher to a department head, assigning grade-level coordinators, or reclassifying support staff.

They are also authorized to suspend, temporarily disable, or permanently deactivate accounts that belong to inactive students, graduated batches, separated employees, or users exhibiting problematic behavior or policy violations, with every action accompanied by detailed notes regarding the reason and exact timestamp, all securely preserved within the audit trail for compliance and historical reference.

This function also facilitates secure password resets and account recovery procedures, utilizing encrypted token-based systems to ensure credentials are restored safely without compromising security.

Furthermore, administrators are responsible for defining detailed role assignments with granular, tiered permissions and rule-based access controls that precisely determine what features, modules, datasets, and functions each user group can view, modify, or execute—for example, strictly restricting parents to view-only access limited exclusively to their own child's records, while granting full editing rights and assessment

capabilities to subject teachers, and reserving system configuration controls only for senior management.

Additionally, this extensive use case encompasses the creation and ongoing management of access groups, departments, committees, and organizational units, allowing for structured grouping of users to simplify permission assignments and communication distribution.

It also involves the configuration of multi-level approval workflows required for sensitive or high-impact actions like budget adjustments, major record modifications, or data exports, ensuring that critical changes undergo proper authorization and oversight before being finalized.

Advanced capabilities include the ability to identify, review, and merge duplicate profiles to maintain data integrity, as well as safely transferring ownership of data, assets, or responsibilities when staff roles change or personnel are reassigned, ensuring continuity and preventing data loss or orphaned records.

Administrators also oversee system-wide security policies, including enforcing organization-wide password complexity requirements, mandatory rotation and expiration rules, and setting restrictions on login locations or device access.

They conduct periodic, scheduled reviews of all access rights and privileges to ensure that permissions remain relevant, necessary, and aligned with current job descriptions, effectively adhering to the principle of least privilege and removing any excessive or outdated access.

Ultimately, this comprehensive functionality serves as the central control mechanism and security foundation that safeguards the entire system environment, ensuring strict data privacy, maintaining accurate and consistent user records, preventing unauthorized access, and enforcing governance standards that are perfectly aligned with institutional hierarchies, educational policies, and regulatory compliance requirements.

## **Configure System**

Representing one of the most exclusive, powerful, and strategically significant administrative functions within the platform, this capability allows fully authorized personnel—such as system administrators and senior management—to define, adjust, and modify the fundamental operational parameters, structural framework, business logic, and behavioral rules that govern the entire platform, effectively tailoring the digital environment to suit the specific context, requirements, and policies of Tigbauan Fundamental Learning Center, Inc.

Administrators access a dedicated, password-protected configuration panel designed to centralize all high-level adjustments, where they manage core institutional details that form the backbone of school operations, including the definition and activation of academic years, precise term dates and grading periods, official holiday calendars and non-working days, standard grading systems and numerical scales, passing and failing criteria, and the complete structural organization of the institution—ranging from departments, grade levels, and sections to subject offerings and curriculum mapping.

Beyond structural definitions, they possess the comprehensive ability to customize module-specific business rules, workflows, and internal logic according to established educational standards and administrative procedures.

This includes setting attendance thresholds for eligibility, promotion, or intervention programs; defining the frequency, timing, and intended recipients of automated notifications and alerts; establishing inventory reorder levels, minimum stock limits, and critical stock alert triggers; configuring the computation formulas for grades and weighted averages; and adjusting the logic and weighting criteria behind algorithms used for class scheduling, resource allocation, student grouping, or performance reporting.

By controlling these parameters, administrators effectively ensure that the system's automated behavior, calculations, and outputs align precisely and seamlessly with the school's unique policies, regulatory requirements, and educational framework.

Furthermore, this function extends deeply into the management of critical security settings and compliance protocols, where administrators establish organization-wide standards such as password complexity requirements, session duration and automatic inactivity timeout limits, data encryption standards, and the frequency, timing, and storage destinations for automated data backup schedules and disaster recovery plans. It also encompasses extensive interface customization options and user experience controls, such as uploading institutional branding elements including official logos, color schemes, and visual themes, arranging menu layouts and navigation structures for optimal efficiency, and selecting or restricting available features and modules displayed to different user groups based on their roles.

This exceptional level of high-level configurability ensures that the system remains highly flexible, fully adaptable, and effectively future-proof, allowing the platform to evolve smoothly alongside the institution's changing requirements, educational reforms, curriculum updates, or administrative restructuring initiatives over time, all without necessitating complex modifications to the underlying source code or technical infrastructure.

Ultimately, this functionality empowers the institution to maintain full control over its digital operations, ensuring that the technology serves the school's processes rather than forcing the school to adapt to the limitations of the technology.



## **Generate Comprehensive**

Administrators utilize advanced, enterprise-grade reporting tools to compile, aggregate, analyze, and export institution-wide data essential for evidence-based decision-making, regulatory compliance, accreditation, and long-term record-keeping, forming a robust backbone for institutional governance and operational excellence.

Unlike standard users who only access personal or limited data views relevant to their specific role—such as teachers viewing only their class rosters or department heads seeing only their unit's budget details—administrators have unrestricted access to generate cross-functional reports that seamlessly combine information from various interconnected modules across the entire system.

This unique capability enables them to perform deep, multi-dimensional analyses that would otherwise be impossible, such as correlating daily and long-term attendance records with academic performance metrics to identify how consistent participation impacts student achievement and retention, analyzing inventory usage patterns against approved budget allocations and procurement schedules to detect waste, optimize stock levels, and ensure financial accountability, reviewing detailed communication logs and interaction histories to measure stakeholder engagement levels and assess the effectiveness of outreach strategies to students, parents, and staff, or auditing comprehensive user activity patterns and access logs to strengthen security protocols, verify compliance with data privacy regulations, and ensure all system actions are transparent and traceable.

The system provides an extensive library of pre-built, standardized templates designed to align with local education policies, national regulatory requirements, and international accreditation standards, covering common official documents including detailed enrollment summaries and demographic breakdowns, attendance percentages, historical trends, and absence tracking reports, inventory audit trails with full transaction histories, student performance analytics ranging from individual progress reports to cohort-wide achievement comparisons, and staff workload assessments that balance teaching duties, administrative tasks, and support responsibilities fairly and accurately.

Beyond these ready-to-use resources, the platform also supports the creation of fully customized reports through an intuitive drag-and-drop interface that requires no specialized technical skills, allowing administrators to select specific data fields, apply precise filters, define custom calculation methods, and arrange visualizations—such as charts, graphs, and comparative tables—according to unique institutional requirements, special project needs, or ad-hoc inquiry requests.

Reports can be configured for automatic generation on daily, weekly, monthly, or annual cycles, with scheduled distribution sent directly to designated stakeholders via secure channels, formatted to strictly meet government or regulatory formatting and content standards to eliminate manual adjustments, and stored in a highly secure, encrypted, and searchable digital archive that preserves data integrity while making historical records easily retrievable for audits, reviews, or future analysis.

This powerful functionality transforms vast volumes of raw, disconnected data into meaningful, actionable insights and clear, understandable visualizations, serving as the solid foundation for strategic administrative planning, informed resource allocation that prioritizes high-impact areas, rigorous policy evaluation to measure effectiveness and guide improvements, and the proactive, forward-thinking strategic management of all school resources and daily operations.

By turning data into intelligence, this reporting system empowers administrators to identify emerging challenges early, recognize opportunities for growth and improvement, demonstrate accountability to governing bodies and the public, and ultimately foster a learning and working environment that is continuously optimized for student success, staff satisfaction, and long-term institutional sustainability and growth.

## **Manage Data and Archives**

To guarantee long-term data integrity, quality, and legal compliance, administrators exercise full control over all aspects of data management, storage optimization, and systematic archiving throughout the entire information lifecycle, from the moment data is captured and entered into the system, through its active use in daily operations, to its eventual preservation or secure disposal.

This comprehensive responsibility includes performing both scheduled, automated backups and on-demand manual database backups, creating redundant copies stored in secure, separate locations to effectively prevent data loss due to unforeseen technical failures, human errors, or unexpected disruptions, thereby ensuring business continuity and uninterrupted access to critical information.

Furthermore, administrators execute secure, thoroughly validated data import and export procedures, facilitating seamless integration with external platforms such as government reporting systems, accounting software, or third-party learning tools, as well as supporting safe and accurate data migration during system upgrades or transitions, all while strictly maintaining data structure and consistency to avoid corruption or loss during transfer.

They also manage the structured archival process for historical records—such as completed academic years, records of graduated students, resolved financial transactions, or finalized administrative documents—effectively moving inactive or non-current data to separate, secure storage repositories; this strategy not only optimizes the performance and speed of the live operational database by reducing clutter and processing load but also ensures that historical information remains fully preserved, indexed, and easily retrievable whenever required for legal inquiries, official audits, accreditation reviews, or long-term institutional reference.

Beyond storage and preservation, administrators oversee regular and systematic data cleansing operations designed to proactively identify and remove duplicate entries, correct inaccuracies or inconsistencies, standardize formats across different data sets, and enforce strict adherence to established institutional data standards and classification rules; this ongoing maintenance is essential to ensure the reliability, precision, and high quality of all information within the system, forming a trustworthy foundation for reporting and decision-making.

Furthermore, they define and enforce rigorous access controls and permission protocols regarding who can view, retrieve, or restore archived data, ensuring that

sensitive information remains protected against unauthorized access and fully compliant with data privacy laws and security regulations.

Administrators are also responsible for establishing formal, written data retention policies that clearly specify how long different types of records must be kept, aligned precisely with local educational regulations, national legal frameworks, and accreditation requirements, as well as overseeing the secure, verifiable, and permanent disposal or anonymization of records once they are no longer operationally necessary or have exceeded their mandated retention period, eliminating unnecessary liability and reducing storage costs.

This use case is absolutely critical for maintaining the overall quality, reliability, legal compliance, and enduring institutional memory of the information stored within the Integrated School Management System, ensuring that data remains a trusted, protected, and valuable strategic asset throughout the entire system's operational lifespan, supporting accountability, transparency, and stability for the institution over many years of service.

## **CHAPTER 4**

### **4.2 Overview of the Developed System**

The Tigbauan Fundamental Learning Center School Management System (TFLC SMS) is a web-based application designed to streamline and automate the management of student records, attendance, academic performance, inventory, and communication between school and home. The system assists school administrators, teachers, and parents in efficiently managing student enrollment, personal information, attendance, grades, school resources, and parent-teacher communication. By replacing manual, paper-based record-keeping with a centralized digital platform, the system improves data accuracy, accessibility, and overall administrative efficiency.

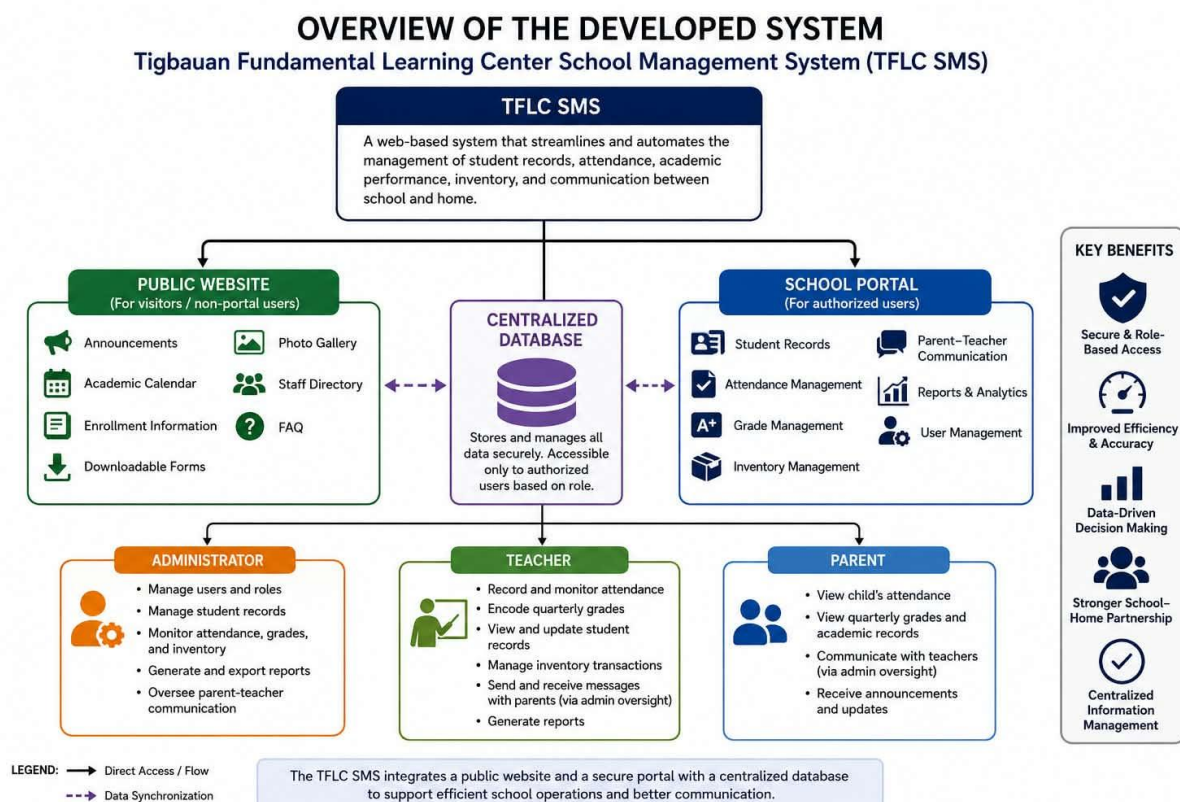
Unlike traditional paper-based student record management, which is often time-consuming and prone to error, the TFLC SMS integrates computerized data management processes that enable faster retrieval, updating, and monitoring of student information. The

completed system consists of three interconnected parts: a role-based Portal for day-to-day school operations, a public-facing Website for prospective and current families, and a shared database layer that keeps both consistent.

The major objectives realized by the developed system are as follows:

1. Automate the recording and monitoring of student attendance through a web-based interface used by teachers.
2. Centralize and securely store student records, accessible only to authorized users according to their role.
3. Provide a grade management module that lets teachers encode quarterly grades and lets administrators and parents view a consolidated academic record.
4. Provide an inventory tracking system for school resources, including a low-stock indicator and a full transaction history.
5. Provide a secure parent-teacher communication portal, with administrator oversight of all conversations.
6. Generate accurate, filterable reports for attendance, grades, inventory, and student records, exportable to CSV/Excel.
7. Provide secure user authentication and role-based access control for Administrators, Teachers, and Parents.
8. Provide a public website (Announcements, Academic Calendar, Enrollment information, Downloadable Forms, Photo Gallery, Staff Directory, and FAQ) for visitors who do not yet have a Portal account.

The development of the TFLC SMS follows modern software engineering principles by incorporating a responsive web interface, secure database management, user authentication, role-based access control, and efficient data processing. The completed system consists of several interconnected modules that work together to manage student information, monitor academic records, attendance, and inventory, facilitate communication, and support both internal school administration and public-facing information needs.



### 4.3 Software Development Methodology

The development of the TFLC SMS adopted the Agile Software Development Methodology, an iterative approach that emphasizes continuous collaboration, incremental

development, and regular evaluation throughout the software development life cycle, consistent with the methodology defined in Chapter 3. Agile was selected because it accommodates evolving user requirements and encourages continuous feedback from school stakeholders, ensuring that the resulting system effectively addresses the operational needs of Tigbauan Fundamental Learning Center.

The Agile methodology enabled the development team to divide the project into manageable iterations. At the end of each iteration, a functional component of the system was developed, tested, and evaluated before proceeding to the next. This approach minimized development risk, facilitated early detection of software issues, and ensured that each module met expected quality standards before integration into the complete system. The development process consisted of the following phases:

**Requirements Gathering.** The team gathered information from the school's administrators and teachers through interviews, observations, and document analysis, identifying existing procedures for student enrollment, record management, attendance monitoring, inventory tracking, and parent communication, along with common challenges such as delays in record retrieval, data redundancy, and record inaccuracies. This information formed the basis for the functional and non-functional requirements defined in Chapter 1 and Chapter 3.

**System Planning.** The team established the project scope, objectives, system specifications, development schedule, and resource allocation, including a feasibility analysis to evaluate the technical, operational, and economic viability of the proposed system.

**System Design.** The team designed the overall architecture of the system, including the database structure, user interface layouts, and system modules (user authentication, student information management, attendance monitoring, grade management, inventory management,

communication, and report generation), producing wireframes and prototypes as a blueprint for construction.

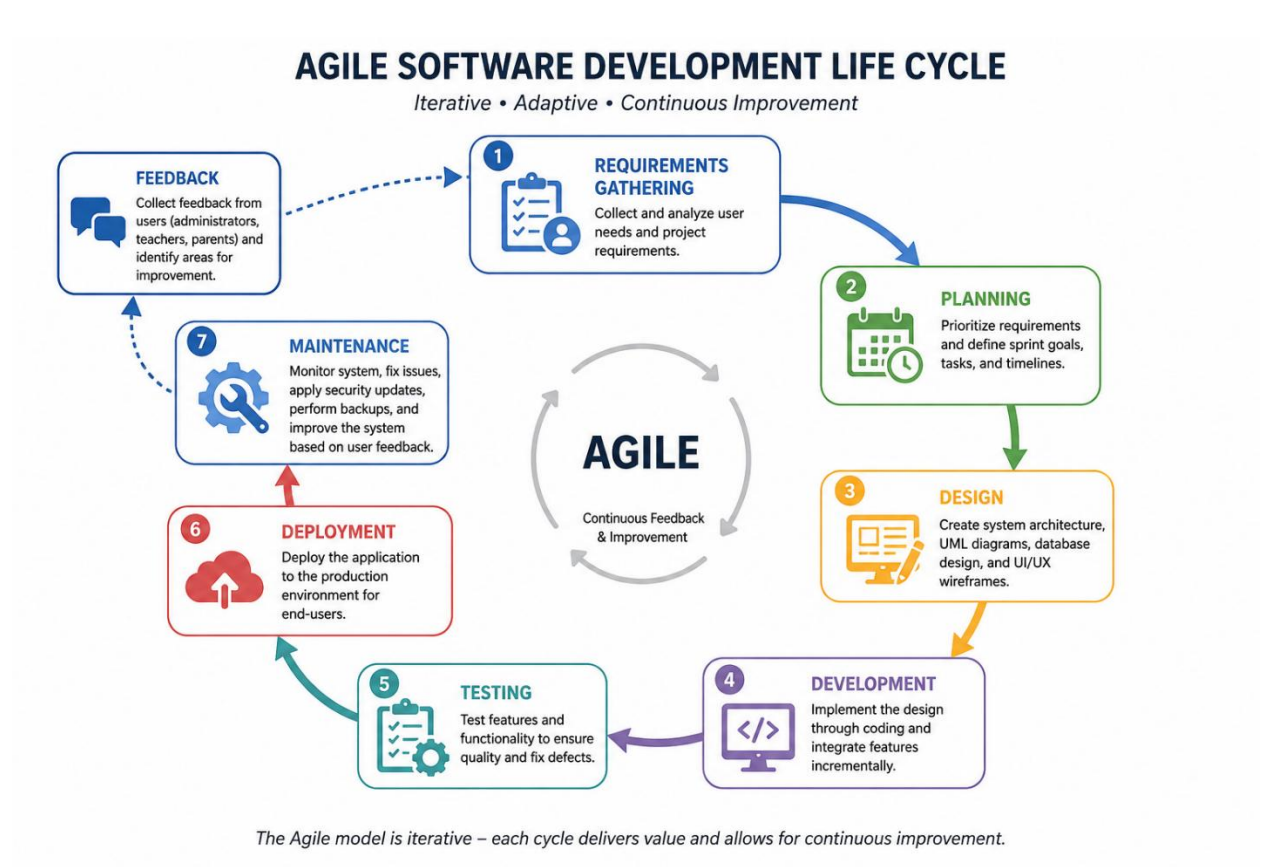
**Development.** The team implemented each module in working code, following the client-server architecture and technology stack defined in Chapter 3.3 and 3.4. Development proceeded module by module (student records, attendance, grades, inventory, messaging, reporting, and the public website), with each module reviewed before the next was started.

**Testing.** Each completed module underwent unit testing (verifying individual components functions correctly), system testing (verifying modules work correctly together), and security testing (verifying role-based access control, input validation, and other safeguards behave as intended). Defects identified during this phase were logged, corrected, and re-tested; the full defect log and testing results are presented later in this chapter.

**Deployment.** After development and testing were substantially complete, the system was deployed to an online hosting environment, making it accessible through a web browser with an internet connection. Deployment involved configuring the web application, uploading system files, setting up the production database, and verifying that all functionalities operated correctly in the live environment. User accounts were configured for authorized administrators, teachers, and parents to ensure secure, role-based access.

**System Maintenance.** Following deployment, ongoing maintenance activities are performed to ensure the continued reliability, security, and efficiency of the application. These include monitoring system performance, fixing reported issues, applying security enhancements, and performing regular database backups. User feedback from administrators, teachers, and parents is collected to identify areas for improvement, and necessary updates are implemented





## 4.4 System Architecture

The Student Management System follows a Three-Layer Architecture composed of the Presentation Layer, Application Layer, and Data Layer. These layers work together to manage student information, attendance, grades, inventory, and communication efficiently and securely.



### *Presentation Layer*

The presentation layer is the part of the system that users directly interact with. It serves as the front-end of the system, where administrators, teachers, and parents access the Portal,

and where visitors access the public website through a standard web browser. This layer is designed to be simple, user-friendly, and easy to navigate on both desktop and mobile devices.

**Functions:**

1. User login and logout
2. Student registration forms
3. Viewing and updating student profiles
4. Attendance monitoring pages
5. Grade viewing and management
6. Inventory management pages
7. Parent-teacher messaging pages
8. Reports and dashboard display
9. Public-facing announcements, calendar, enrollment, forms, gallery, staff directory, and FAQ pages

When a user logs in, enters student information, checks attendance records, or views grades, the request is made through the Presentation Layer. The layer then sends the request to the Application Layer for processing and displays the results back to the user.

**Importance:** The Presentation Layer determines how usable the system is for staff with varying levels of technical comfort. Keeping this layer visually clear and consistent across every module was treated as a priority throughout development, since a confusing interface undermines even a technically correct backend.

***Application Layer***

The Application Layer is responsible for processing user requests and enforcing the system's business rules. It connects the Presentation Layer to the Data Layer.

**Main Functions:**

10. User login verification and role-based access enforcement
11. Student enrollment processing
12. Attendance recording
13. Grade computation and validation
14. Subject and class assignment
15. Inventory stock tracking and low-stock alerting
16. Parent-teacher message routing
17. Report generation
18. User account management

**Responsibilities:**

19. Validates user input before it reaches the database.
20. Processes requests from users and routes them to the correct module.
21. Ensures that only authorized users, based on their assigned role, can access or modify specific features and data.
22. Sends and receives data from the database using secure, parameterized queries.

When a teacher records attendance or enters grades, the Application Layer checks whether the information is valid and whether that teacher is authorized to submit it. After validation, it sends the data to the database for storage. If a user requests information, the Application Layer retrieves only the data that user role is permitted to see and sends it back to the Presentation Layer.

**Importance:** This layer is where the system's security guarantees are actually enforced. Role checks, input validation, and access control all happen here, not merely as an interface

convention. A weakness at this layer would compromise the entire system regardless of how well the Presentation Layer is designed.

### ***Data Layer***

The Data Layer is responsible for storing, organizing, retrieving, and protecting all information within the system. It serves as the central repository where school records are securely maintained using a MySQL relational database.

#### **Stored Data:**

- 23. Student personal and enrollment information
- 24. Attendance records
- 25. Grades and academic performance records
- 26. Class and subject assignments
- 27. Inventory items and inventory transaction history
- 28. Teacher, administrator, and parent account information (stored together in a single Users table, distinguished by a role field)
- 29. Parent-teacher messages
- 30. An audit log of key system actions
- 31. Public website content: announcements, calendar events, downloadable forms, gallery photos, FAQs, and the staff directory

Whenever data is added, updated, or requested, the Data Layer handles the underlying database operations, saving the information and retrieving it whenever needed by the Application Layer.

**Importance:** The Data Layer is what makes the system's records trustworthy over time relationships between tables (for example, linking a parent account to only their own child's

records) are what actually enforce data privacy at the storage level, rather than relying solely on the interface to hide information.

## **4.5 Hardware and Software Requirements**

### ***Hardware Requirements***

The Student Management System requires the following hardware components to ensure smooth operation and reliable performance:

#### **Server/Administrator Computer**

- 32. Processor: Intel Core i3 or higher
- 33. RAM: 4 GB minimum (8 GB recommended)
- 34. Storage: 500 GB Hard Disk Drive (HDD) or 256 GB Solid State Drive (SSD)
- 35. Monitor: 1366 x 768 screen resolution or higher
- 36. Keyboard and Mouse
- 37. Network Connection: LAN or Internet connection

#### **User Computer**

- 38. Processor: Dual-Core Processor or higher
- 39. RAM: 2 GB minimum
- 40. Storage: 120 GB available space
- 41. Monitor: 1366 x 768 resolution or higher
- 42. Keyboard and Mouse
- 43. Internet Connection for accessing the system

#### **Network Equipment**

- 44. Router or Modem for Internet connectivity
- 45. Network Switch (if using a Local Area Network)
- 46. Ethernet cables or Wi-Fi connection

### **Optional Devices**

- 47. Printer for printing student records and reports
- 48. Scanner for digitizing student documents
- 49. Backup Storage Device (External Hard Drive or USB Drive)

## ***Software Requirements***

### **Operating System**

- 50. Windows 10 or Windows 11
- 51. Linux-based operating systems (optional, for server deployment)

### **Web Server**

- 52. XAMPP Control Panel (local development)
- 53. Apache Web Server
- 54. A PHP/MySQL-compatible shared hosting provider (for online deployment)

### **Programming Languages and Technologies**

- 55. PHP — for server-side development
- 56. HTML, CSS, and JavaScript — for user interface structure and interactivity
- 57. Tailwind CSS — a utility-first CSS framework used for responsive, consistent styling across all modules

### **Database Management System**

58. MySQL Database

### **Web Browser**

59. Google Chrome (recommended)

60. Mozilla Firefox

61. Microsoft Edge

### **Development Tools**

62. Visual Studio Code (VS Code)

63. phpMyAdmin for database management

64. Git for version control

These hardware and software requirements ensure that the TFLC School Management System operates efficiently, provides reliable data management, and supports the daily administrative activities of the school.

## **4.6 Database Design**

The database design of the TFLC School Management System serves as the foundation for storing, organizing, and managing school information efficiently. It uses a MySQL relational database, currently composed of eighteen interconnected tables grouped into four functional areas.

**Core school records.** The `students` table stores student details such as student ID, name, gender, birth date, address, and parent contact information. The `users` table stores login credentials for every account in the system —Administrators, Teachers, and Parents alike are stored in this single table, distinguished by a `role` field, rather than in separate per-role tables; this keeps authentication and role-checking centralized and consistent. The `student_parent`



table links a parent account to one or more student records, so a parent's view can be correctly filtered to only their own child. The `classes` table manages class/section information and its assigned teacher.

**Academic records.** The `attendance` table records attendance detail including student ID, teacher ID, date, and status (Present, Absent, Late, or Excused). The `subjects` table lists the subjects offered per grade level, and the `grades` table records each student's quarterly grade per subject, per school year, linked to both the student and the teacher who recorded it.

**Operations and accountability.** The `inventory` table tracks school assets and supplies (item name, category, quantity, minimum stock level, and location), while the `inventory_transactions` table logs every stock change, including who made it and when. The `messages` table stores parent-teacher communications, linked to the relevant student. The `audit_log` table records key administrative actions system-wide (such as adding, editing, or deleting a record) for accountability, and the `login_attempts` table supports login security by tracking failed login attempts.

**Public website content.** The `announcements`, `calendar_events`, `downloadable_forms`, `gallery_photos`, `faqs`, and `staff_directory` tables store the content shown on the public-facing website, managed independently from the login-protected Portal data.

Relationships between these tables—primarily through foreign keys such as `student_id`, `user_id`, and `subject_id`—ensure data consistency and minimize duplication of records. This structure enables fast retrieval of information, accurate report generation, secure data storage, and efficient management of school records, supporting both the daily

administrative operations of Tigbauan Fundamental Learning Center and the public-facing information needs of prospective families.

## **CHAPTER 5**

### **CONCLUSION AND RECOMMENDATIONS**

#### **5.1 Introduction**

This chapter presents the conclusion and recommendations for the Tigbauan Fundamental Learning Center Student Management System (TFLC-SMS). It summarizes the findings of the study, evaluates whether the project objectives were achieved, and provides recommendations for future improvements and implementation. The system was developed to improve the management of student information, enrollment, attendance, grades, and report generation within Tigbauan Fundamental Learning Center.

#### **5.2 Summary of Findings**

The development and evaluation of the Tigbauan Fundamental Learning Center Student Management System produced the following findings:

The system successfully computerized student records, reducing the need for manual filing and paperwork.

Enrollment information can be stored, updated, and retrieved efficiently, improving administrative operations.

Attendance monitoring became more organized and accurate through digital record management.

Grade recording and report generation were completed faster compared to manual processes.

User testing showed that the system is easy to use and helps school personnel perform tasks more efficiently.

The centralized database improved data security, accessibility, and record management.

### **5.3 Conclusion**

The Tigbauan Fundamental Learning Center Student Management System proved to be an effective solution for managing student information and school records. The system successfully met its objectives by providing a centralized platform for student registration, attendance monitoring, grade management, and report generation.

The implementation of the system reduced manual workloads, minimized errors in record keeping, and improved the efficiency of administrative processes. Furthermore, the system enhanced the accessibility and security of student information, enabling school personnel to perform their responsibilities more effectively.

